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UNIVERSITY OF CALIFORNIA,
MERCED

Disambiguating Consciousness:
A Pluralistic Integration of Neuroscience and Philosophy

DISSERTATION

submitted in partial satisfaction of the requirements
for the degree of

DOCTOR OF PHILOSOPHY

in Cognitive and Information Sciences

by

Andrew Proulx

Dissertation Committee:
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2025

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University of California, Merced

2025

Dedication
To my Mom and my Grami,
my Brother,
my Dad and Unc

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Abstract

Disambiguating Consciousness:

A Pluralistic Integration of Neuroscience and Philosophy

The primary and overarching goal of this dissertation is to disambiguate consciousness through a combined approach of physiological experimentation and philosophical investigation. I start with a behavioral experiment in Chapter 2 on inattentional deafness that lays the groundwork for a follow-up EEG paradigm. Both experiments were conducted using a 2×2 design that manipulated perceptual and/or cognitive load while measuring auditory awareness. Our behavioral results aligned with load theory. We found that high perceptual load reliably led to failures in sound detection, while effects of cognitive load were more nuanced. In Chapter 3, I extended this paradigm to explore the neural correlates of consciousness (NCCs) via EEG analysis, with a central focus on how task relevance and the act of reporting may be conflated with perceptual awareness. The neurophysiological results of the experiment included both event related potentials (ERPs) and oscillatory activity. For ERPs, I found a centroparietal N2 and P300, as well as a frontocentral P1–N1–P2–N2 sequence. Aligning with previous literature, the P300 component emerged only in task-relevant (report) conditions. A frontocentral N2 was observed across all conditions, however, it was also modulated by task relevance (reporting). Oscillatory activity results revealed modulations in the alpha, beta, and theta frequency bands. No ERPs were revealed when comparing the no report conditions, however, an atypical frequency band was found in the overlapping theta/alpha range (7-8.5 Hz).

Based on these findings, I propose a graded model of consciousness that bridges empirical data and theoretical perspectives. In this, I emphasize a pluralistic and pragmatic approach to consciousness, in which all theories hold a relatively truthful perspective, and the “right” theory depends on the specific context. Furthermore, in Chapter 4, I challenge the idea of a separation between access and phenomenal consciousness, framing consciousness instead as a temporally extended, relational process. In the final chapter, this perspective is expanded into a classification space for potentially conscious systems across existence. Working from the common definition of consciousness as “subjective experience,” I employ subjectivity and information processing as the two core dimensions of our classification space. Awareness, as I define it, is a fundamental presence and the essence of consciousness. This underlying awareness is unveiled and expressed through the unique architecture of a subjective information processing system. Overall, our work draws from philosophy, neuroscience, evolutionary biology, and dynamical systems theory to offer a pluralistic and integrative account of consciousness.

Chapter 1

Introduction to the Field of Consciousness

Consciousness remains one of the most perplexing phenomena within the scientific community. The overarching goal of this dissertation is to bring greater clarity to the science of consciousness by integrating philosophical inquiry with empirical research. In chapter 1, I map out the philosophical and theoretical landscape, discussing foundational concepts such as phenomenality, accessibility, state/creature consciousness, subjectivity, and attention. In chapter 2, I review the concept of inattentional phenomena and present a behavioral experiment of inattentional deafness. Chapter 3 involves connecting inattentional phenomena to the neuroscience of consciousness, where I present my empirical findings on an inattentional deafness EEG paradigm. In chapter 4, I propose a pluralistic model of consciousness in which cognitive access and phenomenality exist on a continuum. Finally, I apply these insights to systems across nature, synthesizing various fields of cognitive science such as philosophy, neuroscience, evolutionary biology and dynamical systems theory, resulting in a two dimensional classification space for potentially conscious systems.

1.1 Phenomenal & Access Consciousness

Since I will be referencing phenomenal consciousness (P-consciousness) and access consciousness (A-consciousness) throughout the dissertation, I will begin with a review of these concepts. The relationship between these two concepts has evolved over the years, with controversy still existing. Our goal in this introductory chapter is not to argue for or against any position, but to review the ideas and provide working definitions for our framework. In chapter 4, I will discuss in greater detail the relationship between phenomenality and access.

Phenomenal Consciousness

P-consciousness is the intrinsic feeling and "what-it-is-likeness" within experience, involving both the first person point of view and the unique presentation of what you perceive. It is the feeling you have in everything you do and the appearance of how everything looks to you – from eating your favorite food, to listening to music and dancing, watching the sunset, and even dreaming. It is similar, if not identical, to how Nagel (1974) and Chalmers (2023) define consciousness as subjective experience¹. As Block (1995) describes it, P-consciousness refers to the experiential aspect of consciousness, where the totality of the experiential properties of a state constitutes "what it is like" to be in that state. The most fundamental properties of P-consciousness include sensations, but may also include feelings, perceptions, thoughts, desires, and emotions.

¹ A deeper analysis of subjective experience, including its meaning and how it arises, will be expounded upon in the final chapter.

While P-conscious and perceptual experiences often have representational content and can be conceptual, this is not necessarily the case (Block, 1995). However, it should be clarified that conceptual content is still phenomenally conscious, such as having the subjective experience of a thought, daydream, or philosophical musing – there is a what-it-is-likeness and appearance to all of these phenomena.

What-it-is-likeness is widely used to characterize P-consciousness, but its meaning can be deepened by examining the etymology of the word "phenomenal". The term originates from the Greek *phainómenon*, derived from *phainō*, which traces back to the Proto-Indo-European root *bha*, which means "to shine," "to reveal," "to appear or present," or "to become known and manifest" (Oxford English Dictionary, n.d.; Harper, n.d.; Heidegger, *Being and Time*, §7a; Watkins, 2011). In this sense, phenomenal refers to that which appears within or to a subject, inherently involving a unique what-it-is-likeness. The term phenomenology, thereby, is the study (logos) of that which appears (phenomena). This what-it-is-likeness is a distinct mode of presentation, tied to its unique appearance for a specific subject in a given context or environment.

Qualia are also connected to this idea of appearance, which are defined by Block (2007) as the qualities of objects or contents in experience. Qualia are the *ways* things appear, and there are multiple ways in which something may appear, depending on the particular subject and specific context and environment. For example, the redness of red can appear differently across various people (inverted spectrum), and even for the same person, depending on their internal and external environment². Colors may look or feel brighter or duller depending on your state of being, even though the color is the same (Bubl et al., 2010; Johnson, 2019). Qualia can be referred to and described through language, however, it is likely that they are ineffable and unable to be entirely captured and encoded in a symbolic form, requiring instead a type of "dimensionality reduction" (Neuman and Nave, 2010). This ineffability and inherent subjectivity of experience has created significant difficulties for consciousness to be studied in an objective and scientific manner.

Access Consciousness and Cognitive Access

A-consciousness was developed to clear up confusion about the field and to bridge philosophy and cognitive science by providing a quantifiable and scientifically tractable account of consciousness. A-consciousness refers to the mode of experience where a person is actively cognizing about their experience, being able to use such experiential information in a cognitive or rational manner. A-consciousness is a cluster concept (Block, 1995) that emphasizes reportability and the broadcasting of information in the brain, connecting it to "hard science" theories of consciousness, such as Global Neuronal Workspace Theory (GNWT). Verbal or behavioral report is considered the best practical guide or measure of A-consciousness (Block, 2009). A-consciousness involves a form of

² The redness of red is not static or absolute. Inverted spectrum is often used as support for this, but we can also look at our own variable experience of red. Depending on the context, the redness might have a different feel or what-it-is-likeness to it, while still being described as red, and even still corresponding to the same wavelength of light. This is more evident when taking into account our internal environment and mental/emotional state.

centralization in the brain – a global workspace – where information processed by specific regions is shared with other regions. This centralization and broadcasting of information then enables an executive system to control behavior accordingly. Cognitive access has been used to refer to this phenomenon and will be used synonymously with A-consciousness in the remainder of this dissertation (Block, 2011; Mashour et al., 2020).

According to Block (2009, 2011), an A-conscious state consists of having an A-representation, where a representation is A-conscious if it is broadcast for free use in reasoning, cognitive processing and the rational control of action (including reporting). The paradigmatic A-conscious states are propositional attitude states, which are primarily mental by nature, such as thoughts and desires (Block, 2009). A-consciousness involves experiential information that is able to be used in cognitive processing and reasoning, such as looking at five different shapes and ordering them in your head, or looking at a stop sign and pressing the brakes in your car. Accessible content is content that reaches the executive system, which is responsible for the rational control of action and speech (Block, 1995).

Accessed information is practically identical to the contents of working memory, suggesting that "access" roughly implies "working memory" and that working memory serves as the global workspace of accessibility (Block, 2008, 2011; Overgaard, 2018). Working memory is defined as an active representation or temporary buffer of information that can be used and manipulated for cognitive operations (Luck & Vecera, 2002; Luck, personal communication). A-consciousness, therefore, requires some form of a working memory system that acts as a primary mechanism for information to be shared and accessed by other regions within the system. This idea is closely connected to and illustrated by the Global Workspace Theory (GWT), which posits a "brainweb" of fleeting capacity that enables communication between otherwise separate brain processes (Baars, 1997, 2002), and the GNWT, which we will elaborate on in our NCC section (3.3).

The Relation Between Access and Phenomenality

Block (1995) claims that P-consciousness can exist without A-consciousness, which he supports with his pneumatic drill thought experiment. In this experiment, a person is deeply engaged in a conversation while a loud drilling sound occurs outside. Although the noise is not initially noticed, the person later becomes consciously aware of it and takes rational action by closing the window. Block suggests that prior to this realization, the person was phenomenally conscious of the sound without being accessibly conscious of it. Only upon explicitly noticing the sound does it enter A-consciousness, allowing the person to act rationally on the experiential information by closing the window. The dissociation of P-consciousness without A-consciousness was recently supported empirically (Amir et al., 2023). In this study, participants were able to recall some retrospective qualitative aspects of a stimulus, implying a form of p-consciousness, even though the stimulus was not online in a-consciousness during the presentation. However, as the authors explicitly noted, these results only occurred in a few trials in a subsample of participants in very specific conditions, so further research is required to solidify this evidence.

Block (1995) also entertains the idea of A-consciousness existing without P-consciousness and gives multiple thought experiments in support, including *superblindsight* and the philosophical zombie. The philosophical zombie thought experiment is defined by a human-like entity, perhaps a silicon-based, computational robot, that behaves exactly like a human but lacks any intrinsic feeling or what-it-is-likeness. *Superblindsight* builds off of the naturally occurring phenomena of blindsight, which involves the ability of an individual to detect and identify visual stimuli while having little to no perceptual/phenomenal experience of it, usually following lesions to V1 (Overgaard, 2011, Mazzi et al., 2016). The *Superblindsight* thought experiment involves a person with blindsight who has adapted to their condition and trained themselves to effectively use their blindsight knowledge without any prompting (Block, 1995). For example, if a glass of water was in the blind spot of their visual field, a *superblindsighter* would somehow know it is there, reach for it, and drink out of it, even though they have no visual experience of it. A deeper analysis of the relationship between P and A consciousness will be explored in chapter 4.

In this dissertation, we embrace a broad definition of cognitive access. We understand cognitive access to exist on a continuum with two primary levels, both of which are implicitly stated by Block (1995). The first being lower-level access that enables rational action, where rational action is understood broadly to encompass all forms of goal-directed behavior – even including basic lifeforms. The second involves a higher access, which we will refer to as reflective access, which involves propositional attitudes, conceptualization, and reflective capabilities.

In a functional sense, cognitive access is the availability of information to support rational action and goal-directed behavior. Because it can be defined by attentional mechanisms, memory systems, and the ability to utilize information for adaptive behavior or learning, cognitive access can be applied to systems not typically considered sentient or phenomenally conscious. Many such models, including computational models, artificial neural networks, and large language models, exhibit cognitive access in this functional sense (Dehaene et al., 1998; Franklin et al., 2013; Garcez et al., 2019; Xu et al., 2019). With this broader definition, any cognitive or information processing system capable of rational action, including non-sentient machines, can be said to have cognitive access. In this sense, cognitive access refers to the functional access of information, or to the cognitive access of phenomenally experiential content, depending on the system in which it is implemented.

1.2 Attention

We understand attention to be most accurately explained by the common conception given by James, which emphasizes the selectivity, focalization, and concentration of consciousness for functional, effective, goal-oriented behavior (James, 1890). Attention plays a crucial role in cognitive access by facilitating the transfer of sensory or experimental content to the accessible workspace (Baars, 2002; Block, 1995; Mashour, 2020; Smithies, 2011). In our view, attention is not an external or causal mechanism, but an intrinsic process by which consciousness amplifies or filters itself, modulating the accessibility of specific experiences. Attending to something within the field of consciousness reorganizes the entire experiential state, making certain contents more or less available for cognitive and behavioral actions.

From a neuroscientific perspective, attention can be understood as the effect of resolving neural competition, a process naturally emerging within neural systems that serve adaptive purposes (Krauzlis et al., 2021; Wu, 2024). Attentional signals correspond to the enhancement of neural activity associated with a specific stimulus, while concurrently suppressing competing signals. This selective amplification is maintained by persistent neural activity, which corresponds with an enhanced processing and ability to keep information accessible in working memory (Carrasco & McElree, 2001; Mashour et al., 2020). As shown by Koch & Tsuchiya (2007), top-down attention can function independently of consciousness. However, attention is not limited to only top-down mechanisms, and attention-related neurons show combinations of both top-down and bottom-up processing (Krauzlis et al., 2021). This broad view of attention selectively processing preferential information, including both exogenous and endogenous mechanisms, is shared by Desimone and Duncan (1995), Cohen et al. (2012), and Pitts et al. (2018).

A classic example of top down attention is shown in the Cocktail Party Effect (Cherry, 1953), where attention corresponds with the voluntary and intentional enhancement of certain stimuli and suppression of others (Kerlin et al., 2010). Meanwhile, from a bottom-up perspective, a sudden loud sound can automatically and involuntarily capture your attention, which is also selective and preferential. While attention may be deployed differently (voluntary vs involuntary), it remains essentially the same – the modulation of awareness and consciousness. Whether guided by voluntary intention or triggered automatically, attention corresponds to that which is salient within our experience, allowing for more effective and adaptive behavior (Wu, 2024). This selective and limiting functionality is evolutionarily adaptive, both for conserving energy and for guiding survival-relevant, goal-oriented behavior (Chen & Zhang, 2021; Lev-Ari et al., 2022).

Evidence from blindsight patients suggests that attention and awareness may be dissociable, as such individuals can discriminate attended, but not consciously perceived, stimuli at above-chance levels (Schurger et al., 2008). A blindsight patient may accurately guess the location of a visual stimulus without experiencing it. They are

attending to the location and seeing nothing – but are still able to accurately guess its location. However, patients often report a vague “feeling” that something is present, suggesting a minimal form of P-consciousness and perception. A methodological issue also arises from the use of dichotomous yes/no measurements in these studies. These simplistic measurements can lead to inconclusive evidence suggesting that blindsight patients completely lack subjective perception, when in reality they have some minimal sense if asked correctly. Utilizing more nuanced tools, such as the perceptual awareness scale (PAS) or a subjective awareness scale (SAS), shows that it is more accurate to describe their condition as degraded and/or abnormal perception. Blindsight may in fact be an alternative route of “seeing,” bypassing the conventional retino-striate pathway. These individuals retain some phenomenal experience, highlighting the brain’s adaptive plasticity and the flexible nature of consciousness (Overgaard, 2011; Mazzi et al., 2016; Derrien et al., 2022; Aleci and Dutto, 2024).

The Intrinsicity of Attention to Consciousness

In our view, attention is an intrinsic process of consciousness corresponding with the selective transformation and reconfiguration of its contents. It is the natural and dynamic movement of consciousness – consciousness in motion – rather than a causal mechanism. As Wu (2024) posits, attention is the *explanandum*, not the *explanans*. Attention is not some external, causal mechanism acting on the brain or consciousness (*explanans*), but the emergent pattern resulting from the spontaneous organization of consciousness and neural activity (*explanandum*). Like the flow of water in a stream, the flow is not caused by “direction” but arises from the collective organization and motion of H₂O molecules. Likewise, attention does not direct experience, but is the direction in which experience unfolds.

Attention corresponds with the amplification, suppression and filtering of the contents of consciousness, thereby shaping the structure of consciousness from within. This view contrasts with models that treat attention as an external selector or gatekeeper. It is more closely aligned with Aron Gurwitsch’s field theory of consciousness, where experience is illustrated as a field composed of a theme (focus), a thematic field (context), and a margin or fringe (periphery or background) (Gurwitsch, 1964; Magnan, 2007). In this view, attention does not only introduce new contents, but reorganizes the relationships within the field. Yoshimi and Vinson (2015) extend Gurwitsch's field theory by linking it to cognitive science, where they suggest that the field structure is associated with various neural processes such as metastable dynamics and predictive coding. Even in predictive coding models where attention is described in computational terms as precision-weighting of prediction errors, it can still be interpreted as the network's internal modulation of processes encoding salient information. This modulation dynamically reshapes the structure of information in light of prediction and relevance, similar to biological attention (Feldman & Friston, 2010; Friston, 2010). Taken together, these perspectives support the idea that attention is not a separate mechanism layered on top of consciousness, but rather an intrinsic mode of its self-organization and ongoing transformation, rooted in adaptive, predictive, and goal-oriented behavior.

1.3 Creature and State Consciousness

P-consciousness, or what-it-is-likeness, pertains both to an entity itself and to the states it undergoes through its interaction with the world. These are sometimes distinguished as “creature consciousness” and “state consciousness” (Bayne, 2007). Creature consciousness refers to an entity as a whole being phenomenally conscious, the fact that there is something it is like to be that entity, such as being a bat³. Over time, an entity with creature consciousness goes through many individual experiential states. These separate states are what state consciousness refers to, capturing the intuitive idea that one creature goes through many states of experience over time. These different states of consciousness arise from a creature’s internal processing and the interaction of this processing with the external world. It is intrinsically linked with the world and the contents of consciousness, as the shift in contents correlates with a shift in states. States of consciousness include experiences like smelling freshly brewed coffee, seeing a field of flowers in spring, or dreaming about a flying elephant⁴ (Bayne, 2007). While creature and state consciousness are treated as distinct categories, they are deeply interconnected. The continuity of state consciousness and moment-to-moment arising of phenomenal states constitutes and sustains creature consciousness over time, and without a structured flow of states, creature consciousness would be empty or fragmented. Meanwhile, state consciousness presupposes an integrated system, one that is capable of undergoing such states in a coherent manner.

Both creature state consciousness are complex, hierarchical structures that include multiple parts or aspects (what some call “mereological parts”). States are layered within states that are nested within the creature. Various overarching states can include and make up smaller states. For example, you may be in the waking state, while in the state of being excited or joyful, which also involves within it the more ephemeral state of smelling freshly brewed coffee. The changes in overarching states are not necessarily aligned one-to-one with changes in momentary content, however, changes in perceived content always result in some state change. For example, my state of being awake does not change when I drink water, and my state of joy might not either; however, my experiential state shifts to involve the what-it-is-likeness of drinking water, which is different from when I was thirsty and not drinking anything.

On the contrary, some contents may not elicit an immediate, explicit state change, yet still alter the overarching experience. An entity may be very minimally conscious in respect to a particular state at some level, while remaining fully phenomenally conscious at the overall creature level. It is not that the system is unconscious to the stimuli, but rather that the influence of certain stimuli occur below the threshold of conceptual recognition or distinction. However, these stimuli still modulate and influence the totality and character of experience.

³ Being a bat refers to both the species of a bat and individual variations within specific bats.

⁴These examples of state consciousness are very similar to common examples of qualia – the qualities of experience.

This distinction highlights the difference between phenomenally conscious states that are distinct and fully accessible (a-conscious) – those you can notice and report (I am feeling lethargic because I ate too much) – and sub-threshold influences that don't crystallize into their own state, but still affect the quality of consciousness (feeling unwell and not knowing why or consciously recognizing it). In this second case, the person may recognize their experience broadly at a higher level, but cannot properly recognize or distinguish exactly what is causing it, hindering them from fully acting rationally on it. However, we can label this as very minimal cognitive access, as the feeling itself is a subtly accessed state which allows them to take minimal rational action – such as going to lay down to sleep or take general medication.

Although these subprocesses may not be explicit, they subtly shape the ongoing character of conscious experience. A stimulus presented below the threshold of clear perception may not evoke a fully accessible representation, yet can still influence the overall experiential character and following responses. In this sense, the stimulus operates outside the focus of awareness but remains phenomenally relevant. This domain of subtle, non-focal phenomenality aligns with what William James and Bruce Mangan (2007) call “fringe consciousness”; the vague but meaningful felt sense that orients cognition and awareness even when representational content is unclear or less accessible. The fringe and periphery are not the same but often overlap, as the periphery allows the fringe to become more easily revealed (Mangan, personal communication). This domain is not trivial in its influence, for it is the relational matrix through which meaning is grounded, essential to the phenomenal character, the what-it-is-likeness, of experience.

Furthermore, even subliminally processed information may be minimally represented. Although not in the form of a clear percept, it may appear as a subtle shift in experiential tone or phenomenal character. This too becomes integrated into the overall field of consciousness, residing at the outskirts or background of phenomenality and accessibility. In such cases, the individual may still respond rationally or adaptively, revealing both minimal accessibility and phenomenality. For example, a subliminal fearful face may be flashed too rapidly for a clear conscious perception to form, yet it may still be registered indirectly as a vague feeling of unease. This shift in phenomenal character as a feeling of unease constitutes the minimal appearance and accessibility of a representation within experiential space. This subtle feeling can then adaptively guide subsequent behavior, such as causing an increased sense of caution or defensiveness.

We can also consider the experience of background stimuli that are not noticed until that stop, such as a fan or refrigerator. In these cases, the prior presence of a stimulus becomes explicitly known through its sudden absence. This illustrates how background elements that are not fully accessible still affect experience in a meaningful way; if it wasn't, the change would not have been registered at all. These examples show that when a stimulus remains constant in experience, it often fades from explicit awareness. This is not because it ceases to exist or influence our experience, but because the reflective, attending mind deprioritizes unchanging inputs (Clancy, personal communication;

Friston, 2010, Nisargadatta, 1981). In such cases, the phenomenal content is on the outskirts of the periphery, forming the background structure of experience rather than a distinct state within it. This further supports the idea that consciousness involves both explicit states and background influences. In this sense, the stream of consciousness consists not only of clearly articulated states, but a continuum which includes subpersonal influences that modulate a creature's experiential flow.

This understanding of consciousness as a layered, holistic field is supported both scientifically and phenomenologically. Philosopher Jiddu Krishnamurti describes consciousness as a "total field" comprising both surface-level mental activity and deeper, often unexamined layers including desires, fears, intuitions, and subconscious impulses (De Sousa, 2012). These layers, although not explicitly attended, shape the character of experience and behavior. Studies on minimal phenomenal experience (MPE) support the idea that phenomenal content need not be explicitly perceptual or cognitive, but can instead exist in minimal forms including, but not limited to, subtle feelings, dreams, meditative experiences, and dreamless sleep (Josipovic & Miskovic, 2020; Metzinger, 2007, 2019, 2020; Windt, 2015, 2016).

1.4 Consciousness and the Subject/Self

Much of the confusion across theories of consciousness stems not only from divergent definitions of the term but also from differing assumptions about its relation to the self, the subject of consciousness. We do not mean by "self" an awareness of self, which itself has multiple meanings (Millier, 2020), but rather a more ontological notion, concerning "who/what" experience occurs for. This notion of the ontological self/subject is determined by how a system is bound as its own entity to interact with its environment, in which there may be multiple bounded systems nested within each other. These include autopoietic systems as a whole, such as the entire living organism of a human, and layers within that system, which we roughly delineate as the egoic, perceptual, and comprehensive subjects. The stages of subjectivity, boundedness, and agency, and how they arise, are explained in greater detail in chapter 5.

P-consciousness consists of appearances within consciousness, characterized by the what-it-is-likeness quality of both the experienced content and the subject's relation to it. These appearances may range from rich to sparse, strong to faint, focal to peripheral, or from high-resolution percepts to low-resolution sensations. An appearance within consciousness is relative to what it is appearing to; something may appear sparse at one level, yet rich in another. For example, Lamme (2010) posits the idea that we may 'see without knowing' – be perceptually conscious without knowing we are at a higher level. Perceptual richness manifests at one level of processing, yet this same richness may appear sparse or inaccessible when viewed from another. This reinforces the need to ask: to whom do these appearances appear, and what is it that is conscious? The answer hinges on how one postulates the subject of consciousness, and this choice significantly alters what contents and states are considered conscious, along with their respective neural substrates.

The Egoic and Perceptual Subject

The egoic or metacognitive notion of the self refers to the reflective subsystem responsible for verbal report and introspection. If the subject of experience is identified with this subsystem, then consciousness is confined to a small layer of the broader process. In this view, only contents accessible to this reflective self-model are considered conscious. States such as dreaming and deep sleep (unless lucid dreaming) would be deemed unconscious, as they fall outside of this narrow subsystem. With this understanding of the subject, consciousness is explained best by higher-order and global neuronal workspace theories, evidenced physiologically by late-onset neural correlates such as the P300 ERP and frontal brain regions such as the PFC.

Between the reflective self and the system as a whole system lies an important middle layer: the perceiving self. This is the locus of pre-reflective, perceptual subjectivity, where sensory information is processed and experienced without being fully broadcast, reported, or conceptualized. It is manifested during perceptual processing that is subjectively felt, with some form of what-it-is-likeness, yet inaccessible to verbal introspection. This level corresponds to stage three of Lamme's recurrent processing theory (2006), and his notion of 'seeing without knowing', where content achieves local recurrent processing but has not yet been globally broadcast. GNWT (Dehaene, 2011) refers to this as preconscious processing – information that is perceptually available but not yet broadcast globally and consciously accessible. This conception of the subject corresponds with early onset neural correlates of consciousness such as the N2, and more posterior and modality-specific regions.

The Comprehensive Subject

If the subject of consciousness is understood to be the entire substrate of awareness, that is, the integrated system through which adaptive agency and consciousness emerge (the mind-body complex), then even seemingly unconscious processes are conscious to some degree. In this framing, all states, including waking, dreaming, and deep sleep, manifest varying degrees or modes of consciousness. Anything the mind-body complex detects, integrates, or responds to constitutes an appearance within its field of consciousness, even if it is not fully conscious from the perspective of the ego. With this view, it is not just the narrative or reflective self that is conscious, or the perceiving self, but the entire system that instantiates that individual consciousness. There may be varying degrees of consciousness within the system – sub-personal systems – that are not fully accessible but still part of the subjective character of experience. In this sense, consciousness is both dynamic and layered. It is not a discrete threshold but a nested hierarchy of processing, where the most vivid awareness (the "tip of the iceberg") emerges from a deeper and broader overall process.

For example, one may involuntarily respond to an approaching ball without the reflective mind being fully aware of the stimulus, demonstrating how system-level awareness can drive adaptive, rational behavior and shift the experiential field. Even more removed from volitional control – bodily processes such as breathing, heart rate, or digestion shape the overall what-it-is-likeness of the creature. Though these functions are sub personal and involuntary, they modulate the experiential field and contribute to the totality of consciousness, influencing even the most refined layers of attention and self-awareness.

In this view, the conscious system is itself a process of processes. Each nested process is contributing to and shaped by the whole. The subject of experience is not reducible to the focal point of reflective attention, but is embedded within the system's full field of awareness. What appears as the “tip of the iceberg” (the egoic or narrative self) is merely the focal point of this broader consciousness, not its origin or entirety. The entire system, the mind-body complex, can be viewed as the comprehensive subject, a dynamic and adaptive field of consciousness that gives rise to structured, localized appearances within itself. Furthermore, each subsystem within this holistic process is itself a local field nested within the global field of the conscious system. In this way, consciousness is distributed and relational whereby a system is conscious by virtue of being an integrated field of awareness, subjectivity, and information processing, distributed across various layers and dimensions.

Implications of the Subject

In this section, I have elaborated on how consciousness is a relational and graded process involving a multitude of interactions between various subsystems. The perspective one takes on the subject, the *who*, drastically alters the *what, where, when, why, and how* of consciousness, as it fundamentally shifts the underlying relation. Viewing the subject at each of the various levels explained above offers a valuable perspective, however, each perspective is limited by its own structure and point of view. This includes phenomenology itself – the most intimate and direct means of investigating experience. Phenomenological introspection, while invaluable, cannot be said to encapsulate the entirety of consciousness, even if it is all that we know – the same we have learned that our visual experience does not encompass the entirety of the light spectrum, even though it is all we are able to perceive. What we perceive consciousness to be is only another perspective arising within our experience. In this sense, consciousness cannot be fully objectified, because every perception or conceptualization is embedded within the field it attempts to explain. For this reason, our aim is to investigate multiple perspectives and develop a comprehensive, complementary, and pluralistic framework.

Chapter 2

Behavioral Study of Inattentional Deafness

2.1 Inattentional Phenomena

Consciousness can be studied via inattentional blindness or deafness, a phenomenon where perceivable stimuli fail to clearly appear in conscious access due to a lack of attentional resources. Since attention is limited and easily manipulated, inattentional phenomena give scientists a straightforward approach in studying conscious perception (Cohen & Chun, 2017). Inattentional phenomena were first studied by Neisser and Becklen (1975) in their "selective looking" experiment, and later coined inattentional blindness (IB) by Mack and Rock (1998). IB refers to the failure to notice a fully visible, often unexpected, object or event when attention is focused elsewhere. This occurs despite lower level sensory processing, highlighting the limits of visual awareness under selective attention. The phenomenon was further popularized by Simons and Chabris (1999) in their well-known "gorilla" experiment, where they demonstrated inattentional blindness in a more ecologically valid setting.

The auditory counterpart of IB, inattentional deafness, arises when attentional resources are consumed by a competing task, reducing awareness of task-irrelevant or unexpected sounds (Dalton & Fraenkel, 2012). Studies in auditory attention show that both perceptual and cognitive load modulate auditory awareness in varying degrees (Macdonald & Lavie, 2011). Related phenomena have also been identified across sensory modalities, even including inattentional touch (Murphy & Dalton, 2016).

A closely related concept, change deafness, refers to failures in detecting salient changes within complex auditory scenes involving multiple simultaneous sound sources (Backer & Alain, 2012). As Peck (2015) notes, a key distinction between change deafness and inattentional deafness lies in the task structure. Inattentional deafness directs attention away from the auditory stimuli by explicitly instructing the participant to attend to a different task. The design of inattentional phenomena studies generally involve a critical stimulus (target) that is perceptible but potentially missed or seemingly unnoticed due to the attentional attenuation of awareness. Inattentional phenomena are generally explained as the consequence of insufficient attention or a limitation of attentional resources, hindering us from simultaneously processing all the stimuli in our environment at once (Matias et al., 2021).

An overarching theme of this work is that inattentional phenomena are still associated with some diminished awareness and accessibility. That is, certain stimuli fail to reach full accessibility due to the selective nature of attention, however, they are completely unconscious. These stimuli are not able to be categorized or conceptualized in a way that allows distinction or comparison, which Block (2011) calls *inattentional inaccessibility*. There are several ways to understand or explain what occurs when a stimulus is

inaccessible. Two possible explanations are inattentional *amnesia* and *agnosia* (Wolfe, 1999, Simons, 2000). Inattentional amnesia is the idea that the stimulus is present in memory but without attention it decays quickly or is forgotten, inhibiting full recognition or reportability. Inattentional agnosia is the idea that contents are not fully encoded into the working memory, particularly higher-level features necessary for proper discrimination and advanced rational behavior. Because sensory information is not fully processed and interpreted, participants are unable to name or identify stimuli, hence the term “agnosia”.

This is similar to results seen in blindsight patients, where they may have sensed something (blindsight), but were unable to discern exactly what it was. With this understanding, the working memory representation is limited and degraded compared to the experience in the moment of active or online perception, making it less functional for rational action. Both ideas are compatible with the account presented here; inaccessible content still has some minimal appearance in consciousness, but might be too unformed to be described (agnosia), or may quickly decay prior to being available for full report (amnesia).

2.2 Perceptual and Cognitive Load

The study of inattentional phenomena was further advanced by the use of load-induced, continuous paradigms. According to load theory, there is a shared common "resource" or central capacity across perceptual modalities. Perceptual load and cognitive load relate to this capacity in different ways, where perceptual load influences lower-level processing and cognitive load influences top-down working memory and attentional mechanisms (Lavie et al., 1995, 2004, 2005, 2014). In a sense, perceptual load is related to P-consciousness, while cognitive load is similar to A-consciousness.

Manipulating perceptual load implies utilizing various task demands that affect the capacity of perceptual processing (Matias et al., 2021). For example, when a task requires extensive visual processing, stimuli are in competition and fewer resources remain for processing other stimuli, such as auditory oddballs in a change deafness task. Perceptual load is increased simply by increasing the intensity or quantity of sensory information in a task. This idea is supported neurally (Molloy et al., 2015) and behaviorally (Macdonald & Lavie, 2011), where high visual load was found to reduce auditory awareness. While perceptual load suppresses auditory processing in a straightforward and reliable manner, the role of cognitive load is more nuanced.

Cognitive load is related to working memory and executive processing. Tasks such as counting, mental arithmetic, and n-back paradigms (which require comparing the current stimulus to one presented n trials earlier) are classic examples of cognitive load manipulations. Several studies suggest that high cognitive load can increase inattentional deafness by impairing the ability to reorient attention toward unexpected auditory stimuli. For example, Causse et al. (2016) found that high cognitive load led to greater inattentional deafness while high perceptual load did not. However, according to load

theory, the effect of cognitive load is not linear. Under certain conditions, high cognitive load may actually increase auditory detection, especially when central executive resources are taxed to the point that top-down inhibitory control is weakened. In such cases, irrelevant stimuli may "leak in," due to reduced suppression mechanisms, thereby decreasing deafness. This stands in contrast to high perceptual load, which consistently reduces awareness of irrelevant stimuli (increases deafness) by exhausting capacity-limited sensory and perceptual processing.

The interaction between cognitive load, perceptual load, and auditory awareness has not been fully explored or clarified. Evidence from Matias et al. (2021) supports load theory and the reliability of perceptual load in inducing blindness, but he also calls for a greater evaluation of exactly how working memory plays a role in cognitive load. Furthermore, as mentioned above, results from Causse et al., (2016) did not align with load theory, as perceptual load did not increase deafness while cognitive load did. As such, the different effects of load type on inattentional deafness point to a more nuanced understanding of how various perceptual and memory processes relate to auditory perception.

These experimental results on inattentional phenomena can be extended into real-world scenarios, such as having a delayed responsiveness or failure to notice alarms or spoken instructions when engaged in demanding tasks (driving, reading, gaming, etc.). This highlights the importance of distinguishing between different types of load when studying multisensory perception and how it may manifest in everyday experiences (Causse et al., 2016; Dehais et al., 2014; Molloy et al., 2015; Raveh & Lavie, 2015). These cross-modal effects reveal that failures in perceptual awareness can occur between modalities which further supports the idea of a common central resource limitation (Kreitz et al., 2016).

In our current experiment, we investigate inattentional deafness using a load-induced, cross-modal, continuous auditory paradigm. Participants engage in a simple arcade-style visuomotor game designed to induce perceptual load. This design functions as an inattentional deafness task rather than change deafness, since participants are explicitly instructed to focus on a separate task while reporting background auditory stimuli. Perceptual load is manipulated by requiring participants to engage in a visuomotor task that involves visual perception, timing, and coordination. During the task, random pink-noise auditory deviations are embedded in the background white noise. Importantly, we do not introduce any cognitive load demands or working memory manipulations in this experiment. Our current aim and contribution is to isolate the effects of perceptual load, examining how a combination of motor and visual demands impact auditory awareness. This design and paradigm allows us to investigate the role of perceptual load in inattentional deafness without the confound of cognitive or working memory processes, thereby providing evidence for or against load theory.

2.3 Methods

Participants

Participants were between 18 and 30 years of age and were in good general health. They had no history of psychiatric or neurological illness and were not taking any psychoactive medications. 26 young adults were recruited through the University of California, Merced online recruiting database and through recruitment flyers. All participants provided written consent and completed a general questionnaire before experimentation. Participants were compensated 15\$ per hour after completion of the experiment. The experimental protocol was approved by the institutional Review Board at UC Merced. Two participants were excluded because they did not follow instructions on the conditions, forgetting to press the detection key for any stimuli, leaving 24 data sets.

Stimuli

The auditory stimulus consisted of two sounds: one white noise that played continuously through the entire time they were engaged in the game, and pink noise deviations (critical stimulus) which randomly sounded every 3-10 seconds. The pink noise was embedded in the white noise with no gap, so it sounded like a subtle but salient alteration of the noise. The white noise was played at 54dB with the pink noise deviations at the same level. Javascript software was used to control the timing of auditory stimuli and visual gameplay. Participants were seated in a sound-attenuated booth ~1 m away from a 27-in. monitor. Auditory stimuli were presented at ~60 dBA sound pressure level (SPL) through two loudspeakers located to the left and the right of the monitor at a $\pm 45^\circ$ angle, creating the perception that the sound came from the center (0°).

Experimental design and procedure

The load inducing task was an arcade style game (figure 1) with high and low load conditions. The game involved a spaceship dodging harmful objects (meteors) while collecting beneficial ones (stars and diamonds). The meteors would subtract 50 points while the diamonds would add 50 points and stars would add 100 points, with stars spawning the least, meteors spawning the most, and diamonds in between. The high load condition involved the participants actively playing the game, while the low load condition involved the participant watching the game while a pseudo-AI played. During both conditions, the same auditory stimuli was played, only differing in the randomization of pink noise deviations. During the high load condition, participants played by moving the ship left and right with the left and right arrow keys. When a deviation in sound was noticed, participants were asked to press the space bar for detection. During the low load condition, participants did the same in pressing the detection key when the critical stimulus was heard, however they were only watching and not playing the game. Participants were instructed to press the detection key when the sound was heard, but to still engage in the game and attempt to get as high of a score as

possible. Each condition was a ten minute long experimental block with approximately 120 trials (critical stimuli) per block. Before each experimental block, participants engaged in three thirty second practice trials to become accustomed to the flow of the game. All participants underwent an auditory check to ensure they could detect the auditory deviations used in the task. All individuals reported hearing the sound changes during these pre-trial tests.



Figure 1: Game design. Participants move the ship left and right with left and right arrow keys in order to dodge meteors and collect stars and diamonds. The stars were designed to blend in with the background to make them more difficult to see and capture.

2.4 Results and Discussion

Results

The data was analyzed by a Welch's T-test, where a significant difference in deafness between the low and high load conditions was found ($t(23) = 17.07$, $p < .000000001$). The average deafness in the high load condition was 46.49% (SD= 10.97) and the average deafness in the low load condition was 4.40% (SD= 4.42). The deafness in the high load condition reached over 70% and never reached higher than 25% in the low load condition. Figure 2 shows the distribution of deafness between high and low load.

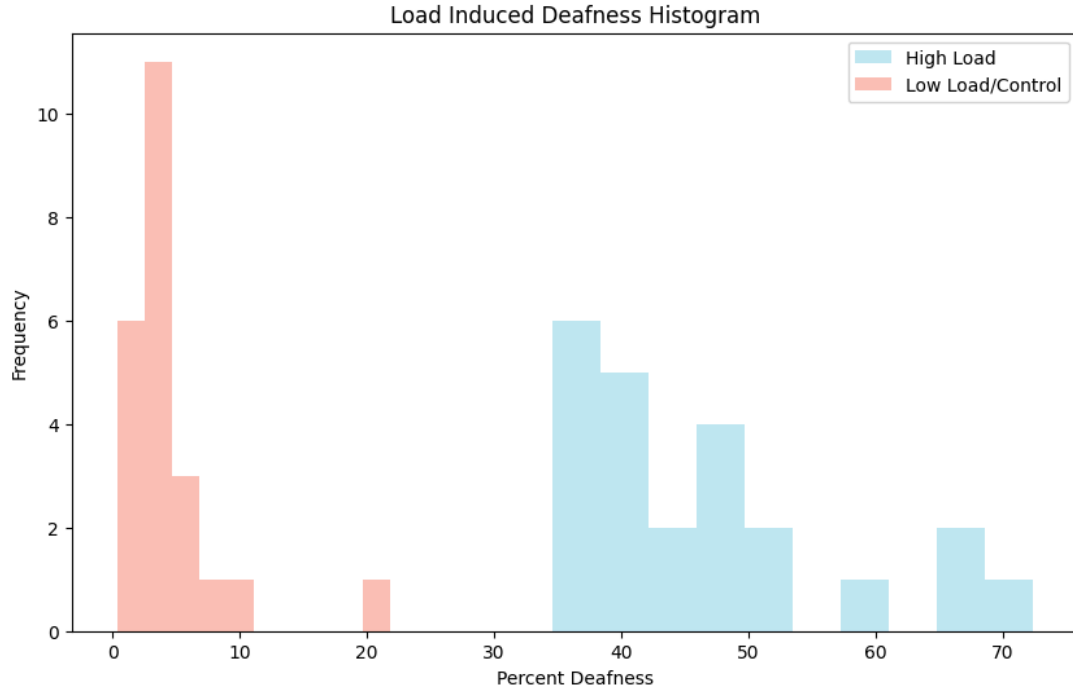


Figure 2: Deafness in the high load condition was between 35% and 73% with an average of 46.49%. Deafness in the low load was between 1% and 23% with an average of 4.40%.

Discussion

The results from our experiment demonstrate inattentional deafness under high perceptual load and may be used for future EEG investigations of auditory awareness. By manipulating perceptual load through a visual-motor task, participants in the high-load condition (Play-Report) exhibited a 46.49% miss rate for auditory deviants, compared to just 4.40% under low load (Watch-Report). This tenfold increase in deafness provides strong support for Lavie's load theory (Lavie et al., 2004), which predicts that high perceptual load reliably and robustly causes deafness. These findings contrast Causse et al. (2016), who reported that only cognitive load suppressed auditory processing, rather than perceptual load. In our study, perceptual load alone was sufficient in inducing inattentional deafness. In our next study, we aim to build off the current design by explicitly incorporating cognitive load through a working memory task. This will allow us to further test the predictions of load theory and potentially identify the neural correlates of inattentional deafness and auditory awareness.

Chapter 3

Investigating the neural correlates of consciousness through an inattentional deafness EEG paradigm

Furthering our investigation of consciousness, we add a neurophysiological component to our previous behavior through the use of electroencephalogram (EEG). Through the use of this inattentional paradigm, we hope to also investigate the neural correlates of consciousness (NCCs). Most research on inattentional phenomena has focused on the visual modality (Koreimann et al. 2014; Conci et al. 2020), however, the auditory domain provides a unique approach for investigating cross-modal perceptual experience.

Load theory (Lavie et al., 1995, 2004, 2005, 2014) provides a strong explanation for such failures of awareness. It proposes that perceptual capacity is limited, and when processing is taxed through high perceptual demands, irrelevant stimuli are more likely to go unnoticed. Perceptual load is predicted to reliably produce a robust negative effect on perceptual awareness. Cognitive load, on the other hand, may have an opposite effect and actually increase the awareness of irrelevant stimuli. This is thought to occur because high cognitive demands increase distractibility by taxing the suppression mechanisms of attentional processing, thereby causing irrelevant stimuli to "leak in".

In order to more comprehensively investigate load theory, we will now add cognitive load to our experimental design. Our goal is to add it at a moderate level (not too high to avoid a reverse effect) alongside perceptual load in order to maximize deafness. Using EEG, we examine how this combined load affects the neural correlates of auditory awareness, including ERP components like the N2 and P3b. These components are not only relevant to attention but have also been proposed as candidate NCCs (Koch et al., 2016, Mashour et al., 2020). We will further include the analysis of oscillatory dynamics including alpha (8-14 Hz), theta (4-8 Hz), and beta (14-30 Hz) bands. Through EEG analysis, we aim to clarify whether or not inattentional deafness reflects a breakdown in perceptual processing (marked by early correlates), or a breakdown in later processes related to reflective access (marked by later correlates).

3.1 Neural Correlates of Inattentional Deafness

Although studies on inattentional deafness are ongoing, the exact neural correlates of inattentional deafness are not fully elucidated. One discrepancy that remains is the role of alpha inhibition. Dehais et al. (2016) reported reductions in both N100 amplitude and alpha-band (10 Hz) power as task difficulty increased and auditory awareness decreased (deafness). This differs from Callan et al. (2023), who found an increase in alpha power alongside a reduction in gamma to be associated with reduced auditory awareness (deafness).

In terms of event-related potentials (ERPs), The N2 and P300 are the primary correlates. Dehais et al. (2016) did not initially observe a significant P300 effect, however, following studies have shown consistent reductions in P3a and P3b amplitude associated with increased deafness (Dehais et al., 2019; Molloy et al., 2015; Scheer et al., 2018). These components are associated with different aspects of attentional processing, with the P3a reflecting stimulus-driven attention and the P3b reflecting working memory updating and reflective access (Polich, 2007). The N2 component has also been connected to auditory perception, with a greater amplitude associated with increased detection and awareness (Puschmann et al., 2013). Overall, these findings suggest that inattentional deafness is primarily associated with reductions in N1, N2, and P3 amplitudes, along with gamma suppression and alpha modulation.

3.2 Neural Correlates of Consciousness (NCCs)

Significant progress has been made recently in the empirical investigation of consciousness, however, there are still many questions regarding both neurophysiology and theory. In this section, we will briefly review the NCCs, with updates on the current most promising candidates. An NCC can be defined as the minimal set of neural mechanisms sufficient for a specific conscious experience, which is further broken down into two categories: content-specific and full (Koch et al., 2016). Content-specific NCCs refer to the neural activity associated with particular states of experience, such as seeing a face or hearing a tone. The full NCC refers to an underlying neural substrate or process necessary for consciousness as a whole, independent of the specific content. Current leading candidates for content-specific NCCs include the P3b component, the perceptual awareness negativity (PAN), and the hybrid N2 (N2). The primary candidate for the full NCC is the cortical "hot zone" – a posterior network implicated in conscious perceptual integration (Cohen et al., 2024; Koch et al., 2016). In the following sections, we will briefly review each of these candidates.

The cortical hot zone, as defined by Koch et al. (2016), refers to a temporo-parietal-occipital region of the posterior cortex that contains the best current anatomical candidates for both content-specific and full NCCs. Content-specific NCCs may consist of particular subsets of neurons within this zone that support specific phenomenological distinctions, such as the experience of seeing shapes or hearing noises. Unlike event-related potentials (ERPs), which reflect the temporal dynamics of brain activity measured at the scalp, the cortical hot zone represents a cortical network that indicates where conscious experience is likely generated or integrated. However, activity

within this region has been linked to oscillatory dynamics. Specifically, during dreaming (a state of consciousness) compared to non-dreaming sleep, a decrease in low-frequency power (1–4 Hz) and an increase in high-frequency power (20–50 Hz) have been observed within this hot zone (Siclari et al., 2017). This suggests that in general, diminished low-frequency and elevated high-frequency activity originating in this region are positively correlated with conscious experience.

The P3b ERP is a late onset positivity (300-700ms) found in fronto-central-parietal regions, and is associated with a variety of function including priming, cognitive processing, memory storage, working memory, context updating, decision processes, reactivation of stimulus–response (S–R) links, and the production of conscious representations (Verleger, 2020). The P3b has been a promising and popular candidate, especially as evidence for the Global Neuronal Workspace Theory (GNWT) (Dehaene & Changeux, 2011; Mashour et al. 2020). However, due to improper experimental design involving reporting and task relevance, the P3b has been criticized for being conflated with post perceptual processing of experience (Dembski et al., 2021, Pitts et al., 2018). The issue with experimental design and no report paradigms will be further elucidated in a section 3.4.

The perceptual awareness negativity (PAN) is an umbrella term for an awareness related negativity found to occur in visual (VAN), auditory (AAN) and somatosensory (SAN) modalities (Dembski et al., 2021). The PAN is an early onset negativity that occurs between 100-300 ms and is closely related to the N1 and N2 ERPs. It is a difference wave, meaning, it is isolated by calculating the difference in neural signals between consciously perceived and non-perceived stimuli.

The hybrid N2 occurs between 250-400 ms after stimulus onset and seems to be an ERP with mixed features of the VAN and P3b. It is a later negativity compared to the PAN, but earlier and less broad than a P3b. This hybrid N2 can be interpreted as conscious perception occurring in a stage of processing between low-level sensory encoding and higher level broadcasting across global neuronal networks, requiring a minimal degree of attention (Cohen et al. 2023, 2024).

The primary remaining ERP candidates for the NCCs most related to my current study are the PAN, the P3b, and the N2 (Figure 3). While debate continues, current consensus increasingly favors earlier components, such as the AAN and N2, as more likely to reflect the minimal neural mechanisms required for conscious perception. Later components like the P3b, have been more connected to post-perceptual processes such as decision-making, report, or cognitive access. The present study examines auditory content-specific processing using ERPs and time-frequency dynamics (4–25 Hz) to distinguish neural signatures of awareness under varying attentional and task demands.

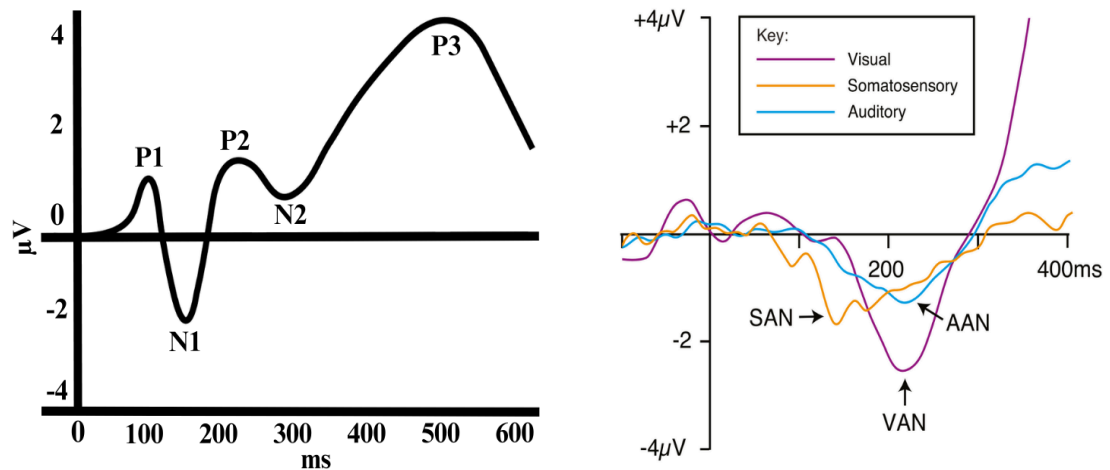


Figure 3. Canonical ERP components and the Perceptual Awareness Negativity (PAN) difference wave. (A) A standard ERP waveform showing typical early and late components (P1, N1, P2, N2, and P3) in response to sensory stimuli. (B) The Perceptual Awareness Negativity (PAN) plotted as a functional difference wave (aware minus unaware trials) across visual (VAN), auditory (AAN), and somatosensory (SAN) modalities. The PAN emerges as a modality-specific negativity peaking between ~150–250 ms, reflecting early neural activity associated with perceptual awareness independent of post-perceptual processing. Reprinted from *Trends in Cognitive Sciences*, Vol. 25, No. 8, Dembski, J., Koch, C., Pitts, M., & Lamme, V., Visual, auditory, and somatosensory awareness negativities, figure 3, p. 666, Copyright (2021), with permission from Elsevier.

3.3 Theories of Consciousness

The ultimate goal of a theory of consciousness is to not only identify neural correlates, but build a framework with explanatory power. This framework would be capable of addressing fundamental challenges such as the explanatory gap and the hard problem of consciousness. Disputes about the NCCs may stem less from disagreement over the neural correlates themselves and more from differences in how consciousness is defined, or to whom the subject of consciousness is, as explained in the first chapter. In this sense, different NCCs may be capturing distinct aspects or layers of consciousness depending on how the concept is framed, whether as sensation, perception, or reflective access.

This metaphorical elephant of consciousness, with its various parts explored by different approaches and frameworks, is slowly but steadily taking shape as a cohesive whole. The partial insights yielded by various investigations have produced a number of ingenious and influential theories, many of which are now beginning to interact and (hopefully) complement one another in the collective pursuit. One key theme emerging across these perspectives is that consciousness is a multidimensional and graded process (Chalmers, 2023; Cohen et al., 2023). This view challenges traditional dichotomies and aligns with recent efforts to model degrees of consciousness, both between different organisms or systems and within different cognitive states of a single agent.

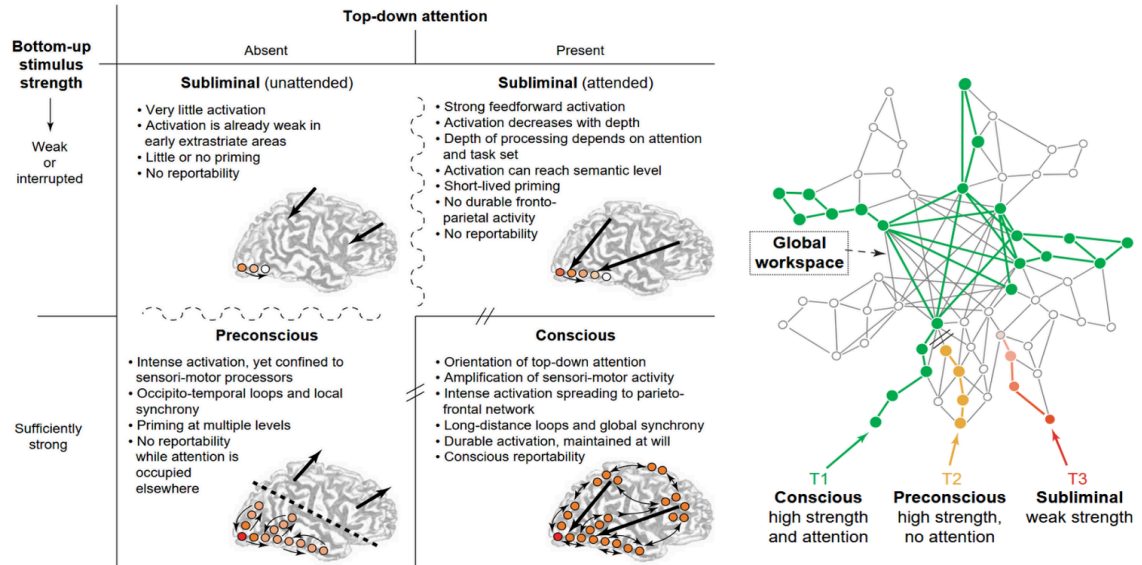
In the following sections, I will outline the theories that are most relevant to my current work. Each will be introduced in terms of their core framework, assumptions, and relation to the NCCs. I will further connect the theories to the distinction between P-consciousness and A-consciousness (Block 1995) and explore how they are related to inattentional phenomena.

Global Neuronal Workspace Theory

The Global Neuronal Workspace Theory (GNWT) is a neuroscientific expansion of Bernard Baars' Global Workspace Theory (GWT) (Baars, 1997). According to GNWT, consciousness corresponds to information that is “on the mind” – that is, content that is currently held in working memory and attended to. Attention and working memory jointly enable certain neural representations to be globally broadcast across specialized brain systems. This widespread broadcasting is what makes a representation conscious. When a representation becomes the focus of attention and is sustained by persistent neural firing, it enters the global workspace and becomes consciously accessible (Mashour et al., 2020).

GNWT posits a distributed network of specialized processors connected by workspace neurons that both receive bottom-up sensory information and transmit top-down signals. Information in the workspace is not only broadcast but also amplified and sustained via recurrent thalamocortical loops. Attention serves to amplify and stabilize representations, making them available to higher-order cognitive systems. These loops are essential to consciousness as such amplification enables information to reach broader systems, while sustainability through attentional maintenance keeps the information “online” in working memory (Mashour et al., 2020). Conscious access, therefore, depends on a combination of bottom-up stimulus strength and top-down attention. When bottom-up input is strong but attention is absent, the content remains preconscious. Perceptual processing occurs through occipito-temporal loops with local synchrony, but global ignition fails to occur due to lack of activation in frontoparietal regions (Dehaene et al., 2006, 2014). With sufficient stimulus strength and/or attentional signaling, global ignition and broadcasting is achieved: a rapid, widespread activation involving regions such as the prefrontal cortex (PFC). This ignition process unfolds over time, which is why GNWT is associated with late electrophysiological markers such as the P300 component.

The PFC is believed to be an essential hub in the global neuronal network, and activation in the PFC is often used as evidence for or against the presence or absence of consciousness. The PFC plays a crucial role in this model of global consciousness, but is not an exclusive region *where* consciousness occurs. The importance of the PFC is twofold; 1) its increased neuronal density is believed to be necessary for global broadcasting of information. 2) its vast web of connections to other regions of the brain, allowing it to dynamically modulate activity in various regions and function as a central hub or workspace. Certain regions such as the dorsolateral PFC and the inferior parietal cortex are hypothesized to have a specialized role (Mashour et al. 2020). Dehaene et al. (2006) further delineate their framework into a taxonomy of consciousness including subliminal, preconscious, and conscious, modulated by degrees of stimulus strength and attention (figure 4).

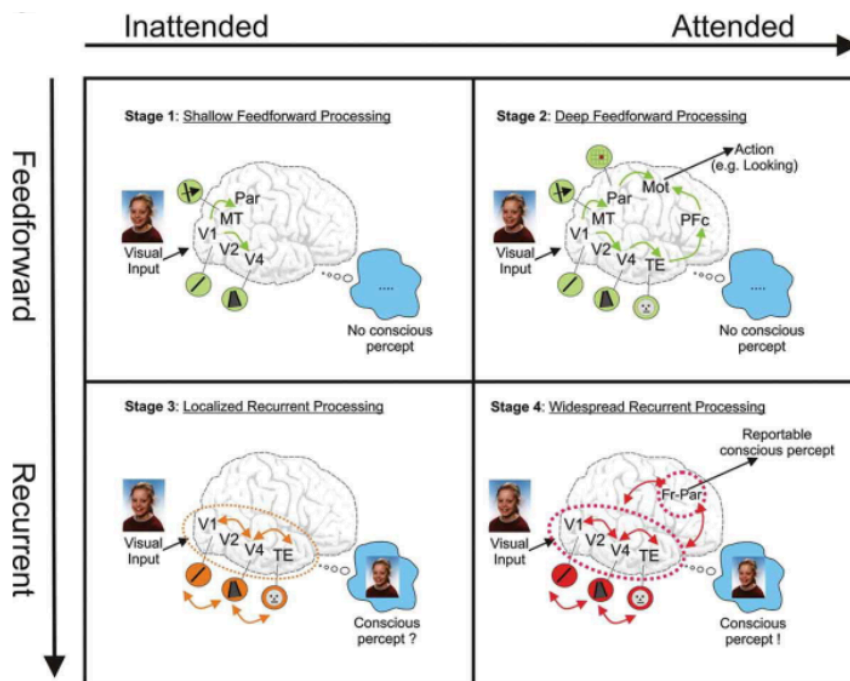


Figures 4: Subliminal, preconscious, and conscious in the GNWT. Subliminal implies a stimulus lacks the strength to connect to the workspace, even if with top down attention.. Preconscious activity correlates with strong enough stimulus strength, but lacks top down attention to be connected with the workspace. Adapted from *Trends in Cognitive Sciences*, Vol. 10, No. 5, Dehaene, S., & Changeux, J.-P., *Experimental and theoretical approaches to conscious processing*, figures 1 and 2, pp. 206 & 208, Copyright (2006), with permission from Elsevier. An additional category might include conscious experiences driven by extremely high stimulus strength, which engage bottom-up attentional mechanisms even without top-down input.

Although grounded in neuroscience, the GNWT raises important questions about what qualifies as conscious experience. Dehaene and colleagues believe that subjective experience, P-consciousness, always accompanies cognitive access, so by studying the latter you are studying the former (Dehaene et al., 2006). This may hold true, however, it perhaps does not fulfill the minimality condition of the NCCs, as proposed by Koch et al. (2016). For the GNW theorist, it seems that what counts as conscious is what is currently accessible to the egoic, reflective self: the subsystem responsible for introspection, verbal report, and deliberate control. Anything not available to this subsystem is treated as unconscious or, at best, preconscious. In this framing, consciousness is effectively equated with cognitive access: the ability to manipulate information in working memory for goal-directed behavior. The mind becomes conscious only when content reaches this high-level, supervisory workspace. For the GNWT consciousness is intertwined with a refined, late-emerging self-model, one built on metacognition, narrative, and reportability. This view sidelines the possibility of pre-reflective or non-conceptual forms of awareness, confining consciousness to a narrow range of what can be accessed, reflected upon, and used by the executive system. It is a model of consciousness centered on a specific subset of experience, namely, that which is accessible, reportable, and cognitively integrated by the reflective mind. In our account, the GNWT is primarily a theory of higher level conscious access, where lower levels of experience, especially in regards to perceptual consciousness, can be explained sufficiently by other theories that root their evidence in earlier physiological signatures of consciousness.

Recurrent Processing Theory (RPT)

In 2021, Rafael Malach provided evidence that experiential content is correlated with local patterns of neuronal activity independent of activity in frontal regions of the brain. The brain imaging showed evidence that frontal-parietal networks are specialized for higher order and meta-cognitive functions such as reporting and decision, and a lack of evidence that these networks are for conscious perception (Malach, 2021). Neural activity for conscious perception was found to occur locally in the visual cortex with activity in the frontal cortex associated with higher order functions such as decision making and reporting. This understanding that local sensory cortices are sufficient for consciousness through recurrent processing, and frontal areas are activated for post conscious functions such as reporting aligns with the Recurrent Processing Theory of Consciousness (RPT) presented by Victor Lamme (2006). Lamme posits that we can 'see without knowing': that is, experience perceptual content without fully accessing, conceptualizing, or reflecting upon it, aligning with the pre reflective, perceptive view of the subject. Contents of perception can be considered conscious without the mind/self knowing or accessing them, and with little to no attention. These contents influence and are at least minimally part of our experience, although we might not fully realize their presence or effects. Lamme first defines the information flow from visual to motor areas as the feedforward sweep (FFS), in which activity spreads within 100-120 ms from the V1 to the dorsal and ventral areas, including the motor cortex and prefrontal regions. Once FFS has reached a certain area, connections reach more distant cells and feedback connections start sending information from higher level areas back to lower, even all the way down to V1 (Lamme 2010). Recurrent processing (RP) is supported by the activation of these feedback connections, which are triggered by the FFS. Lamme further delineates neural processing into four stages (figure 5).



*Figure 5: Stages of recurrent processing. Stage 1: (0-150ms) Superficial processing during FFS. A stimulus is not attended and is masked, where the stimulus does not penetrate deeply into the ventral stream hierarchy. Stage 2: (100-200ms) Deep processing during FFS. A stimulus is attended and is masked, and it may travel through to motor and prefrontal areas to influence behavior as in unconscious priming. Stage 3: (200-300ms) Superficial processing of a recurrent nature (RP). A stimulus is not masked and can evoke RP, but is not attended to (example - inattention blindness, change blindness, attentional blink). Stage 4: (300ms +) Deep, widespread (global) RP. RP spreads through low level sensory to high level executive areas. The stimulus has time to evoke RP and is attended. This level of processing is similar to entering the global workspace. Lamme argues that consciousness or “phenomenality” can be found in stage 3 alone, when recurrent processing is initiated but not yet widespread, and is limited to activation in specific local circuits such as the visual areas for visual consciousness. In stage 4, RP spreads through motor and prefrontal areas which allows the information to directly influence functions necessary for reporting and other higher cognition. Reprinted from “How neuroscience will change our view on consciousness: Neuroscience and consciousness,” by Victor A. F. Lamme, *Cognitive Neuroscience*, 1(3), 204–240, © 2010, reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com>*

Lamme claims that in cases such as inattention blindness, it is not the case that we fail to see and experience a stimulus, but rather it was not yet stored in working memory to be consciously accessed. Unlike GNWT where behavior (subjective report) is taken as the primary evidence of conscious experience, Lamme believes the neuroscience of consciousness necessitates the possibility of a perceptual experience occurring without being fully known or reportable, since it has not been stored in working memory. Just because something is not remembered does not mean it was not experienced in the moment. With this, an experience can occur through recurrent processing within sensory cortical areas without late activity reaching the motor or prefrontal areas. RPT is sufficient for and correlates with P-consciousness or subjective experience, but it is not sufficient for conscious access or awareness of an experience occurring. In this view, which is compatible with the current framework, an organism can have perceptual experience without *fully* knowing it is, and this is correlated in human experience with stage three localized recurrent processing. Cognitive access, the awareness or conscious knowing that an experience is occurring, is associated with stage four widespread recurrent processing where the recurrent processing is able to spread to frontal areas associated with working memory to be fully encoded.

3.4 The Issue of Report and Task Relevance in Experimental Design

Tsuchiya et al. (2014) and Pitts et al. (2014) emphasize a critical methodological issue in the search for the NCCs: the potential confound of report and task relevance. Many studies on consciousness require participants to explicitly report their experience in order to measure detection. Although simple and reliable, this act of reporting unfortunately introduces unrelated neural activity which may be conflated with the experience itself. This creates a risk of conflating the neural mechanisms of experience with those required to report that experience. In such paradigms, the neural activity associated with experience (correlate x) may be followed or even overshadowed by activity linked to the

decision to report (correlate y), obscuring the minimal neural mechanisms that underlie consciousness itself. As a result, isolating the true NCC requires experimental designs that can distinguish between the neural correlates of experience and those of report-related processes.

The issue of the report is actually part of an even deeper potential confound – task relevance. Stimuli that are directly relevant to a task elicit different patterns of brain activity compared to those that are task-irrelevant, so it is not clear if we are identifying the neural correlates of task relevancy or experience. Consequently, traditional NCC studies may be identifying neural correlates of task engagement rather than consciousness *per se*. The primary goal is to isolate consciousness as much as possible from other processes, specifically attentional, executive, and working memory, and if possible, accurately measure the degrees of each sub-process involved in experience.

For example, although a one-back task does not require an overt verbal report, it still imposes task relevance, as stimuli must be actively encoded and maintained in working memory to allow comparison with subsequent inputs. This sustained cognitive engagement of working memory recruitment introduces a confound. The neural activity may reflect task relevant, working memory processes, rather than the perceptual experience itself. Consequently, while the one-back task minimizes some report, it does not fully eliminate task-related influences and may therefore be insufficient for isolating the true NCCs. Pitts et al. (2018) advocate for the use of no-report, task irrelevant paradigms, which infer conscious perception through indirect measures such as eye tracking and neural decoding, thereby isolating NCCs from subjective report and task-related confounds.

This distinction is especially relevant to research on inattentional deafness, where subjective report is commonly used. Recognizing this challenge is essential for disentangling the neural correlates of experience from the cognitive demands of experimental tasks. That said, it may be neither realistic nor conceptually valid to attempt a complete dissociation of consciousness from attention or working memory. These processes may be intrinsically intertwined in the instantiation of conscious experience. Varying degrees and types of attentional or cognitive engagement may shape different expressions or levels of consciousness, as illustrated in figures 6 and 19.

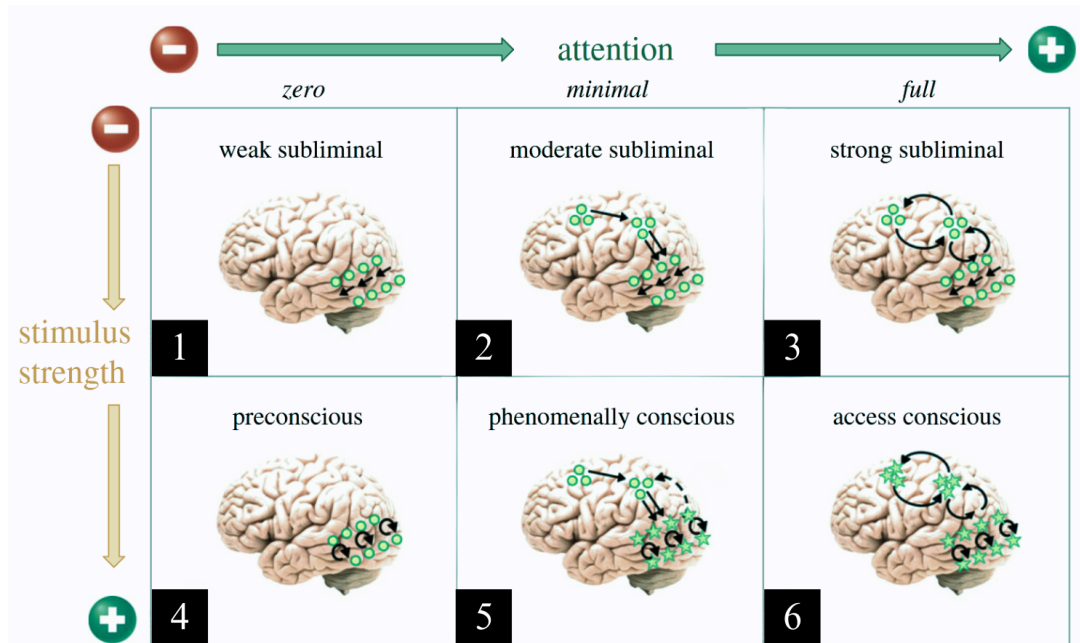


Figure 6: An updated taxonomy of consciousness involving stimulus strength and attention, which integrates ideas from both GNWT and RPT. Panels 1, 4, and 6 map to the GNWT taxonomy, while panels 1, 2, 4, and 6 map to RPT. Panels 3 and 5 are not featured in RPT or GNWT, and represent intermediate stages of attentional and sensory processing. This taxonomy differs from GNWT and is aligned with RPT in that it separates P-consciousness from A-consciousness in different stages. However, it differs from RPT and is aligned with GNWT in that attention, in some form, is still necessary for consciousness. Adapted from "The relationship between attention and consciousness: an expanded taxonomy and implications for 'no-report' paradigms," *Philosophical Transactions of the Royal Society B*, vol. 373, issue 1755, article 20170348 (2018), by Hillyard, Lutsyshyna, & Pitts. © The Royal Society. Licensed for reuse under Copyright Clearance Center Order License ID 1627274-1.

The goal is to develop task-irrelevant paradigms that can still differentiate between conscious and non-conscious stimuli (Pitts et al., 2018). In the context of EEG research, the P3b is particularly susceptible to such conflation. Many studies identifying the P3b as an NCC do not employ no-report, task-irrelevant paradigms, raising the possibility that the P3b reflects post-perceptual cognitive processes rather than consciousness itself (Pitts et al., 2014). Recent task-irrelevant experiments suggest that the hybrid N2 (Cohen et al., 2024), and cortical hot zone may serve as the true NCCs of consciousness (Cohen et al., 2024; Koch et al., 2016). In connection to P-consciousness and A-consciousness, the PAN, NN2, and hot zones may correspond to P-consciousness, whereas the P3b is more closely associated with reflective access. Moreover, the PAN and NN2 seem to reflect content-specific NCCs, while the hot zone is proposed as a candidate for the full NCC of consciousness (Dembski et al., 2021; Koch et al., 2016).

The Aim of our Paradigm

The present study introduces a novel no-report EEG paradigm to dissociate task-relevant and task-irrelevant conditions in the study of auditory awareness. Leveraging the phenomenon of inattentional deafness, we employ two reporting conditions with different cognitive loads to first establish that high-load conditions produce greater deafness (i.e., reduced auditory awareness) than low-load conditions. We then apply these same conditions in a no-report context – where participants are not asked to report or attend to the critical stimulus – to investigate the neural correlates of task-irrelevant auditory awareness. Because the conditions have been behaviorally validated to differ in auditory content awareness, we can infer that neural differences observed in the no-report conditions reflect content-specific NCCs, uncontaminated by report or task demand. Our focus is thus on identifying the NCCs of auditory content, rather than the full NCC of consciousness itself.

To further verify Lavie's load theory and build off of our previous experiment, we included both cognitive and perceptual load. The perceptual load stayed relatively the same, while the cognitive load was increased to a moderate level, with the goal of increasing deafness. Cognitive load was not designed to be too high, as this may, according to load theory, result in lower deafness which is counter to our overall objective. We expect the mix of moderate cognitive load and perceptual load to result in greater deafness than our previous experiment. The 2x2 experimental design includes four conditions:

Playing and Reporting (high load)	Watching and Reporting (low load)
Playing and not reporting (high load)	Watching and not reporting (low load)

The high load condition is expected to have increased deafness and decreased awareness, while the low load condition is expected to have decreased deafness and greater awareness. The reporting conditions are task relevant, although they are secondary tasks. The primary task is to play the game and get as high a score as possible while keeping track of high value targets collected, where reporting deviant sounds is a secondary task added on top of this, still requiring post-perceptual processes such as decision making and working memory. The non reporting conditions are void of any task relevant involved processes surrounding the auditory critical stimulus, such as decision making, top-down attention and working memory. Our expectations for the neural correlates are that in conditions of reporting, we will see increased P3b compared to non reporting, alongside increased theta power and decreased alpha power. In the watch, low load conditions (greater awareness), we expect to see an increased N2 and P3b when compared to the play, high load conditions (lower awareness).

3.5 Materials and Methods

Participants

Participants were between 18 and 30 years of age, and were in good general health. They had no history of psychiatric or neurological illness, and they were not taking any psychoactive medications. 25 young adults were recruited for this experiment through the University of California, Merced online recruiting database. All participants provided written consent and completed a general questionnaire before experimentation. Participants were compensated 15\$ per hour after completion of the experiment, or rewarded SONA credits. The experimental protocol was approved by the institutional Review Board at UC Merced.

Stimuli

The auditory stimulus consisted of two sounds: one white noise that played continuously through the entire time they were engaged in the game, and pink noise deviations (critical stimulus) which randomly sounded every 2-10 seconds. The pink noise was superimposed on the white noise with no gap, so it sounded like a salient alteration of the noise. The combined noise segment was created using Adobe Audition. Each critical stimulus pink noise deviation was 150ms long. The Pink noise deviations within each white noise segment were randomized for each condition and for each participant. The auditory stimulus (white noise + pink noise deviations) played the entire time a condition was occurring.

Procedures

EEG was recorded as participants were seated in a sound-attenuated booth ~1 m away from a 27-in. monitor. Auditory stimuli were presented at ~55 dBA sound pressure level (SPL) through two loudspeakers located to the left and the right of the monitor at a $\pm 45^\circ$ angle, creating the perception that the sound came from the center (0°). The study involved 4, 7-minute long conditions, consisting of ~86 critical pink noise trials. Participants had 3 gameplay practice trials, and 1 auditory practice trial to ensure they could detect the auditory deviations used in the task. All individuals reported hearing the sound changes during these auditory practice trials.

Presentation software (NeuroBehavioral Systems Inc., Berkeley, CA) was used to control the timing of auditory stimuli and Javascript was used in controlling the visual gameplay. EEG was recorded using a high-impedance 64-channel Biosemi System (Active Two system, 10–20 Ag-AgCl electrode, with Common Mode Sense and Driven Right Leg passive electrodes serving as grounds, A/D rate: 1,024 Hz). The block order for the 4 conditions – watch report (WR), play report (PR), watch no report (WNR), play no report (PNR), was randomized for each participant. After each block, participants were paused and told to wait for the researcher to come in. During this time, participants answered a game related question and were prepared for the next condition. During WR conditions, participants were asked to sit and watch a pseudo AI with simple game mechanics play the game while they attended to and reported sound deviations. In the PR condition, participants were asked to play the game while reporting sound deviations. In the PNR

condition, participants were told to only play the game, and in the WNR condition, participants were told to passively watch the pseudo AI play. Participants used their left hand to move their in-game ship left and right using the "A" (left) and "D" (right) keys. During the play conditions, participants were also required to keep track of how many valuable targets (aliens, see figure 7), they were able to capture. This increased cognitive demand by requiring players to seek out these objects and keep track of them in working memory. The average number of aliens captured was 52 (low: 25, high: 81), meaning the average number participants had to count up to was 52. In watch conditions, participants sat passively while staying engaged in watching a pseudo AI with simple game mechanics play, and did not keep track of the aliens. In reporting conditions, participants reported hearing a critical stimulus by clicking the mouse with their right hand.



figure 7: Players controlled the ship at the bottom by moving left and right using the "A" and "D" keys. The goal of the game was to dodge the meteors, while collecting the stars and aliens. The aliens were rare and valuable targets (worth significantly more points), and were kept track of as part of the game task.

3.6 Data Analysis

Behavioral

Behavioral data were analyzed using a custom MATLAB script to extract signal detection metrics from log files generated during the experiment. Each participant's responses were logged in condition-specific folders (WR and PR) and parsed to identify responses to target auditory deviations. Only participants with valid data in both the Watching + Report (WR) and Playing + Report (PR) conditions were included in the analysis. For

each participant and condition, the following metrics were computed. Hit rate (HR) was the rate of correct responses occurring within two seconds of an auditory cue (the window). False alarm rate (FAR) was the rate of responses that occurred outside the correct window and was normalized by the total duration of non-target noise. Sensitivity (d') was calculated as $z(\text{HR}) - z(\text{FAR})$, using a log-linear correction to handle extreme values (Macmillan & Kaplan, 1985). Response bias (criterion c) was computed as $-0.5 \times [z(\text{HR}) + z(\text{FAR})]$, quantifying the participant's overall tendency to respond or withhold responses. Reaction time (RT) was defined as the mean latency of correct responses (hits), and precision was calculated as the proportion of hits among all responses, indicating how frequently responses were correctly aligned with target stimuli. Reaction time (RT) was the time it took for a participant to detect and respond to a target, with precision indicating how frequently responses were correct.

Responses were parsed sequentially, and only the first response following each audio cue was considered a hit if it occurred within the 2-second response window. Any responses outside of this window were counted as false alarms. Participant data was put together into condition-specific tables and sorted to ensure consistent ordering across subjects. A paired-samples t -test was conducted on d' values between the WR and PR conditions, followed by computation of Cohen's d for effect size and the 95% confidence interval for the mean difference. Similar analyses were applied to reaction time and hit rate comparisons. Summary results were visualized using grouped bar plots with error bars representing the standard error of the mean (figure 8).

EEG

EEG analyses were conducted using ERPLAB (Lopez-Calderon & Luck, 2014), EEGLAB (Delorme & Makeig, 2004), and FieldTrip (Oostenveld et al., 2011) toolboxes along with in-house MATLAB code. Data were recorded from 64 scalp electrodes using a BioSemi layout. Processing steps included: Continuous EEG recordings from individual blocks were downsampled to 512 Hz and concatenated into a single dataset per participant. Each dataset was epoched from -200 to 1000 ms relative to the onset of each noise deviation and baselined to the entire epoch (mean removed). Independent component analysis (ICA) was performed on each dataset, producing 64 components. Components reflecting ocular artifacts were removed (mean of 2 per subject), and noisy channels (maximum of 2 per subject) were interpolated using EEGLAB's spherical interpolation. Data were then re-referenced to the average of all electrodes.

For analysis, all data were time-locked to the onset of the deviant auditory stimulus. To better reflect perceptual alignment, we applied a uniform temporal shift of -75 ms to all time axes, such that 0 ms corresponds to the acoustic midpoint of the 150 ms deviant sound. All time windows reported in the results refer to this midpoint-aligned scale; onset-aligned timings can be obtained by adding 75 ms. Datasets were subsequently processed along two parallel pipelines: one for event related potentials (ERPs), and one for time-frequency analysis of oscillatory activity.

ERP

ERP data were analyzed using EEGLAB and FieldTrip toolboxes in MATLAB. Preprocessed EEG datasets were converted from EEGLAB to FieldTrip format using a custom wrapper, and were organized by condition for each participant. ERPs were computed for four experimental conditions: Watch + Report (WR), Watch + No Report (WNR), Play + Report (PR), and Play + No Report (PNR). For each condition, time-locked grand averages were computed using `ft_timelockgrandaverage`. Pairwise statistical comparisons were conducted between selected condition pairs (e.g., WR vs WNR, PR vs PNR), with each contrast tested independently using cluster-based permutation tests (Maris & Oostenveld, 2007). A two-tailed within-subjects t-test (statistic = 'depsamplesT') was used for each comparison. Clusters were formed based on spatiotemporal adjacency (minimum of two neighboring electrodes), and significance was assessed via 2000 Monte Carlo permutations, with a cluster-forming threshold of $p < 0.05$. Electrode neighborhoods were defined using a standard BioSemi64 layout and computed via FieldTrip's `ft_prepare_neighbours`. Significant clusters were visualized using custom topographic and time-series plotting functions. ERP variance estimates were computed using within-subject standard error of the mean (SEM) across trials. All statistical outputs and figures were saved for interpretation and reproducibility.

Difference Wave Analysis

To assess condition related differences in ERP amplitudes, we computed difference waves between Watching + Report (WR) and Playing + Report (PR) conditions. EEG data were analyzed in MATLAB using EEGLAB and FieldTrip toolboxes. Preprocessed ERP datasets for each subject were loaded from condition-specific .ERPset files and converted to FieldTrip format. All waveforms were time-shifted by 75 ms to align with the acoustic midpoint of the auditory deviant. Grand average ERPs were computed across participants for each condition using `ft_timelockgrandaverage`, preserving individual subject data for statistical testing.

Analyses focused on a predefined frontocentral region of interest (ROI) composed of nine electrodes: F1, Fz, F2, FC1, FCz, FC2, C1, Cz, and C2. This specific ROI was selected based on prior work from Cohen et al. (2024), who identified the hybrid N2. ERP amplitudes were averaged across this ROI within a 200–400 ms time window, corresponding to the expected latency of the N2 component. This time window was chosen based on prior literature showing that N2 typically peaks between 200 and 350 ms, but may extend to 400 ms depending on the cognitive demands of the task (Folstein & Van Petten, 2008). A paired-samples t-test was conducted on the subject-wise difference in ROI-averaged amplitudes between conditions. Bayesian t-test was also computed using the default Cauchy prior ($r = 0.707$) to estimate the Bayes Factor (Rouder et al., 2009). To complement the point-estimate analysis, a custom cluster-based permutation test was performed on the full frontocentral ROI time series between 200–400 ms. Suprathreshold clusters were identified by applying a t-value threshold corresponding to $p < 0.05$ (two-tailed), and cluster-level mass statistics were computed. Significance was assessed via 2000 permutations using random sign-flips across subjects, and a cluster-level p-value was obtained by comparing observed cluster masses to the null distribution. Topographic maps of the significant difference wave (WR – PR) were

generated using FieldTrip's `ft_topoplotER`, highlighting electrodes in the frontocentral ROI and averaging over the significant cluster time window. Finally, ERP waveforms for each condition were plotted with standard error ribbons, along with the resulting difference wave, to visualize group-level effects over time.

Time frequency

Time-frequency (TF) analysis to generate figures of inter-trial phase-coherence (ITPC) and Event-Related Spectral Perturbation (ERSP) was implemented by the `timef.m` function of EEGLAB. The TF analysis spanned 4-30 Hz and used a sliding Hanning-windowed sinusoidal wavelet-based discrete Fourier transform (DFT) of the time-domain signal with a step size of 10 ms and frequency increments of 1 Hz. Post-stimulus activity was baselined to the pre-stimulus interval (pre-pink noise onset) of -1250 ms to -250 ms to avoid smearing of pre-stimulus activity with the post-stimulus activity. The sliding window was 500 ms in size at the lowest frequency (4 Hz, 2 cycles) and decreased in size, while the number of cycles increased linearly with frequency, reaching 125 ms in size (3.7 cycles) at the highest frequency (30 Hz).

Statistical analysis

Nonparametric permutation tests were used to evaluate time-frequency differences in inter-trial phase coherence (ITPC) and event-related spectral power (ERSP) across conditions. These tests offer an advantage over traditional parametric methods by avoiding assumptions about the underlying population distribution. Instead, they rely on repeated random resampling of the data to generate a null distribution under the assumption that no true condition differences exist (Good, 2000). By randomly reassigning condition labels across participants, a distribution of values representing chance-level effects is obtained. Observed differences are then compared against this null distribution to determine statistical significance. To address the multiple comparisons problem inherent in EEG data, we employed a max-statistic correction approach, in which the maximum value from each permutation is used to build the null distribution (Holmes et al., 1996). Permutations were applied across all 64 scalp electrodes, and the null distribution was derived from a 500 ms pre-stimulus baseline period (-1000 to -500 ms), following procedures recommended by Chau et al. (2004) to enhance sensitivity while controlling for false positives.

The following steps were conducted for the permutation tests: 1) Compute the mean ERSP or ITPC difference between condition 1 and condition 2. 2) Resample. In each resampling step the subjects are randomly assigned to either condition 1 or condition 2. The number of subjects in each condition remains unchanged during resampling. 3) For each resampling in step 2, compute the ERSP or ITPC spectrogram from the pre-stimulus data for all electrodes and time-frequency points. 4) Record the maximum absolute mean of ITPS/ERSP difference between the resampled conditions of step (3). 5) Following all the resamplings ($n = 2000$), the recorded maximum absolute differences are pooled together to generate the maximal null distribution. 6) Determine the threshold value for a p value of (0.05) from the maximal null distribution. 7) Identify significant group differences in the mean ERSP or ITPC computed in step (1) based on this threshold.

3.7 Results and Discussion

Behavioral Results

Figure 8 displays auditory target detection performance, in which performance declined significantly under high-load conditions compared to low-load conditions (high load = PR, low load = WR). Signal detection sensitivity (d') was significantly higher in the WR condition than in the PR condition, $t(25) = 7.77$, $p < .001$. Participants exhibited a mean d' of 3.78 (SD = 0.79) in the WR condition and 2.61 (SD = 0.62) in the PR condition, yielding a large effect size (Cohen's $d = 1.52$) and a mean difference of $\Delta d' = 1.18$, 95% CI [0.86, 1.49]. Behavioral accuracy mirrored this sensitivity drop, as participants correctly detected 90.14% (miss rate = 9.86%) of auditory targets in the WR condition versus 74.56% (miss rate = 25.44%) in the PR condition. The miss rate under high load was 2.6 times greater than under low load. Reaction times also significantly increased under high load, rising from a mean of 0.621s in WR to 0.770s in PR ($\Delta RT = 0.149$ s), suggesting delayed access to auditory signals. Task engagement was supported by the results from the working memory task, where participants had a mean capture rate of 76.56% (captured/spawned), and a mean accuracy of 87.05% (answered/captured).

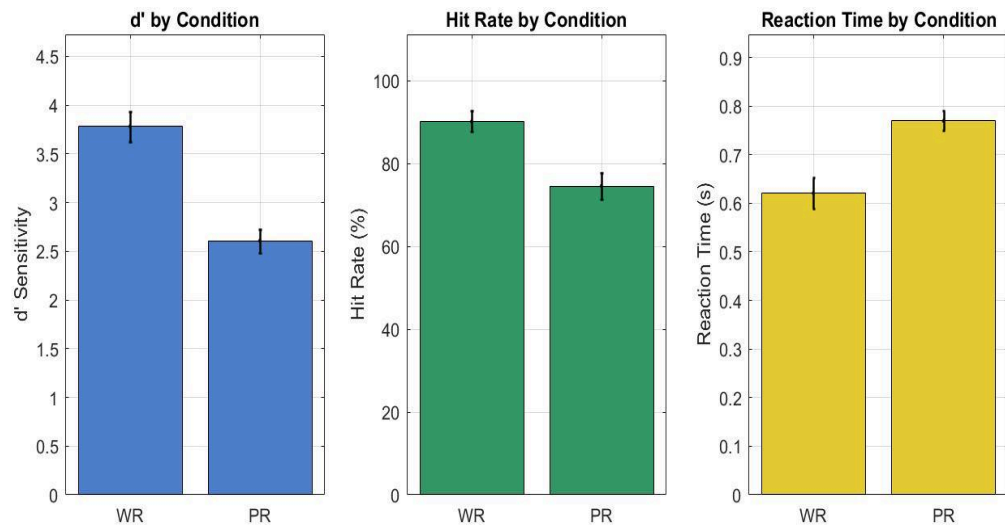


Figure 8. Behavioral performance by condition. Left: Signal detection sensitivity (d'). Middle: Hit rate (% correct). Right: Mean reaction time (s). Error bars reflect ± 1 SEM.

EEG Results – ERP

Consistent with prior research (Dehais et al., 2019; Molloy et al., 2015; Puschmann et al., 2013; Scheer et al., 2018), the N2 and P3b components were significantly reduced in conditions associated with greater inattentive deafness. Specifically, the presence and amplitude of the N2 and P3 were positively associated with task relevant, auditory awareness. Conditions in which participants reported the deviant sound (WR, PR) showed clear N2 and P3 responses, while non-reporting conditions (WNR, PNR) did not exhibit these components. Under high-load conditions (PR), the N2 and P3 components

were reduced in amplitude and delayed in latency relative to low-load conditions (WR), suggesting that increased perceptual and cognitive demands impair or delay the neural signatures of conscious access. Significant ERP responses were observed only in conditions where the auditory stimulus was task-relevant and reportable. No robust P3 component was detected in the task-irrelevant, non-report conditions, indicating that conscious reportability is a necessity for the reliable elicitation of these later components.

Looking at the early components, we see a strong P1-N1-P2-N2 sequence, even in the non-reporting conditions. In both non-report conditions (PNR and WNR), visual inspection of the grand-average ERP revealed consistent early sensory responses across frontocentral electrodes. As shown in figure 9, a stereotyped sequence of P1–N2–P2–N2 components emerged between approximately 50–300 ms, most prominently over frontal and central channels (e.g., Fz, FCz, Cz). Both non-report conditions (WNR and PNR) exhibited robust early auditory ERP components at FCz (and surrounding components), including a clear sequence of P1 (~50-100 ms), N1 (~100-150 ms), P2 (~150-250 ms), and N2 (~250-400 ms). No late positive deflection resembling the P3 component was observed in either task-irrelevant condition.

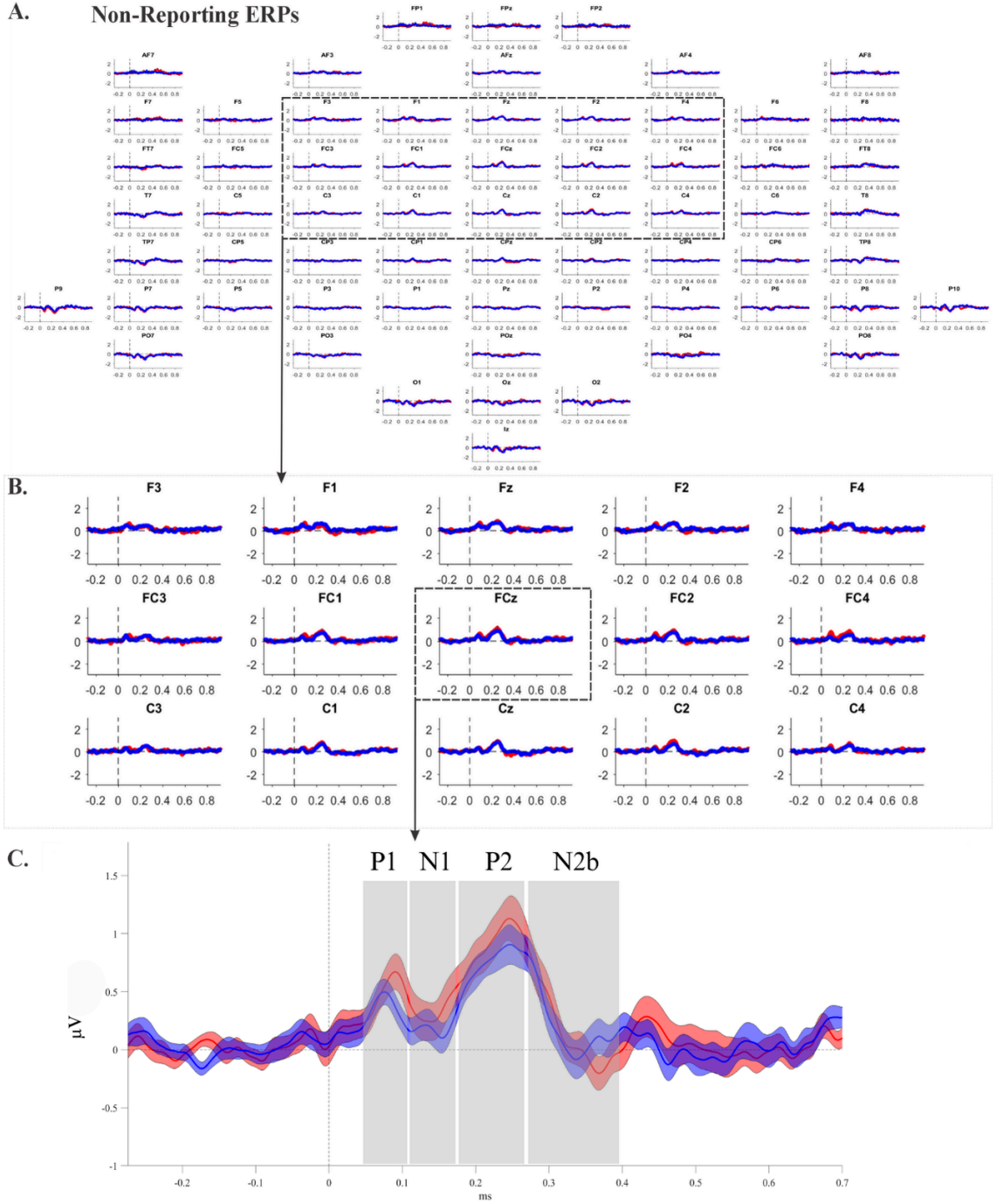


Figure 9: Grand-average ERPs for the non-reporting conditions (WNR: Watch–No Report; PNR: Play–No Report). (A) Scalp-wide ERP waveforms from –200 to 800 ms across all electrodes, comparing WNR (red) and PNR (blue) conditions. A P1–N1–P2–N2 sequence of early auditory components is observed over frontocentral electrodes and is consistent across all conditions. (B) Expanded view of selected frontocentral electrodes, highlighting the waveform in the 0–400 ms range. (C) ERP waveforms at FCz illustrating the respective individual positive and negative deflections emerging after 0 ms relative to a pre-stimulus baseline. A P1–N1–P2–N2 sequence is visible in both conditions, with no late positivity (P3) observed. All waveforms are time-locked to the onset of the deviant stimulus and adjusted by –75 ms to reflect its acoustic midpoint.

WR-WNR

Cluster-based permutation testing revealed two significant ERP waveforms. An early negative cluster was observed between 75–257ms post-midpoint (150–332ms post-stimulus onset), with a topography maximal over right centroparietal electrodes. The negative deflection was most likely an N2 component, peaking around 200ms post-midpoint. No distinct N1 peak was observed, and this may have been due to a smearing of the two components.

A second positive centroparietal cluster centered around CPz and Pz emerged between 280–638 ms post-midpoint (355–713 ms post-onset). Waveform plots (Figure 10) show that the WR condition elicited a larger early negativity and a pronounced late positivity compared to WNR, which exhibited relatively flat waveforms in both time windows.

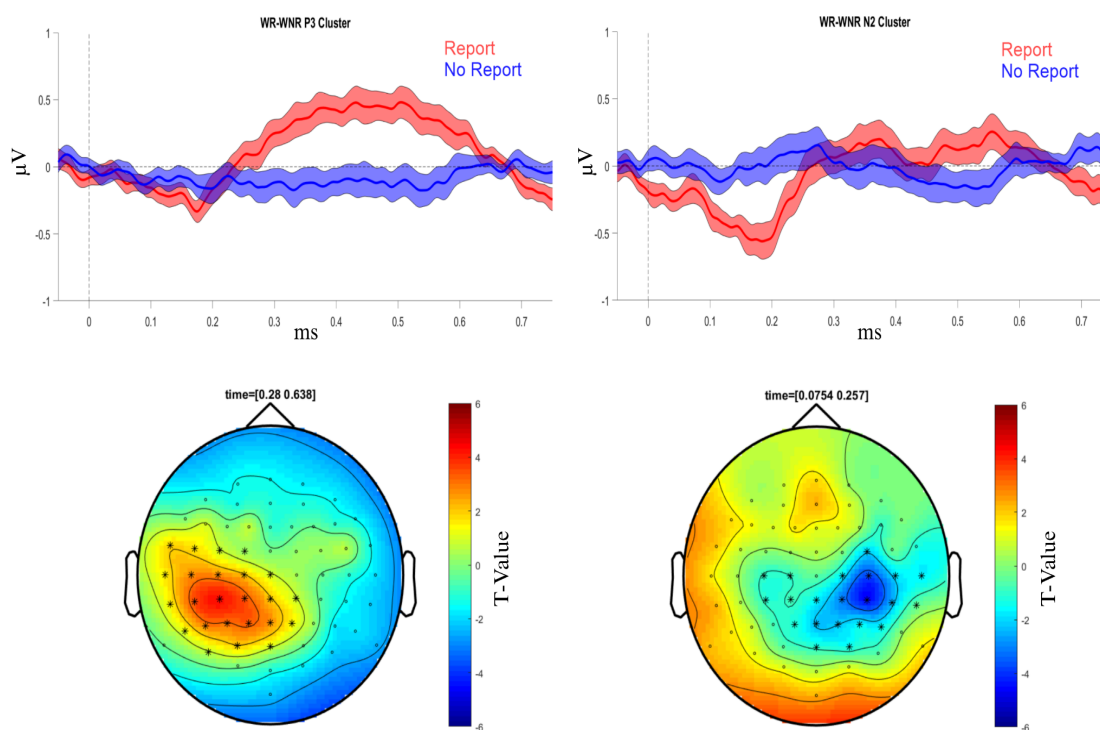


Figure 10: Event-related potentials (ERPs) comparing the report (WR) and no-report (WNR) conditions. (Top Left) ERP waveforms over the centroparietal cluster show a late positive deflection in the WR condition (red) relative to WNR (blue), spanning 280–638 ms post-midpoint (355–713 ms post-onset). This positivity exhibits a classic P3b morphology and distribution. (Top Right) An early negative deflection is observed in the WR condition between 75–257 ms post-midpoint (150–332 ms post-onset), peaking near 200 ms. This deflection is maximal over right centroparietal electrodes and may reflect a dominant N2 component, with a smeared or overlapping N1. (Bottom) Scalp topographies show the spatial distributions of the two significant clusters. The late cluster is left-lateralized, while the early cluster is right-lateralized. All waveforms were time-locked to stimulus onset and shifted by –75 ms to align 0 ms with the acoustic midpoint of the 150 ms auditory deviant.

PR-PNR

In the comparison between the high-load report (PR) and no-report (PNR) conditions, two significant clusters were identified. An early negative cluster with a central distribution was observed between 128–269 ms post-midpoint (203–344 ms post-onset), indicating an N2 component. A later, positive centroparietal cluster emerged between 407–683 ms post-midpoint (482–758 ms post-onset), indicating a P3 (Figure 11). The P3 topology was almost exactly the same as the watch condition, however the N2 component was more centralized here, while it was more right lateralized in the watch condition.

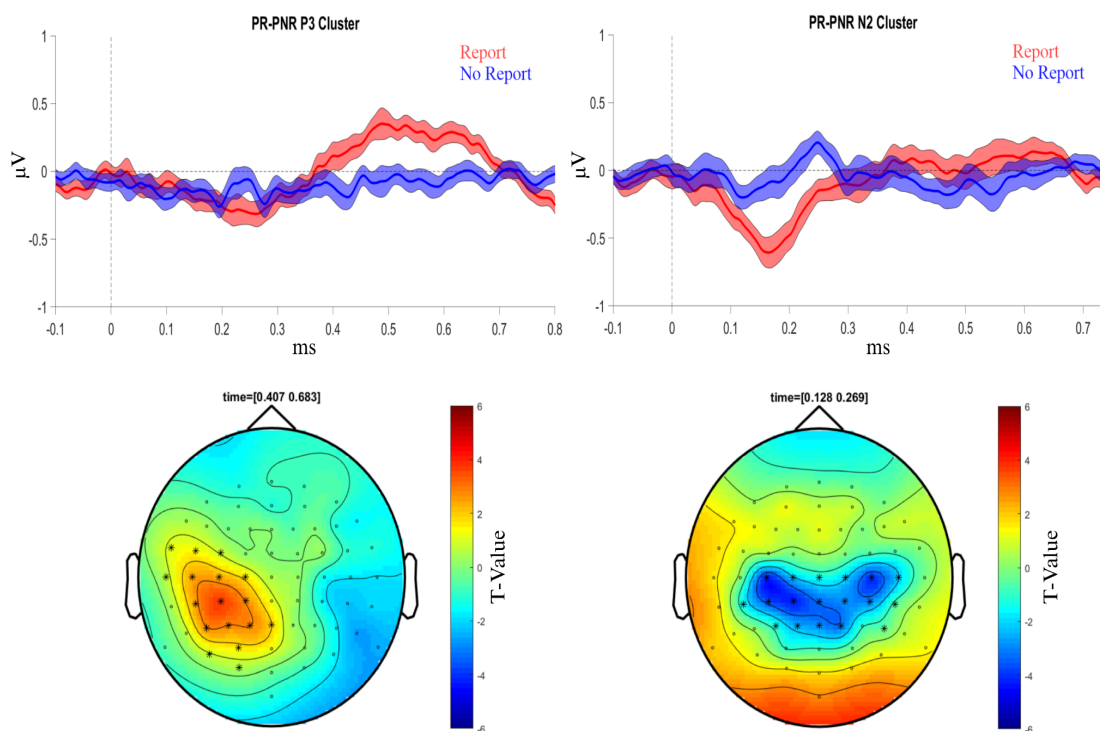


Figure 11: Event-related potentials (ERPs) comparing the report (PR) and no-report (PNR) conditions under high cognitive load. (Top Left) ERP waveforms over the centroparietal cluster reveal a late positive deflection in the PR condition (red) relative to PNR (blue), spanning 407–683 ms post-midpoint (482–758 ms post-onset). (Top Right) ERPs over a central cluster show an early negative deflection in the PR condition from 128–269 ms post-midpoint (203–344 ms post-onset), peaking near 200 ms. (Bottom) Scalp topographies show the spatial distributions of the two significant clusters. The P3b (left) is left-lateralized, while the N2 (right) exhibits a centrally distributed topography. All waveforms were time-locked to stimulus onset and shifted by –75 ms to align 0 ms with the acoustic midpoint of the 150 ms deviant sound.

WR-PR

A significant late positive cluster was identified in the comparison between the low-load report (WR) and high-load report (PR) conditions (Figure 12). This effect spanned 228–517 ms post-midpoint (303–592 ms post-onset) and exhibited a centroparietal distribution centered around Pz. The WR condition elicited a larger sustained positivity relative to PR across this window, consistent with modulation of the P3b by task load.

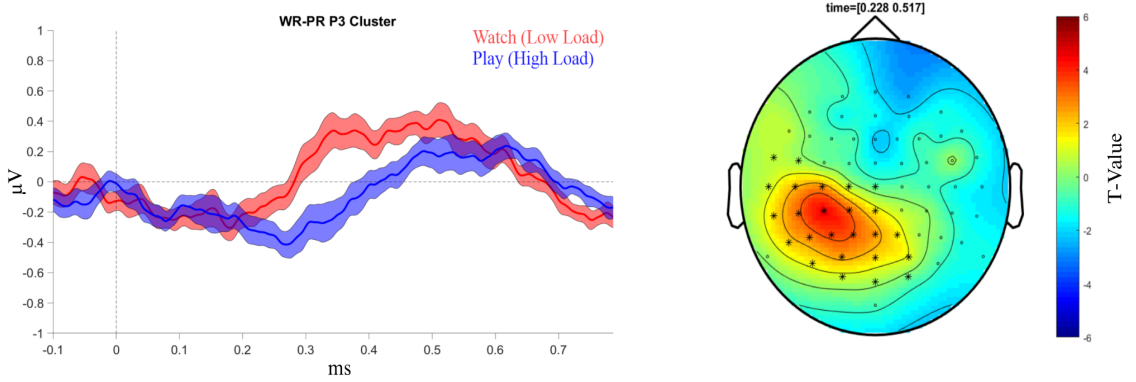


Figure 12: Event-related potentials (ERPs) comparing the low-load report (WR) and high-load report (PR) conditions. (Left) ERP waveforms over the centroparietal cluster show a larger sustained positivity in the WR condition (red) compared to the PR condition (blue), spanning 228–517 ms post-midpoint (303–592 ms post-onset). (Right) Scalp topography illustrates the spatial distribution of this significant cluster, with maximal activity centered over parietal electrodes. All waveforms were time-locked to stimulus onset and shifted by -75 ms to align 0 ms with the acoustic midpoint of the 150 ms auditory deviant.

To analyze the frontocentral activity and evaluate the hybrid N2 found by Cohen et al. (2024), we computed a difference wave between Watch-Report (WR) and Play-Report (PR) conditions (figure 13). A frontocentral region of interest (ROI) was defined a priori, including electrodes F1, Fz, F2, FC1, FCz, FC2, C1, Cz, and C2. Mean ERP amplitude was extracted between 200–400 ms post-stimulus midpoint. A paired-samples t-test comparing average amplitude between conditions did not reach significance, $t(24) = -1.42$, $p = 0.1687$, with a mean difference of $-0.228 \mu V$ (WR – PR). However, a cluster-based permutation test over the same ROI and time window revealed a significant negative cluster, $p = 0.0360$, spanning 335–392 ms post-midpoint. This reflects a localized frontocentral difference between WR and PR within the 250–400 ms latency range, illustrated by a negative deflecting difference wave in figure 13.

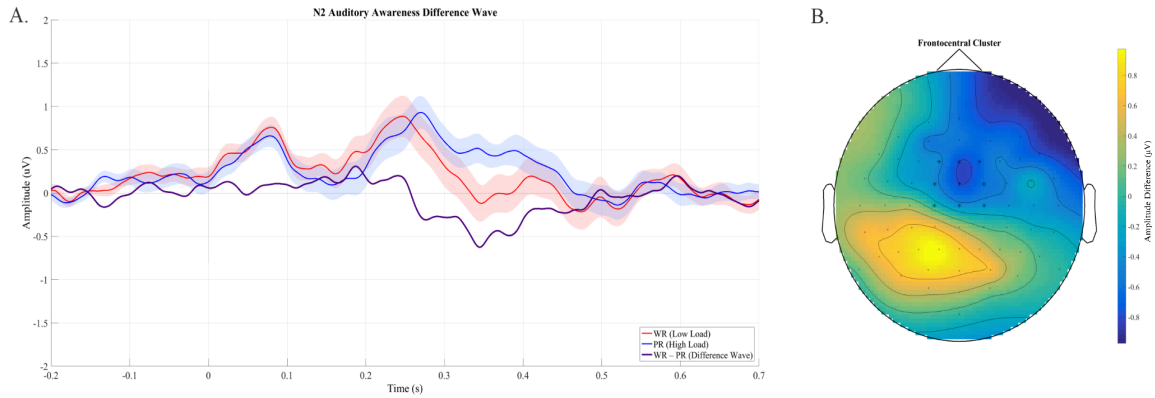


Figure 13: Auditory awareness-related N2 component differences between low-load (WR) and high-load (PR) reporting conditions. (A) ERP waveforms at a frontocentral cluster showing the N2 component for the WR condition (red), PR condition (blue), and their difference wave (WR–PR; purple). The N2 peaked around 300–350 ms, with greater negativity under low-load conditions (WR), indicating enhanced auditory awareness. (B) Scalp topography of the WR–PR difference in the N2 time window (250–400 ms post-midpoint) reveals a frontocentral distribution, consistent with typical N2 sources in auditory paradigms. All data were time-locked to stimulus onset and shifted by –75 ms to align 0 ms with the acoustic midpoint of the auditory deviant.

Time Frequency

WR vs WNR – Power

Time-frequency analysis of oscillatory power revealed significant condition-related differences between WR (Watch + Report) and WNR (Watch + No Report) trials. Using permutation testing ($p < .05$, corrected), we identified significant differences in theta (4–6 Hz), alpha (9–13 Hz), and beta (18–24 Hz) frequency bands. As shown in figure 14, the WR condition exhibited a significant beta-band (18–24 Hz) power suppression from 300–600 ms over left centroparietal electrodes (e.g., C3, CP5; Panel A), as illustrated in the topographical plot (Panel C). A second significant cluster revealed alpha-band (9–13 Hz) suppression over left posterior regions, also occurring between 300–600 ms (Panel D), perhaps reflecting cognitive access. A theta-band (4–6 Hz) power increase was observed from 200–350 ms over frontocentral electrodes (e.g., FCz, Cz), as shown in Panel E. This early theta enhancement may reflect task engagement or attentional processing associated with sound detection.

Raw power differences between conditions are presented in Panel B, while Panel A shows the statistically thresholded cluster masks across time-frequency plots at each electrode. These findings demonstrate that conscious auditory reporting is associated with increased early frontal theta power and reduced late alpha/beta synchronization over posterior scalp regions.

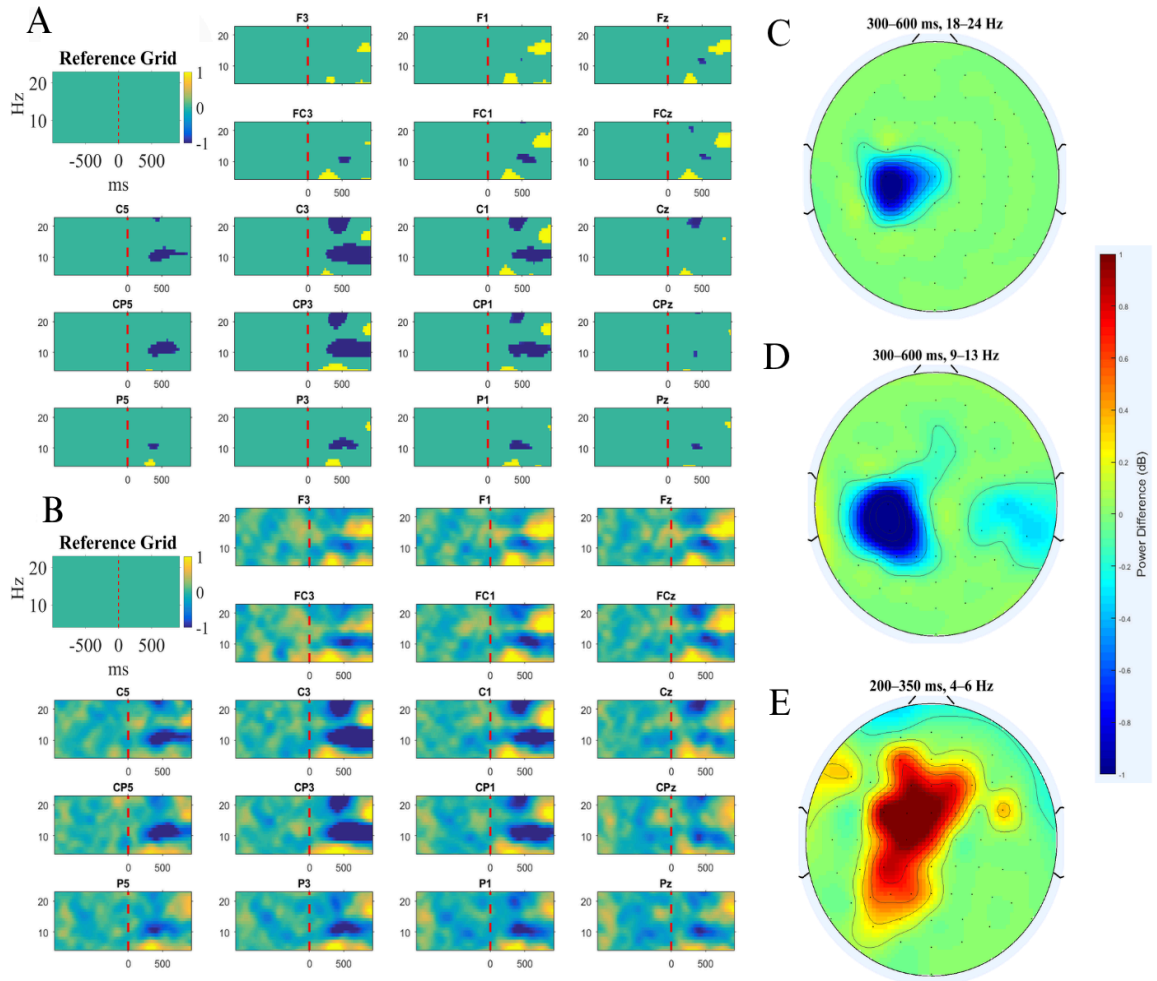


Figure 14: Time-frequency power differences between Watch + Report (WR) and Watch + No Report (WNR) conditions. (A) Time-frequency cluster masks from a permutation test ($p < .05$, corrected), indicating significant power differences at individual electrodes. (B) Raw condition differences in event-related spectral perturbation (ERSP) power across time (0–800 ms) and frequency (4–24 Hz). (C) Topography of significant beta-band (18–24 Hz) power suppression in WR from 300–600 ms, localized over left centroparietal electrodes. (D) Topography of alpha-band (9–13 Hz) suppression during the same window, spanning left centroparietal regions. (E) Early theta-band (4–6 Hz) power enhancement from 200–350 ms over frontocentral areas. Color scales reflect normalized differences (WR – WNR). A miniature Reference Grid is shown in the upper-left corner to clarify axes and units: time (ms) on the x-axis and frequency (Hz) on the y-axis. Red vertical lines mark the acoustic midpoint of the auditory deviant stimulus.

WR vs WNR – Phase

Analysis of inter-trial phase coherence (ITPC) in the theta band (4–5 Hz) revealed differences between WR and WNR conditions. A permutation test ($p < .05$, corrected) identified a sustained significant cluster from 100 to 400 ms post-stimulus. As shown in figure 15, the effect displayed a clear spatiotemporal progression in theta phase-locking across the scalp. From 100–200 ms, increased theta phase-locking in the WR condition was observed over central electrodes, including Cz and CP1. In the 200–300 ms window the topology shifted toward frontal electrodes such as AFz and Fz. By 300–400 ms, the

theta enhancement was most prominent over left parietal electrodes. These results suggest that conscious auditory reporting in the WR condition is associated with a temporally evolving pattern of theta phase alignment, beginning centrally, advancing frontally, and culminating parietally.

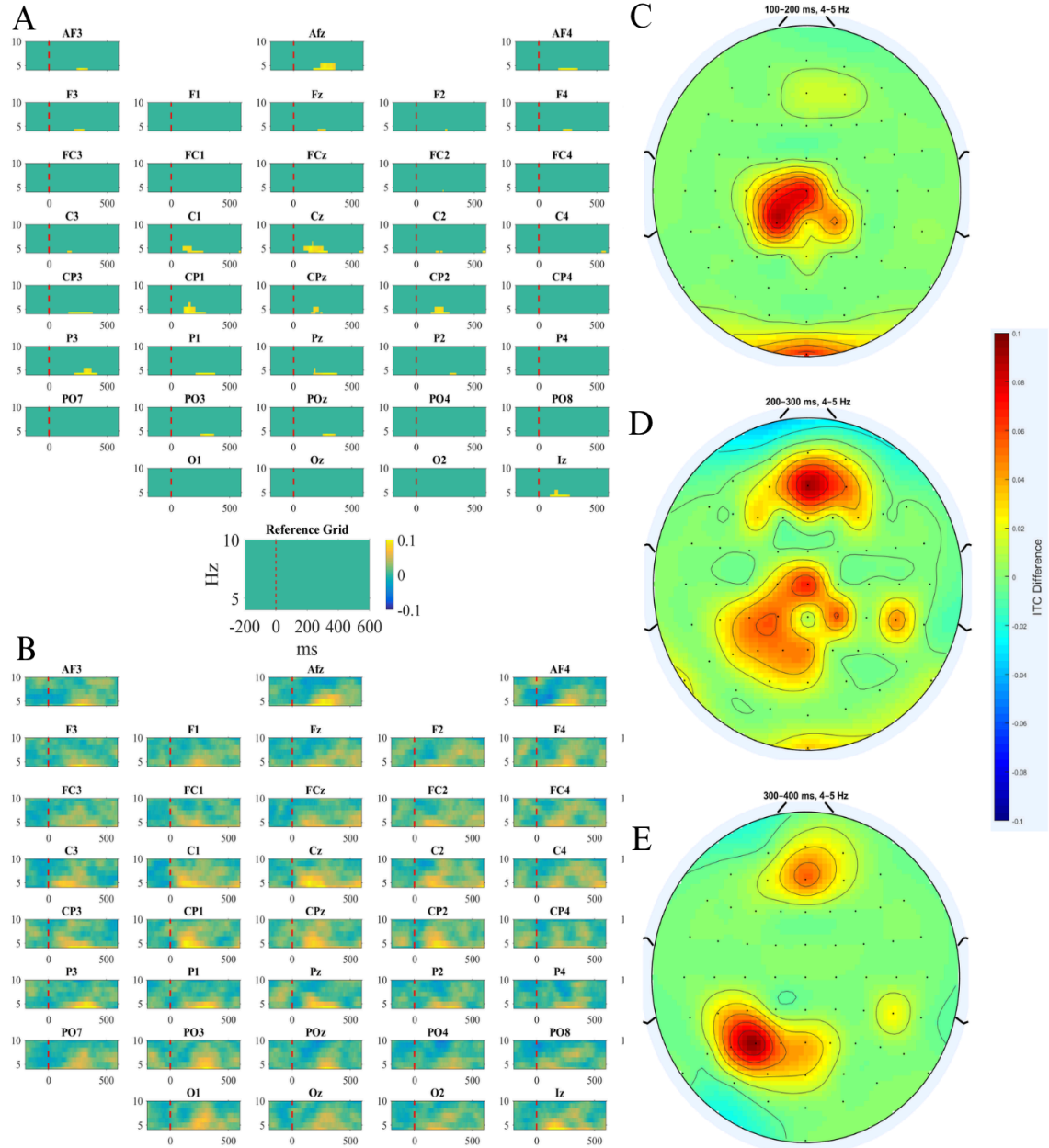


Figure 15: Theta-band (4–5 Hz) phase-locking differences between WR and WNR conditions. (A) Time-frequency cluster masks from a permutation test ($p < .05$, corrected), showing significant differences in inter-trial phase coherence (ITPC) between WR (Watch + Report) and WNR (Watch + No Report) conditions across electrodes. (B) Corresponding unmasked raw ITPC difference maps (WR – WNR) for each electrode. A miniature Reference Grid is shown in the upper-left corner to clarify axes and units: time (ms) on the x-axis and frequency (Hz) on the y-axis. The color scale reflects normalized phase-locking differences. Red vertical lines mark the acoustic midpoint of the auditory deviant stimulus. (C–E)

Topographical maps display the spatiotemporal progression of theta-band phase-locking enhancements in WR trials. At 100–200 ms (C), increased phase-locking is observed over central electrodes (e.g., Cz). At 200–300 ms (D), the effect shifts anteriorly to frontal regions (e.g., AFz). By 300–400 ms (E), theta synchronization becomes more localized over left parietal and midline frontal electrodes. Together, these results reveal a dynamic temporal propagation of theta phase consistency associated with auditory detection and reporting.

PR vs PNR – Power

Time-frequency analysis of induced power revealed significant differences between the PR (Play + Report) and PNR (Play + No Report) conditions across theta (4–6 Hz), beta (20–25 Hz), and alpha (9–14 Hz) frequency bands. Permutation testing ($p < .05$, corrected) identified three temporally and spatially distinct clusters. A significant theta-band enhancement (4–6 Hz) emerged first, from 300–600 ms, localized to left parietal electrodes including P5, CP5, and CP3. This was followed by a beta-band suppression (20–25 Hz) from 350–650 ms over left centroparietal electrodes, centered near C3 and CP5 (figure 16). Finally, a sustained alpha-band suppression (9–14 Hz) was observed from 350–750 ms, overlapping spatially with the theta and beta effects but extending slightly posteriorly. Compared to the WR–WNR contrast, these effects were not only weaker in amplitude and more spatially restricted, but also delayed in onset. Whereas WR–WNR differences began as early as 200–300 ms, PR–PNR effects did not emerge until approximately 300–400 ms, indicating a slower recruitment of oscillatory mechanisms under increased cognitive load.

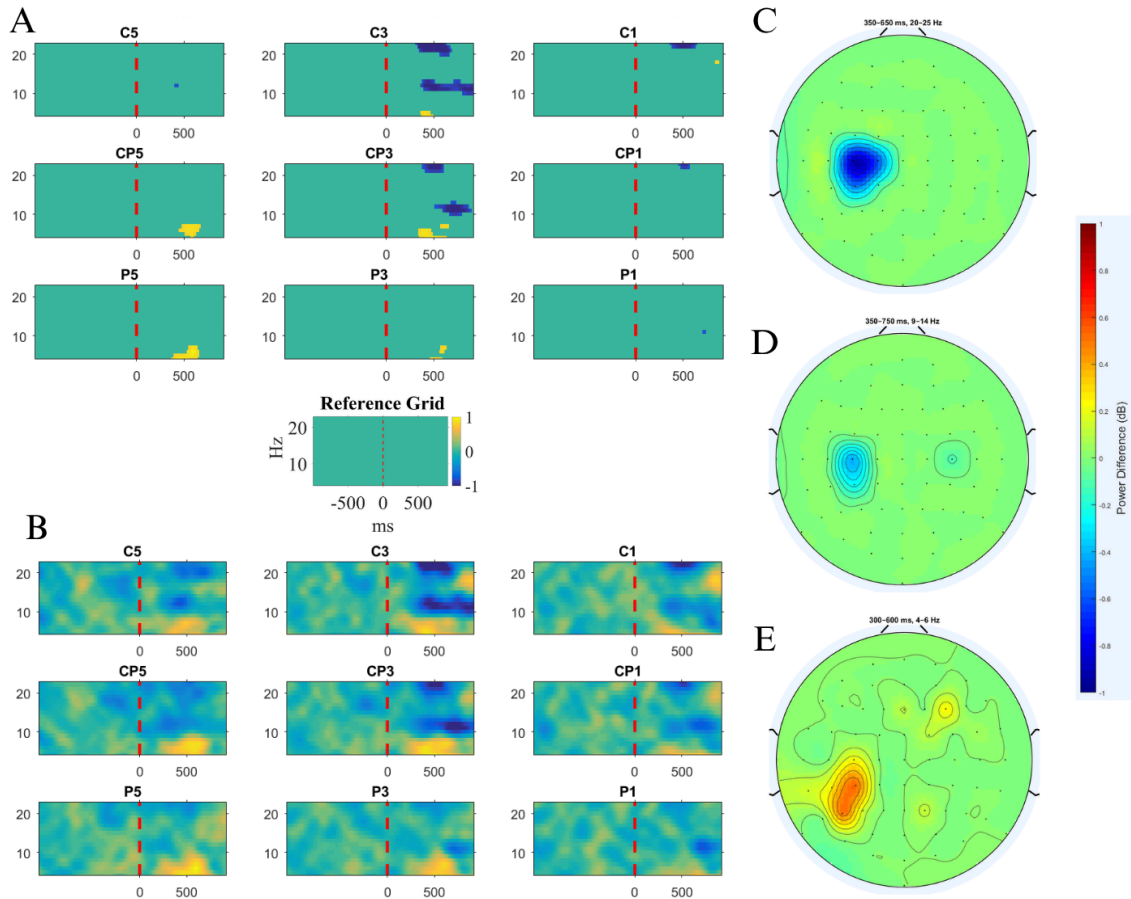


Figure 16: Time-frequency power differences between PR and PNR conditions. (A) Time-frequency cluster masks showing significant differences ($p < .05$, corrected) between PR (Play + Report) and PNR (Play + No Report) trials across a subset of centroparietal and posterior electrodes. (B) Raw ERSP difference maps (PR – PNR) for the same electrodes. A reference grid inset (center) illustrates axis orientation (time in ms on the x-axis; frequency in Hz on the y-axis). The color scale represents normalized power differences, with negative values indicating suppression in PR. (C–E) Topographical distributions of significant clusters derived from the permutation analysis. (C) Beta-band (20–25 Hz) suppression from 350–650 ms over left centroparietal regions. (D) Alpha-band (9–14 Hz) suppression from 350–750 ms, localized to left-lateralized central electrodes. (E) Theta-band (4–6 Hz) enhancement from 300–600 ms primarily over left parietal sites. Compared to the WR–WNR condition, these effects were similar in form but weaker in magnitude, more spatially restricted, and delayed in onset by approximately 100 ms.

WR vs PR – Power

Time-frequency analysis of induced power in the alpha band (9–14 Hz) revealed a significant difference between the WR (Watch + Report) and PR (Play + Report) conditions. A permutation test ($p < .05$, corrected) identified a significant alpha-band suppression in the PR condition relative to WR, occurring from 300–600 ms post-stimulus. This effect was most prominent over left centroparietal electrodes, including CP3, C3, and P3. No significant clusters were observed outside this time-frequency window or scalp region.

WR vs PR – Phase

Inter-trial phase coherence (ITPC) analysis in the theta band (4–5 Hz) revealed a significant difference between WR (Watch + Report) and PR (Play + Report) conditions. A permutation test ($p < .05$, corrected) identified a significant increase in theta phase-locking for the WR condition between 200–300 ms post-stimulus. This effect was observed over mid frontal and frontocentral electrodes, including AFz, Cz, CPz, and C2. As shown in the topographical map, theta phase-locking in WR was strongest over central regions and extended frontally, with peak activity centered over Cz and FCz.

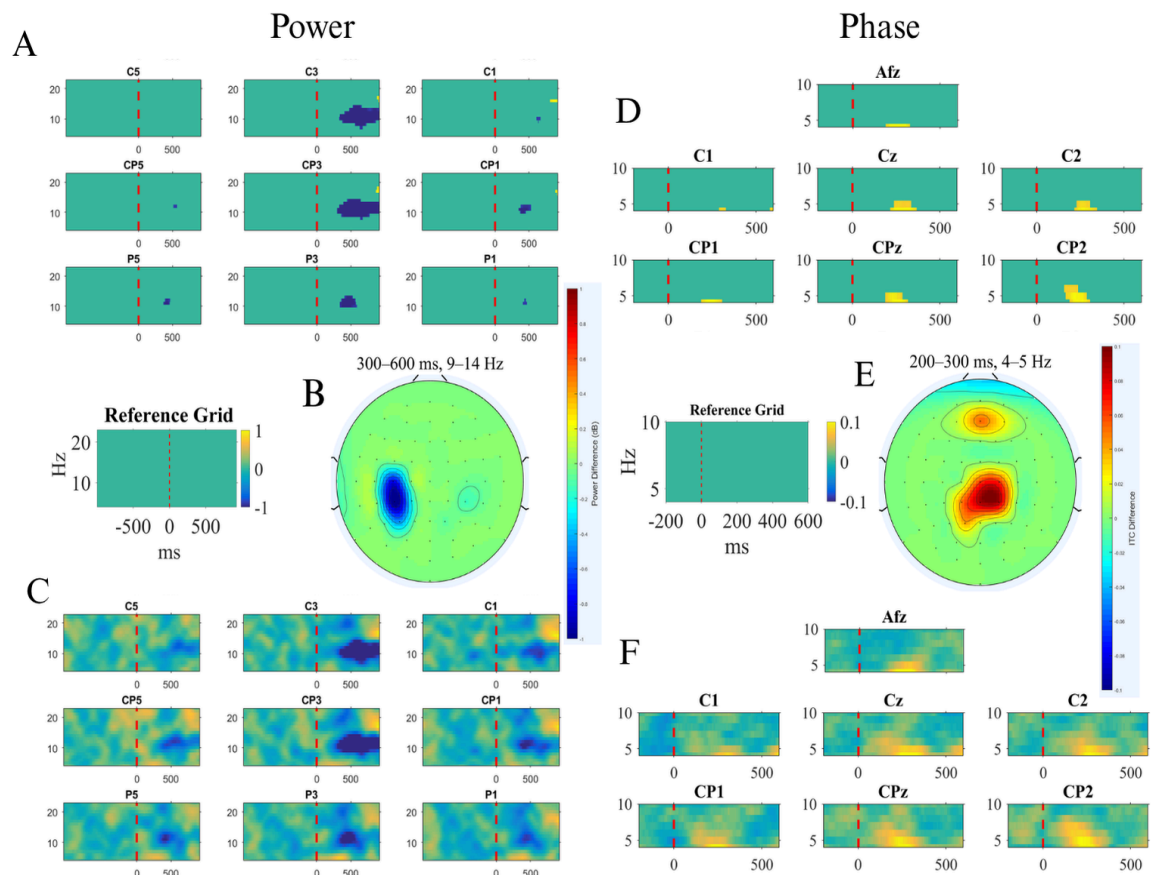


Figure 17: Time-frequency power and phase-locking differences between WR (Watch + Report) and PR (Play + Report) conditions. (A) Time-frequency clusters from a permutation test ($p < .05$, corrected), reveal a reduction in alpha-band power (9–14 Hz) in PR relative to WR from 300–600 ms over left centroparietal electrodes. (B) Topography of alpha suppression centered over CP3 and posterior electrodes. (C) Raw ERSP difference maps (WR – PR) for the same electrodes. The reference grid indicates axes (time in ms, frequency in Hz), and the color scale reflects power differences. (D) Time-frequency cluster masks showing significantly greater theta-band phase-locking (4–5 Hz) in WR compared to PR from 200–300 ms, primarily over midfrontal and central electrodes. (E) Topographical map of theta phase-locking differences, peaking at CPz and spreading to frontocentral sites. (F) Raw ITPC difference maps (WR – PR) at electrodes corresponding to the significant cluster.

WNR vs PNR – Power

Time-frequency analysis revealed a significant difference in power between WNR and PNR conditions (Figure 18). A permutation test ($p < .05$, corrected) identified an increase in power in the WNR condition. This frequency band occurred within the 7.5–8.5 Hz range at 400–500 ms post-stimulus, positioned at the boundary between high theta and low alpha. The topography was localized to left-lateral electrodes, FT7, T7, and C5. No other significant differences in power, phase-locking, or ERP amplitude were observed for this contrast.

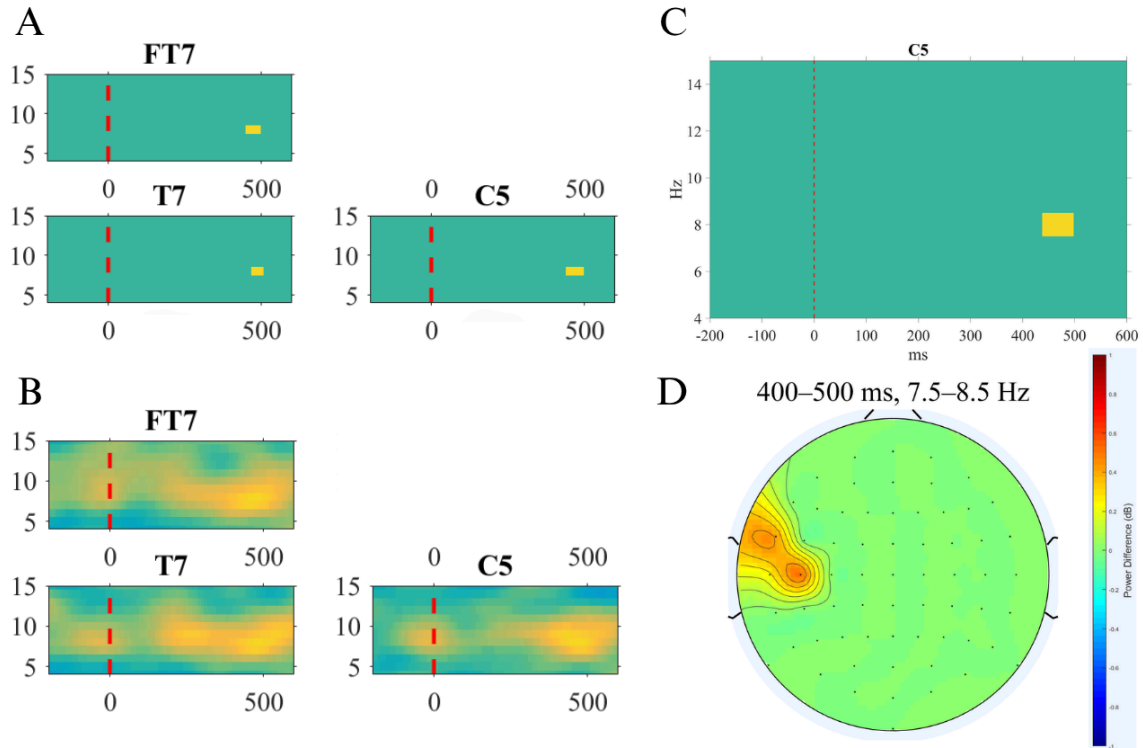


Figure 18: Induced power differences between WNR and PNR conditions (WR – PNR). (A) Time-frequency cluster masks showing significant power increases in WNR relative to PNR within the 7.5–8.5 Hz range between 400–500 ms at FT7, T7, and C5 ($p < .05$, corrected). (B) Raw ERSP difference maps (WNR – PNR) for the same electrodes. (C) Time-frequency cluster detail at C5. (D) Topographical map of the significant power increase from 400–500 ms in the 7.5–8.5 Hz band, showing a left-lateralized distribution. The color scale reflects normalized ERSP differences, with positive values indicating greater power in the WNR condition.

3.8 Discussion

This experiment had three primary aims: first, develop a no-report, crossmodal, continuous paradigm capable of inducing inattentional deafness and testing predictions of load theory, Second, clarify the neural signatures associated with inattentional deafness, and third, extend these findings toward the investigation of consciousness. We successfully induced inattentional deafness at a moderate rate, though the magnitude was somewhat lower than in prior studies, likely due to the introduction of a high cognitive load. These findings support load theory by demonstrating that perceptual load suppresses auditory awareness and that excessive cognitive load may paradoxically increase distractibility rather than reduce it (Lavie et al., 2004). The EEG data found in this experiment yielded a very large set of results. Many of which were consistent with our expectations, however, there were some unexpected patterns that emerged under specific conditions.

For clarity, we have summarized the four experimental conditions below:

- WR (Watch Report): Low load, task-relevant – lower inattentional deafness, greater awareness
- PR (Play Report): High load, task-relevant – greater inattentional deafness, lower awareness
- WNR (Watch No Report): Low load, task-irrelevant – lower inattentional deafness, greater awareness
- PNR (Play No Report): High load, task-irrelevant – greater inattentional deafness, lower awareness

Behavioral Discussion

Compared to our previous study, we found a shift in results related to inattentional deafness. Participants exhibited more misses in the low-load condition (WR) and fewer in the high-load condition (PR). In our prior study, the average deafness in the high-load condition was 46.49% (SD = 10.97), compared to just 4.40% (SD = 4.42) in the low-load condition. In the current experiment, miss rates were lower in the high-load condition at 25.44% and higher in the low-load condition at 9.86%. Two key methodological differences may account for this divergence. First, the auditory target in the present study was slightly less salient, which could have contributed to increased misses even under low cognitive-perceptual load. Second, the high-load task in this experiment involved a combination of perceptual, motor, and cognitive demands: participants actively played a game that required rapid visuomotor responses (perceptual/motor load) while simultaneously tracking and remembering how many objects they captured (cognitive load). In contrast, the previous study manipulated only perceptual load. This addition of a cognitive layer may have increased participants' susceptibility to distraction rather than suppressing auditory processing. The cognitive load, although intended to be moderate, may have been too intensive, thereby increasing susceptibility to distraction by taxing executive resources. This interpretation is consistent with load theory, which posits that high cognitive load impairs executive, top down inhibitory control and increases distractibility by irrelevant stimuli. It also suggests that combining perceptual and

cognitive load is actually less effective in inducing deafness compared to perceptual load alone. However, further analysis is warranted, particularly with a design that has combined perceptual load with a very minimal amount of cognitive load, or perhaps varying degrees of both.

Our findings, along with load theory, seem to be ecologically valid. Tasks that are overly engaging, especially those with high cognitive demands and forced effort, may paradoxically increase distractibility. For example, many of us (especially those with ADHD) may have experienced that when we are trying too hard to focus, we actually become more easily distracted by irrelevant stimuli. This may be due to an overexertion or misregulation of attentional resources. Individual differences in attentional control may have further influenced our results due to our participant pool consisting entirely of younger-generation college students. Recent studies have found a strong effect of social media and attentional control within young adults. Frequent engagement with social media has been associated with reduced sustained attention and increased distractibility (Chaudhuri & Imran, 2023; Haliti-Sylaj & Sadiku, 2024; Firth et al., 2019). These findings are especially relevant for our participant pool and emphasize the need to account for baseline attentional variability in designing inattentional deafness paradigms. We hope to further refine our paradigm by maximizing perceptual load while minimizing cognitive demands. In doing this, we may come across a “sweet spot” of combined load, where the task is engaging and maximizes both sustained attention and the probability of deafness occurring.

Neural correlates of inattentional phenomena

Previous research and evidence, alongside our current results, indicate that inattentional phenomena are marked by reduced N1/N2 and P300 amplitudes, suppressed gamma-band activity, and increased alpha-band power. However, the issue of task relevance and reporting is still a concern. In the no-report comparison (WNR vs. PNR), both conditions showed a frontocentral P1–N1–P2–N2 sequence, yet were not at all modulated by load, but instead by task relevance and report. This suggests two possibilities: either inattentional deafness was not strongly induced in these conditions, or, more plausibly, these neural signatures rely on top-down attentional amplification to become detectable. Without task relevance, this amplification may be very weak or absent, obscuring potential NCCs.

In contrast, the WR–PR comparison (low vs. high cognitive load during reporting) revealed ERP differences that mirrored behavioral deafness rates. In the PR condition, which is associated with greater inattentional deafness, a reduced and shortened P3b amplitude over centroparietal electrodes was found. This supports the role of the P3b in conscious access and post-perceptual processing (Polich, 2007; Sergent et al., 2005) and illustrates how these processes degrade under conditions of high load. Furthermore, a delay was found between the onset of the P3b in the PR condition (407 ms) compared to WR (280 ms). These findings suggest that increased load not only elevates the likelihood of inattentional deafness but also delays stimulus processing and accessibility. These cross-modal interactions are seen in real-world situations, such as listening to music or a

podcast while driving. In these cases, many people experience delayed responsiveness and reactivity to driving related stimuli, which is supported empirically by our results.

Additionally, PR showed a weaker and shorter-duration N2 response (250–400 ms) in the difference wave derived from frontocentral electrodes, relative to WR. This later N2-like negativity likely reflects early attentional engagement or the detection of auditory change (Puschmann et al., 2013), and its attenuation in PR suggests that increased perceptual and cognitive demands interfere with auditory processing, both in task relevant and irrelevant paradigms. The absence of a significant N2 difference between WNR and PNR further supports the idea that this activity may indirectly depend on task relevance and reportability for signal amplification rather than bottom-up salience alone.

Time-frequency analyses supported and extended these ERP results in the PR-WR comparison. In the PR, we found an increase in alpha-band power and a decrease in theta phase-locking. Alpha power increases emerged over left-lateralized centroparietal sites between 300–700 ms, overlapping both spatially and temporally with the P3b. This overlap suggests that alpha power may serve as an oscillatory analog of the P3b ERP, where elevated alpha power indicates inhibition of sensory processing under high task demands. Alpha and Beta suppression also reflect differences in motor output between report and non-report conditions, where reporting causes greater motor recruitment. Theta phase locking was reduced in PR between 200–400 ms at frontocentral electrodes, reflecting an impaired ability to track and encode auditory input. This theta activity overlapped in both timing and topography with the frontocentral N2 observed in our ERP difference wave, supporting the interpretation that theta phase-locking may be an oscillatory analog of the early frontoparietal N2. These findings indicate that high load disrupts both early and late processing across both ERP and oscillatory domains.

In the no report, task irrelevant condition, we found one difference in neural activity: a decrease in alpha/theta overlap activity (7.5-8.5 Hz) in the PNR (high load) condition. Although this activity is atypical in terms of frequency, timing and topography, we still hold it to be a significant result due to its manifestation over multiple electrodes. We interpret this component as high theta rather than low alpha. Interpreting it as alpha would not make sense in light of the previous literature and our current results that found increased alpha associated with increased deafness.

Taken together, our findings align with and extend prior work, in which inattentional deafness is marked by reductions in N2 and P3 ERP components, decreased theta power, and increased alpha activity, particularly under high load and task-relevant conditions.

Neural Correlates of Consciousness

A central aim of this experiment was to extend the findings of inattentional deafness to the ongoing search for the NCCs. The NCC literature has traditionally emphasized the N2 and the P3 ERPs, creating a schism of early and late theories. The P3b has traditionally been the front runner in terms of NCC candidates, especially within the framework of the GNWT. However, a growing body of evidence suggests the P3b may instead reflect post-perceptual or task-relevant processes rather than perceptual experience. In this view,

the P3b may correlate with consciousness but fails to meet the “minimality” criterion of an NCC, in which a neural event is both necessary and sufficient for the conscious experience itself, rather than an auxiliary process (Aru et al., 2012; Koch et al., 2016).

As the P3 has lost some popularity as the primary NCC, the N2 family, including the earlier Perceptual Awareness Negativity (PAN) and the later "novel" N2, has gained traction. These components have been linked to perceptual experience even in the absence of explicit report or task relevance, making them strong candidates for satisfying the minimality criterion of an NCC. Our experiment, which incorporates both task relevant and task irrelevant conditions under varying load, has yielded a rich data set that adds to the literature. To support our interpretation of these findings and aid reader accessibility, we will start by summarizing the full temporal sequence of neural dynamics observed across ERP and oscillatory domains.

Time Window	ERP Findings	Time-Frequency Findings	Report/Task Relevant	No Report/Task Irrelevant
50–300 ms	P1–N1–P2–N2 sequence, Frontocentral		✓	✓
100–300 ms	N1–N2 sequence, centroparietal	Theta phase locking (4–6 Hz), centroparietal, Frontocentral	✓	
250–400 ms	N2, Frontocentral, (Part of P1–N1–P2–N2 sequence)	Theta power (4–6 Hz), Frontocentral	✓	
400–500 ms		High theta power (7–8.5 Hz), Left-lateral Frontotemporal to Central		✓
300–700 ms	P3b, Left-lateralized centroparietal	Alpha suppression (9–14 Hz), Left-lateralized centroparietal	✓	

P3b and Task Relevance confirmed

Our study provides strong evidence for a connection between the P3b and task relevance in multiple ways. To start, a robust P3b component was observed only in the two report conditions (WR and PR). Furthermore, there was a 127ms delay in the onset of the P3b in

the PR condition compared to WR. This delay aligns with the 149ms delay found in the behavioral data of the same conditions, where they had to press a button to report. This temporal alignment between neural and behavioral responses supports the connection between the P3b and report.

Beta power suppression also provides support for the link between the P3b and task relevance. Beta suppression is commonly associated with motor preparation and task engagement (Engel & Fries, 2010; Khanna and Carmena, 2015, 2017; Molloy et al., 2015; Pfurtscheller & Lopes da Silva, 1999), and was found in both report conditions within the same time window as the P3b (300-600ms). Since participants were required to press a button for report, beta suppression likely reflects the motor readiness to execute this action. In this way, beta suppression bridges perception and action, and supports the link between the P3b and report.

The N2 family

The N2 appeared twice in our results, both times as part of a broader cascade of neural activity. Most prominently, it was observed within the classic N2–P3 complex in all report conditions, peaking at approximately 200ms over centroparietal electrodes. One of our initial hypotheses was that the N2 would remain significant even in non-reporting conditions. However, the centroparietal N2, alongside the P3b, was greatly reduced in the non-reporting conditions, suggesting a stronger connection between the N2 and task relevance than we initially expected.

Another N2 component was observed in all conditions, as part of a P1–N1–P2–N2 sequence. This sequence spanned over frontocentral electrodes and likely reflected late level sensory processing and early perceptual processing. We expected that this component would show greater magnitude in both conditions associated with heightened awareness (WR and WNR), but it was amplified only in the WR condition. These findings provide partial support for the hybrid N2 from Cohen et al. (2024). A key difference is that while they found the late N2 predominantly in no-report conditions, we observed it only in report conditions, with no reliable modulation by awareness in non-report trials.

While a frontocentral N2, as part of the P1–N1–P2–N2 sequence, was present in non-report conditions, it did not differ between high and low load (PNR vs. WNR), and thus cannot be conclusively interpreted as reflecting awareness in the absence of task relevance. Importantly, however, participants completed a pre-experimental sound check in which they were exposed to the same auditory deviants used during the task. In this low-load context, all participants clearly reported hearing and recognizing the sound changes, providing subjective confirmation that the stimuli were perceptually accessible.

This pretest serves as a baseline indication that participants were aware of the auditory stimuli when task demands were minimal, reinforcing the interpretation that any later differences, or lack thereof, reflect changes in attentional focus and task engagement, not failure to perceive the sounds themselves. Overall, these findings support a precautionary stance, in which even early ERP components may be modulated by task relevance and attention and thus should be interpreted critically in no-report paradigms (Bola & Doradzińska, 2021).

Alpha and Theta Activity

As outlined above, alpha suppression mirrored the P3b ERP in both topography and timing. Since alpha is inhibitory, alpha suppression is associated with heightened receptivity and attentional engagement. Theta, which we found across all conditions, has been linked to divided attention and cognitive control (Keller et al., 2017; Cooper et al., 2019). It was most prominent in the report conditions, where it displayed similar topography and timing to the N2 ERP. Theta phase-locking was increased in the WR condition compared to PR, which implies greater neural synchrony and alignment of neurons to a stimulus (Shahin, personal communication). Greater theta phase locking in this context shows greater alignment and tracking of the sound in the low-load condition, which is consistent with decreased deafness.

In the WNR–PNR comparison (awareness in non-task-relevant conditions), we identified only one neural correlate: an increase in high-theta power (7-8.5 Hz) between 400–500 ms over left frontotemporal to central electrodes. Although this frequency range falls on the border between classic theta and low alpha, prior studies have linked high-theta activity to working memory engagement, pre-sleep sensory awareness, and even anomalous detection states (Anderson & Horne, 2003, 2004; Booth & Persinger, 2009; Jensen & Tesche, 2002). We interpret this late, narrow-band theta activity as corresponding with a state of minimal awareness that appears in the absence of explicit task demands. It may be similar to a daydreaming state where you can perceive external stimuli without fully capturing or accessing them in your experience. This interpretation makes sense within this context, as some participants noted feeling sleepy in this condition since there was no task requirement.

Consciousness as a Process

Although both the P3b and N2 components emerged only in task-relevant, report conditions, we do not interpret this as grounds to fully discard them as potential NCCs. Instead, we suggest that consciousness is a temporally extended and multidimensional process. Within this process, different neural correlates align with different behavioral and phenomenological states of experience. Although the P3b does not meet the minimality criterion of an NCC, it may still reflect a conscious state corresponding with fully crystallized perception and explicit accessibility. This higher-level processing may be closely intertwined with adaptive task demands and attention yet still instantiate a form of consciousness.

High level accessibility, which we label reflective access, involves the conceptual manipulation of information. These processes are themselves experiential and likely rely on different neural substrates, including more frontal and global cortical areas. While early components like the N2 may be closer to the minimal neural correlates of sensory or perceptual consciousness, later components like the P3b may reflect higher-level states. Consciousness is not limited to sensory or perceptual experience alone, and there are various levels within perceptual processing that correspond with various phenomenological states. As highlighted by Chalmers (2023), consciousness has many different dimensions, ranging from sensory, to affect, to cognitive and agential. Beyond sensory-perceptual experience exists a myriad of potential states, which all may have their own unique set of neural correlates.

From an evolutionary and ecological perspective, consciousness evolves to serve adaptive functions, which implies it will manifest to the greatest degree when organisms are actively engaged in their environment. From this view, completely separating consciousness from related mechanisms such as attention and task-relevance may be impossible and misguided. Consciousness is not a single object or mechanism, but rather a holistic process that encompasses and coordinates multiple mechanisms. Evidence from individuals with hydranencephaly (Merker, 2007), cortical reorganization in sensory loss (Merabet & Pascual-Leone, 2010), and compensatory pathways in blindsight or cortical lesions (Aleci & Dutton, 2024; Derrien et al., 2022) support the idea that the brain may implement multiple routes to conscious experience. This high plasticity and adaptability of the brain suggest that there may be no single, fixed neural correlate of consciousness (NCC).

Consciousness is better understood as existing on a continuum in which different neural markers reflect different stages and degrees of processing. Treating consciousness as a binary mechanism, where it is either fully on or off, is not supported by empirical evidence and fails to capture the richness of phenomenology. For example, participants may be minimally aware of a stimulus even when they report no awareness in a binary task paradigm. This discrepancy between phenomenology and behavioral report highlights the importance of using graded measures of awareness (Cohen et al., 2023; Derrien et al., 2022; Mazzi et al., 2016; Overgaard, 2011). Furthermore, the scientific models and theories we create are pragmatic, metaphorical tools to facilitate our own understanding (Levin, 2024). They have functional utility, but do not fully capture the complexity of the phenomena they describe. This holds true for consciousness, in which a pragmatic and pluralistic approach is optimal. This implies a model that recognizes multiple levels of processing and accommodates the diverse stages, expressions, and neural signatures of consciousness.

Figure 19 illustrates this approach to consciousness in the context of our data, where neural signals at different times correlate with different features and degrees of experience. It is pluralistic by integrating key theoretical frameworks and existing taxonomies, and shows how they align with the empirical findings from our experiment. The figure illustrates how consciousness unfolds across time through distinct cognitive stages, neural processing mechanisms, and measurable neural signatures.

The top row depicts stages/types of consciousness: subliminal (sensory consciousness), preconscious, phenomenal consciousness (perceptual consciousness), and access consciousness (reflective consciousness). Each later stage adds a structure to the prior one: access occurs on top of (or in addition to) phenomenal consciousness, and phenomenal consciousness builds on preconsciousness. In the next chapter we will elaborate on the access and phenomenal consciousness, expanding on their graded nature (from that perspective the labels of this diagram are a bit misleading; for example we allow a small amount of access even in the middle part of the figure). This view involves a continuum rather than discrete categories, shaped by stimulus strength, attention, and task relevance.

The second row illustrates the neural mechanisms proposed by Recurrent Processing Theory (Lamme, 2010). This process occurs through transitioning from feedforward processing (represented by unidirectional arrows) to local and global recurrent processing (represented by bidirectional arrows), corresponding to the formation and access of perceptual content. Recurrent processing is a proposed neural mechanism that plays a key role in various theories of consciousness, but in different ways and regions.

Both bottom rows are data from our experiment, time-locked to the acoustic midpoint of the auditory stimulus. The third row displays oscillatory EEG activity, including theta phase-locking (4–5 Hz, 100–300 ms), theta power (4–5 Hz, 200–350 ms), high-theta activity (7.5–8.5 Hz, 400–500 ms), and alpha suppression (9–14 Hz, 300–600 ms). The fourth row shows event-related potentials (ERPs), including P1–N1–P2 (50–250 ms), a frontocentral hybrid N2 (250–400 ms), and a centroparietal N2 (150–300ms) and P3b (300–700 ms). ERP plots reflect a comparison between low load (higher awareness; red) and high load (lower awareness; blue) task conditions.

3.9 Conclusion

Overall, our results support consciousness being a temporally extended and graded process, where neural activity reflects increasing levels of awareness, access, and task engagement. The precise point at which one draws the line between conscious and non-conscious processing is not objective but varies across theoretical frameworks. Recurrent Processing Theory (RPT; Lamme, 2010, 2018) posits that consciousness emerges with local recurrent processing. Pitts et al. (2018) believe that consciousness arises slightly later, where minimal attention bridges local recurrence. Dehaene and colleagues within the GNWT camp posit that consciousness does not emerge until attentional and sensory regions fully connect through global recurrent processing. Our view is that these theories each capture a valid aspect of the overall trajectory of conscious processing in the brain, with each pointing to different phases of a continuous, overlapping sequence. Pragmatically, the most appropriate theory is completely situational and contextual. We shouldn't ask which theory is objectively right, but rather, which is the most informative and useful given the specific context. This further supports and emphasizes the need for a pluralistic and integrative perspective of consciousness – one that recognizes the nuances and complexities of consciousness as a whole.

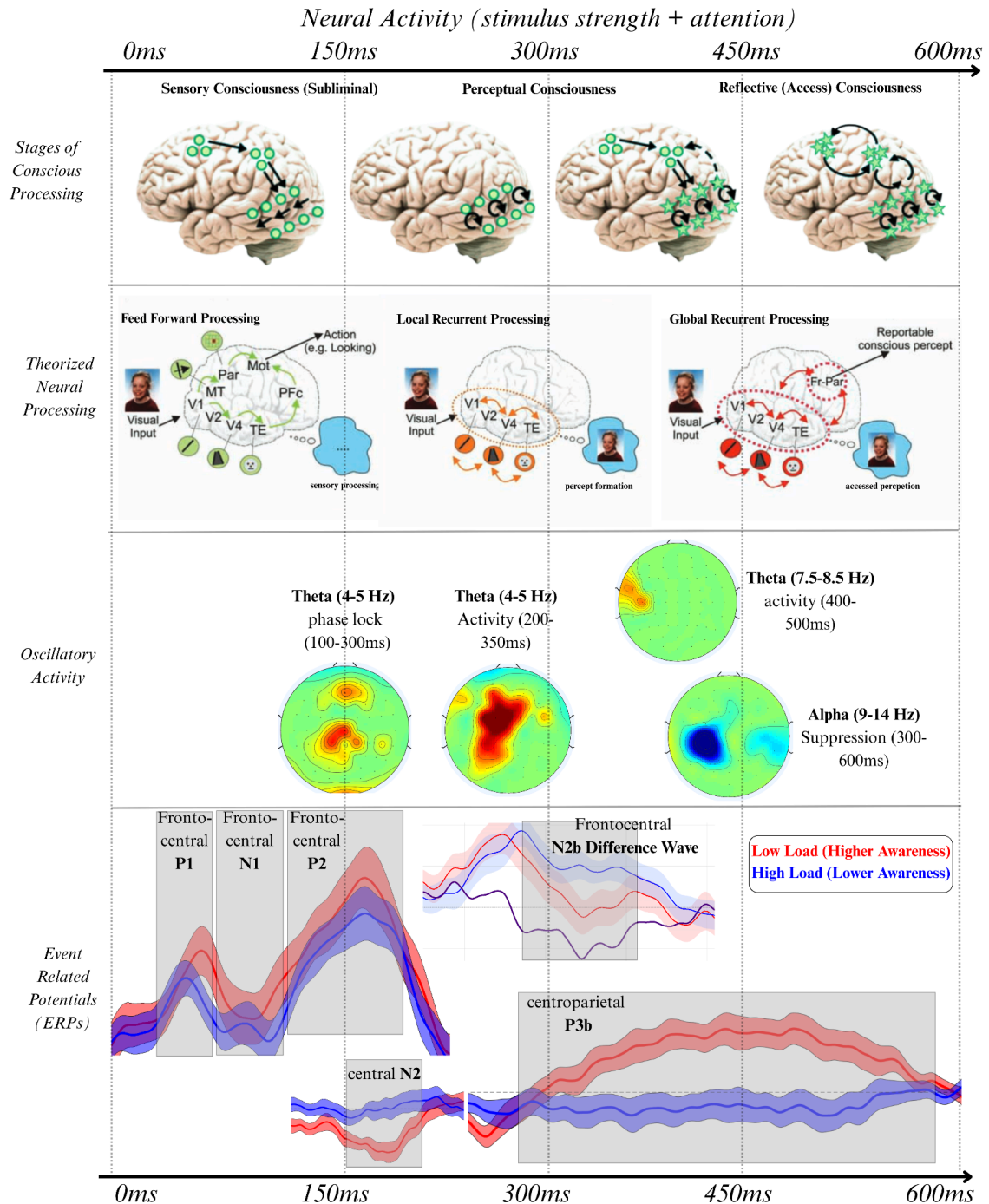


Figure 19: An overall illustration of the graded process of consciousness and how various theories discussed in this chapter relate to the EEG structures identified, which also illustrates a pluralist approach to NCC theories. Adapted from "How neuroscience will change our view on consciousness: Neuroscience and consciousness," by Victor A. F. Lamme, *Cognitive Neuroscience*, 1(3), 204–240, © 2010, reprinted by permission of Informa UK Limited, trading as Taylor & Francis Group, <https://www.tandfonline.com> and adapted from "The relationship between attention and consciousness: an expanded taxonomy and implications for 'no-report' paradigms," *Philosophical Transactions of the Royal Society B*, vol. 373, issue 1755, article 20170348 (2018), by Hillyard, Lutsyshyna, & Pitts. © The Royal Society. Licensed for reuse under Copyright Clearance Center Order License ID 1627274-1.

Chapter 4

Consciousness on a Continuum: Exploring the Relationship between Phenomenality and Cognitive Access

4.1 Introduction

In the previous chapter, our empirical findings suggested that consciousness is a graded, multidimensional process with various neurophysiological signatures scaling with perceptual clarity and task relevance. In this chapter, we aim to expand upon this view, particularly in the relationship and interaction between phenomenal and access consciousness.

In 1995, Block advanced consciousness research by introducing a foundational distinction: phenomenal consciousness (P-consciousness) and access consciousness (A-consciousness). He defined P-consciousness as subjective experience, rooted in a "what-it-is-likeness", with its primary (but not only) properties being sensations. A-consciousness was accurately distinguished from this, being defined as information made available to rational control, with its primary states being propositional attitudes. A-consciousness has evolved through the years to require global broadcasting of information (Block, 2008), and has more recently been relabeled as "cognitive access", involving a representation being made available for cognitive processing (Block, 2011). There are two primary ambiguities in relation to this distinction we wish to address and clarify:

- 1) What constitutes cognitive access? What exactly counts as rational action or cognitive processing?
- 2) What is the temporal status of accessibility? Does access occur in the moment of perception, or only after the perception occurs?

These ambiguities will be examined through two main phenomena: blindsight and phenomenal overflow. We will begin by detailing each ambiguity in greater depth, followed by an analysis of overflow and blindsight and how they relate to the conceptual challenges being discussed. Some ideas within this analysis of A and P consciousness may already be shared by Block, while others may be novel and distinct. Our goal in this chapter is to take steps to clarify the confusion regarding the function of consciousness while providing a theoretically and empirically grounded framework that may more accurately model the phenomenon.

4.2 The First Ambiguity

In 1995, Block defined A-consciousness as information made available for reasoning and the control of action. He later refined this view in 2008 by emphasizing the necessity of information being broadcast within the brain, a move that aligned A-consciousness with global workspace theory (GWT) and its neural counterpart, the global neuronal workspace theory (GNWT). Block (2011) later relabeled the term as "cognitive access" and defined it as a representation that is made available to cognitive processing. Taken together, cognitive access can be defined as information that is broadcast throughout a system enabling cognitive and rational behavior. Two central features of cognitive processing are attention and working memory, both responsible for holding and manipulating information over short durations. Working memory enables information to be sustained, updated, and used flexibly for reasoning, decision-making, and behavioral control, making it a core mechanism of cognitive access. This strengthens the definition, whereby accessibility manifests in various degrees and forms depending on a system's attentional and working memory capacities (Block, 2004).

While this definition appears relatively clear on the surface, a critical ambiguity remains: what precisely qualifies as "cognitive processing" or "rational behavior" in these definitions of A-consciousness? Is access limited to metacognitively reflected, reportable content, or can it include instinctive, non-reflective behaviors that are nevertheless rational and goal-directed? A substantial body of evidence supports adaptive, rational behavior across a variety of lower-level organisms in nature. For example, a fly is able to receive sensory information and process it to such a degree that allows it to respond quickly and intelligently to its environment. Both a fly evading a swat and a human reflexively swatting at a fly exhibit rational behavior that serves adaptive, survival-oriented goals, even despite lacking deliberative or higher-order cognition (Simon, 2001). Cognitive and adaptive behaviors are evidenced throughout nature, including a variety of organisms including animals (Matzel & Kolata, 2010; Zorina, 2004), plants (Calvo et al., 2017; Garzon & Keijzer, 2009; Trewavas, 2016, 2017), and even single celled organisms (Baluska & Levin, 2016a; Jacob et al., 2004; Levin, 2019, 2021, 2023; Ostovar & Boedicker, 2024). Even in the most primitive systems, there exists some minimal cognition where information is functionally retained and used for adaptive, rational action.

If A-consciousness is defined too narrowly and restricted to conceptual reasoning or linguistic thought, it excludes the vast majority of cognitive systems found in nature that display this rational behavior. Block (1995, 2011), however, does acknowledge this. He states that he does not want to completely rule out non-linguistic animals such as chimpanzees from having A-consciousness, and expresses that even bees display a minimal working memory capacity. Although not explicitly stated, this suggests an inclusive, graded interpretation which better captures the continuum of cognitive sophistication across biological systems and within human consciousness.

We propose that metacognitive, reflective consciousness, characterized by symbolic language, propositional reasoning, and introspective report, represents a highly developed form of a-consciousness, which we call reflective access. This reflective accessibility is

not only cognitively sophisticated but also experientially rich, with its own distinct what-it-is-likeness and likely distinct neural correlates. However, simpler forms of adaptive, goal-directed behavior should also be regarded as involving minimal degrees of accessibility. In this view, A-consciousness is continuous, extending along a spectrum of sensorimotor intelligence evident even in the simplest biological agents. Furthermore, this continuum is not only observed across species but also within individuals, as levels of conscious access can vary dynamically depending on situational context. Because A-consciousness is defined in terms of cognitive processing and rational action, it is inherently tied to task relevance; what counts as accessible and its degree depends on how and for what purpose the information will be used.

4.3 The Second Ambiguity

The second ambiguity concerns the overflow argument, where we analyze the temporal relation between cognitive access and perceptual and phenomenal consciousness. Phenomenal overflow is the claim that P-consciousness exceeds cognitive access – that we experience more than we can access, report, or utilize for reasoning (Block, 2007, 2011). The Sperling experiment (Sperling, 1960) is the first and main source of evidence used to support overflow. In paradigms like this, participants are asked to report their experience after a stimulus is quickly presented and removed, causing them to depend only on a post-perceptual memory representation. These designs raise an important question regarding cognitive access – does it occur concurrently with perception, or is it strictly limited to post-perceptual processing and memory representations?

Since working memory serves as the functional workspace of accessibility (Block, 2011; Overgaard, 2018; Mashour et al., 2020), memory and A-consciousness are deeply intertwined and perhaps impossible to completely disentangle. Memory is always susceptible to decay and distortion, even in its earliest phases of encoding (Kuuramo et al., 2022; Lentoor, 2023; Ricker et al., 2016). If cognitive access, as Block seems to suggest, refers solely to a memory-based representation, then phenomenal overflow simply restates the obvious, that memory is a degraded version of active perception. This makes the overflow argument relatively uncontroversial and unavailing, as it is widely accepted that memory is not a perfect replica, and does not accurately preserve the richness of perceptual experience.

In the framework moving forward, we will argue that cognitive access does not only involve post-perceptual representations but also includes the moment of active perception. We argue that the appearance of overflow may stem from a temporal mismatch. That is, comparing the richness of P-consciousness during perception to the degraded contents available for access at a later time results in a methodologically flawed interpretation. From this perspective, standard paradigms that remove the stimulus and assess access solely through degraded memory representations offer an incomplete empirical foundation. At the end of this chapter, we will introduce a more nuanced model of A and P consciousness, where phenomenality and accessibility are temporally aligned and arise in tandem.

4.4 The Illusion of Overflow

The prime example of overflow is associated with the Sperling partial-report paradigm (figure 20), in which participants briefly view an array of letters. In the full-report condition, they can recall only a limited subset of the array. However, in the partial-report condition, where a cue indicates which row to report *after* the stimulus disappears, participants can consistently recall any cued row, implying that iconic visual memory has a vastly larger capacity than working memory. Interpreted through the lens of consciousness research, more of the array was initially present in (sensory/phenomenal) consciousness than could be accessed for cognitive report. Block argues that this indicates a dissociation between phenomenal experience and access, in which all letters are phenomenally conscious but only a fraction are accessed. What we feel and subjectively experience while perceiving the array of letters exceeds what we are able to report or utilize in reasoning for the memory representation (access).

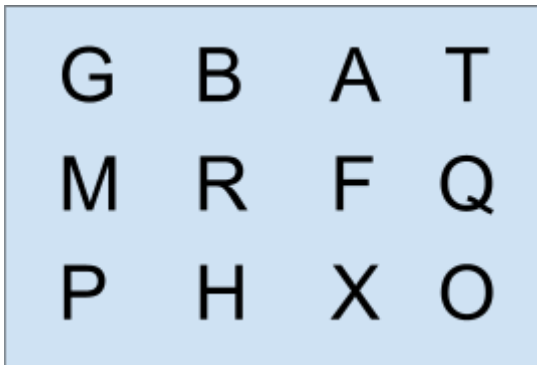


Figure 20. Illustration of the classic Sperling (1960) partial-report paradigm. A 4×3 letter array was presented for 50 ms and then removed. Participants were either asked to report all letters (full report) or a single cued row indicated by a tone (partial report). Superior performance in the partial-report condition suggested that more information was briefly available in sensory memory than could be accessed through full report.

Herein lies the temporal ambiguity: is cognitive access limited to the moment after the array of letters is removed, or does it also occur while the letters are present and being actively perceived? The phenomenal overflow argument maintains that active perception is richer than the later post-perceptual memory representation, in which the stimulus is no longer present. However, as shown in figure 21, it is both methodologically and conceptually flawed to compare P-consciousness at t_1 (during active perception) with cognitive access at t_2 (after the stimulus has disappeared). Such comparisons risk conflating distinct stages of processing and overlooking the natural decay that occurs as perceptual content transitions into memory. When both phenomenality and accessibility are examined at the same timepoint, during active perception, it becomes clear that they are proportional and aligned (there is a gradient from high to low A-consciousness and another gradient from high to low resolution P-consciousness) and no overflow is apparent. The core question remains: is cognitive access only the post-perceptual working memory representation of a stimulus, or is it also operative during the act of perception itself?

From our perspective, cognitive access occurs both during the moment of active perception and in the subsequent memory representation. Working memory is the primary workspace of cognitive access, but access also involves other transient systems such as iconic and sensory memory. While the stimulus is still present, working memory representations are dynamically updated in real time to preserve perceptual fidelity. Once the stimulus disappears, however, this updating ceases, and the memory trace that remains becomes highly susceptible to distortion. The phenomenality associated with this memory representation is likewise diminished relative to the original perception. Thus, between t_1 and t_2 , P and A consciousness degrade in tandem. Just as access is not only in the post perceptual memory representation but also in the moment of perception, P-consciousness is also not only in sensation and perception, but also exists in the experience of a memory representation. The appearance of overflow may instead result from a temporal mismatch in how these forms of consciousness are evaluated. When phenomenal and access consciousness are considered concurrently, there is no compelling evidence for overflow.

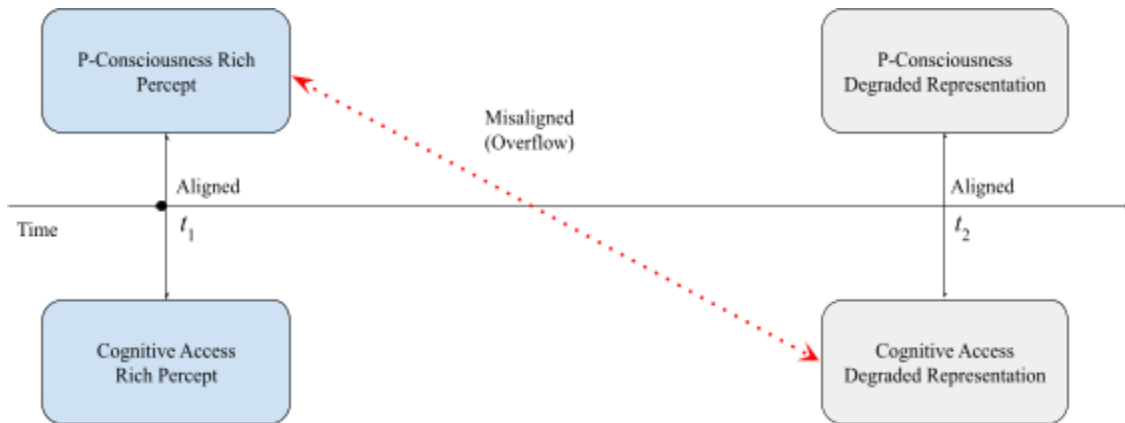


Figure 21. Illustration of the temporal alignment between phenomenal consciousness and access consciousness. Within each percept there is a gradient from focus to periphery. At t_1 , both p-consciousness and cognitive access are rich, and at t_2 , both are degraded. The overflow argument incorrectly compares phenomenal experience at t_1 with degraded access at t_2 (see the red dotted line), producing a misinterpretation.

Sperling Experiment as Support for a Continuum

The Sperling task can further support the view that phenomenal experience and cognitive access exist on a continuum. When used as evidence for overflow, it is interpreted and assumed that all letters are either fully phenomenally conscious or not. However, it is more plausible that all letters were present in experience, but with varying degrees of phenomenality, clarity, and accessibility. Participants may not have had access to the exact identity of each letter, yet they likely accessed lower-level perceptual features. These might include things like whether or not it was letter or number shaped and that it

occupied a particular spatial location. This level of access would be sufficient for an easier version of the task, such as asking participants to distinguish between seeing letters or shapes. On the other hand, if task difficulty were increased then accuracy may decline significantly. For example, we could ask participants to not only identify the letters, but to also decide the size and angle of each one. Both of these scenarios support the idea that cognitive access is shaped by task difficulty and situational context. Furthermore, participants frequently describe the *impression* of seeing all the letters, followed by the inability to recall them in full detail. This impression does not imply that they saw everything in perfect detail, but rather that they had a fully encompassing view of the letter array. Similar to observing the scene of a movie or an image in a painting, you may feel like you see the entire picture, but the details depend on exactly where you are focusing.

Rather than interpreting this as evidence of phenomenality exceeding access, we propose that it reflects a gradient of accessibility. Some letters may have reached fully structured, reportable experience, while others remained less differentiated, where they were still experienced and accessed to some extent, but not sufficiently encoded into working memory to support detailed report. Recent replications further support this view. Bradley-Garcia et al. (2023) carried out an extensive replication of Sperling's paradigm and found lower recall levels than previously reported. When participants were prompted to view three rows of letters and cue a specific row within that, they recalled only 1–2 letters. This number is significantly lower than the 3–4 in the original study. Furthermore, when the display was reduced to two rows, recall improved significantly (1.5–2.5 letters). This suggests that the phenomenal field of awareness varies with task demands and is modulated and structured by a focus and periphery. With a smaller scene and fewer competing elements, attention could be distributed more effectively. This results in enhanced cognitive access that is in proportional alignment with greater phenomenal clarity.

Example

A practical and intuitive example illustrates both of these ambiguities. While writing this paper, I noticed a small pineapple plant beside my computer (Figure 22). Depending on where I directed my attention, different aspects of the plant had greater clarity, yet the overall percept felt coherent and accessible. If, while actively perceiving the plant, I was asked, "Are there more than 10 leaves on the plant?", I could confidently respond yes. The percept was both phenomenally rich and fully accessible. However, if the plant were removed before the question was asked, then I would be forced to rely on my decayed and outdated working memory representation. This would cause me to be far less confident and accurate in my answer. This failure, however, would not imply that the original percept had high phenomenality and low access. Rather, it reflects the inherent limitations of post-perceptual memory representations, especially when fine-grained detail is required.

Conversely, if I were asked a simpler question, such as “Did the plant have more than three leaves?”, I could still answer with confidence, even when the image is removed. This illustrates how task difficulty modulates the reliability of post-perceptual access, rather than revealing a strict dissociation between P and A consciousness. This suggests a graded field of consciousness – as the scope of attended content broadens, the clarity of each individual percept tends to diminish. Put simply, the more narrowly attention is focused, the more detailed and accessible that specific content is, which is an effect observed across all sensory modalities.



Figure 22: Pineapple plant. While briefly viewing the pineapple plant, both phenomenality and accessibility are relatively high. Although the level of detail may vary depending on where one directs their focus, shifting the field of phenomenality and accessibility, the entire plant remains within the visual field and is perceived as a unified whole. If the plant were suddenly removed and a cognitively demanding question were asked, the accuracy of my response would depend on the complexity of the task. This example illustrates the gradual nature of phenomenality and cognitive access, where certain aspects of perception remain more accessible than others depending on attentional distribution and post-perceptual cognitive demands.

Blindsight and Blindsight

Blindsight further supports our position regarding the first ambiguity, specifically, that both P and A consciousness exist on a spectrum shaped by task demands. Blindsight is generally defined as the ability to detect or discriminate visual information in the absence of visual perception, typically due to damage to the primary visual cortex (V1) (Overgaard, 2011; Derrien et al., 2022). Block (1995) has invoked blindsight as evidence for cognitive access without phenomenal consciousness, in which patients may perform above chance in forced-choice tasks while denying any conscious experience of the stimuli. Particularly, he uses its imagined, extreme form, *superblindsight*, where a subject is able to consistently and rationally use his blindsight knowledge without prompting.

This interpretation can be challenged both conceptually and methodologically. First off, traditional dichotomous measures of awareness are too absolute to capture the subtleties of experience. When utilizing these more nuanced measurements such as the PAS or SAS, patients report some minimal phenomenality. This has led some researchers to reinterpret blindsight as degraded or abnormal vision rather than a complete absence of awareness (Overgaard, 2011; Mazzi et al., 2016; Derrien et al., 2022). This can be interpreted as "blindsense" – a minimal form of subjective experience that involves a vague sense or feeling that something is occurring. In relation to our previous experiment, blindsense could correspond with the overlapping high theta, low alpha activity found in task irrelevant, non reporting conditions. Participants may have experienced a felt recognition of the stimuli which never fully crystallized into a clear percept (Aleci & Dutto, 2024; Overgaard, 2011; Mazzi et al., 2016). In this view, patients have some minimal degree of phenomenality which aligns with the degree of accessibility. The phenomenal experience is faint and degraded, and the degree of accessibility is proportionally low and insufficient for complex reasoning. Simple cognitive actions such as stimulus discrimination, however, may still be possible.

Lamme (2018) presents a similar perspective in comparing the visual processing systems of a fly and human. He proposes that abnormal or low levels of consciousness in humans (sleep, blindsight, or inattentional blindness), may be equivalent to the fully conscious states of simpler organisms. As mentioned above, a fly dodging a hand attempting to swat it away illustrates adaptive, cognitive, and rational behavior, even in the absence of reflection or report. Rather than demonstrating a clean dissociation between access and phenomenality, blindsight may instead illustrate their shared continuum, revealing how both can be present in attenuated and atypical forms.

4.5 A Continuum of Consciousness

As we have been insinuating throughout this chapter, phenomenality and access co-vary along a continuum, with weaker percepts supporting correspondingly weaker access. Phenomenality refers to how something appears (see section 1.1), shaped by sensory and perceptual processing. Cognitive access concerns how such experiential information is utilized, shaped by working memory capacity and task relevance. This accessibility is proportional to the phenomenality of experience. In this sense, they can be seen as two sides of the same coin, each constituting an essential aspect of the overall process of our consciousness. As sensory information progresses through stages of processing, it becomes more perceptually structured, high-resolution, and phenomenally rich – qualities that align with increased cognitive accessibility and a greater capacity to use that information in complex, goal-directed tasks. In contrast, minimally processed information and coarse-grained phenomenality is less functional and accessible, supporting only simple behaviors.

As illustrated in the pineapple example, what is cognitively accessible shifts depending on both where one is attending and the task demands within the situation. Something on the periphery of visual experience will be less accessible in a cognitive sense, as the

information will be less beneficial for potential task demands. For example, while you are reading this, there may be words or content on the side of your screen that are in your visual periphery. You do not know the exact details of the words unless you shift the focus of your attention to them. It may "feel" like you see the whole screen, similar to how participants felt in the Sperling experiment (1960), and this feeling is not exactly wrong. You do see the whole screen, but with varying degrees of clarity depending on the focus and periphery of your attention. Similar to issues with blindsight and dichotomous, binary measures, participants may have answered differently if they were asked in a more nuanced way about their subjective perception, such as, "Did you see the entire array of letters and each letter with full clarity, or were some letters more or less clear?"

Stimuli in your periphery correspond with minimal access, and this access may have some cognitive utility if the task demands are simple enough. Attentional focus structures the experiential field, and items in the periphery appear blurry, less detailed, and lower resolution. Shifting attention to these peripheral elements will increase their phenomenal clarity while simultaneously increasing accessibility. As Block (1995) notes, "Bringing something from the periphery to the center can affect one's phenomenal state. The attention makes the experience more fine-grained, more intense..." (p. 241). P-consciousness and cognitive access can reinforce and amplify one another via attentional modulation. Access to aspects of the experiential field (cognitive access) enhance and intensify their phenomenal qualities (P-consciousness), thereby shifting the overall phenomenal state. This shift, in turn, influences attentional modulation, further determining what becomes accessible. This interplay reveals the graded nature of consciousness and its relationship to attention. As attended content moves deeper into the accessible workspace, it becomes increasingly fine-grained, phenomenally rich, and in turn, more functionally accessible for complex rational action.

Working Memory as the Workspace

Working memory serves as the primary workspace for cognitive access, but not all cognitively accessible contents are necessarily fully encoded working memory representations. Moreover, the quality of experience associated with different degrees of access, whether brief perceptual glimpses or abstract symbolic thought, may vary phenomenologically and neurophysiologically. Just as perceptual consciousness has identifiable NCCs, reflective consciousness may involve frontal and parietal regions engaged in symbolic representation, conceptual manipulation, and executive control. Perceptual contents that are attended to more closely and/or sustained for longer durations are more likely to be accurately encoded into working memory, making them more readily available for voluntary rational action. However, even less encoded information can still contribute to rational behavior, albeit in a more limited or less precise manner. For example, when briefly glancing at a street sign while walking, a person may not encode the exact wording into working memory but it may have been enough for a general sense of direction. Working memory is defined as an active representation or temporary buffer of information that can be used and manipulated for cognitive operations (Luck & Vecera, 2002; Luck, personal communication). This aligns with A-consciousness being characterized by the availability of information for rational thought and action (Block, 2008). Attention acts as a mechanism to modulate contents in

working memory, which serves as the global workspace of accessibility (Bae & Luck, 2017; Block, 2008; Mashour et al., 2020; Overgaard, 2018).

The Graded Model of Access and Phenomenality

Figure 23 illustrates the relationship of P and A consciousness. The transition from light to dark represents the gradient nature of cognitive access and phenomenality. The entire circle corresponds to the field of consciousness, where darker regions represent the periphery. Peripheral experiences are less accessible but still vaguely present, such as fleeting impressions, background emotions, or implicit sensations (the feeling of one's posture or the hum of distant noise). The center of this gradient represents the focus of attention and consciousness. This bright central region involves aspects of experience that are most accessible, clear, well-formed, and cognitively processed, such as explicit thoughts, perceptions, and attended sensory input. This region can be viewed as the workspace of consciousness where contents are most cognitively accessible. The overall field is modulated by both top-down processing and bottom-up stimulus strength (Dehaene, 2006), where strong stimuli can drive accessibility even with minimal attention, while weak stimuli may require focused attention to reach clarity. Subliminal stimuli, for example, may receive high attention but remain on the far outskirts of accessibility due to low stimulus strength.

Our findings in chapter 3, along with Recent findings from EEG studies on inattentional phenomena, support this graded view. P3b amplitude has been found to scale with the level of task engagement, indicating varying degrees of conscious access depending on the overall context (Dehais et al., 2019; Molloy et al., 2015). Oscillatory activity also supports this view, as increased alpha power was associated with inattentional deafness in high workload conditions (Callan et al., 2023). These results align with findings from our own experiment, where different levels of conscious access were associated with graded neural signatures. Greater awareness (in easier, lower-load conditions) corresponded with stronger P3b amplitudes, enhanced N2 components, reduced alpha power, and increased theta activity. Taken together, this body of work supports a graded and pluralistic model of consciousness, in which both phenomenality and access scale with perceptual clarity, attentional load, and cognitive demands. Each mode or stage of consciousness may involve its own experiential quality and neural signature, together forming a unified but multidimensional field of conscious experience and processing.

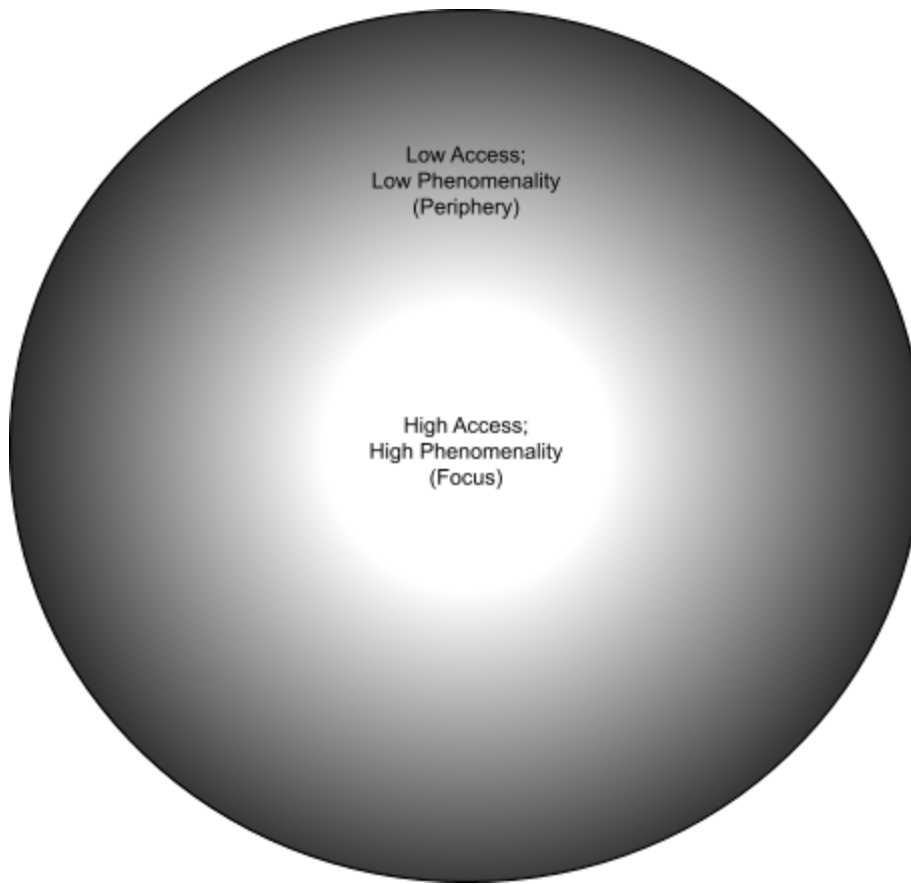


Figure 23: Graded model of consciousness. The inner region represents high cognitive access and vivid phenomenality, while outer regions reflect faint, degraded, or peripheral experience. This model illustrates the co-scaling of phenomenal and access consciousness as shaped by attention, stimulus strength, and task demands.

4.6 Language and Reflection

At the highest levels of cognitive access lies the capacity not only to utilize information for rational action, but also to engage in reflection – an abstract form of meta-awareness. Reflection entails an entity's ability to conceptualize its own experience, possessing not only awareness of itself and the external world, but also of its internal models, experiential content, and subjective character. This level of access aligns with Block's (1995) original conception of access consciousness as involving propositional attitudes – beliefs, judgments, intentions, and other mental states. This capacity seems to be deeply rooted in symbolic language, which enables experiences to be symbolically encoded, retained, and manipulated in working memory, thereby supporting advanced reasoning and introspection.

Experience and qualia can be thought of as a richly structured domain that cannot be fully accessed by the experiencing subject beyond the immediacy of the present moment. Through language and conceptual abstraction, this multidimensional experiential field is reduced and reconstructed to preserve its essential features – a distillation process. A helpful analogy is found in cartography: just as the curved surface of the Earth must be projected onto a two-dimensional map, the original phenomenal experience is too rich to represent directly, and must be ‘flattened’ into a symbolic form. As Neuman and Nave (2010) describe, the cartographer does not perform a one-to-one transformation, but instead slices the surface into manageable segments that can be reformed into a meaningful representation. This re-created experience is inevitably distorted but still preserves the most essential features.

Reflective consciousness and symbolic language are deeply intertwined and operate together within the workspace of cognitive access. As language evolves from gesture, to symbols, to complex grammatical systems (Neuman & Nave, 2010), the workspace becomes increasingly expressive and capable of mirroring internal experience. In this sense, the workspace is both a lens through which experience is accessed, and a mirror by which it is reflected. This recursive processing represents the most sophisticated form of cognitive access, allowing an entity not only to access its experiences but reflect and reconfigure its own states.

We can appreciate the depth of this capacity by reflecting on the fact that words are, at their core, nothing more than vibrations in the air. These sound waves are received and transduced by the sensory system into bioelectrical signals for further processing. Depending on the context, a single word can elicit a variety of states (joy, fear, sorrow, etc.) which can alter both our physiology and consciousness. This transformation, from seemingly meaningless vibration to deeply meaningful experience, illustrates the power of language to influence and reshape the field of consciousness. This process of recursive access and representational abstraction is illustrated in figure 24.

Sensory content, such as the sight of a sunset, activates the visual cortex, where raw sensory input is processed. Through long-range recurrent connections with the prefrontal cortex – associated with the global workspace (Mashour, 2020) – this content becomes accessible, reportable, and conceptually structured. Language plays a critical role in conceptualizing experience, distilling the high-dimensional qualia into a form that can be manipulated, stored, and reflected upon. The feedback loop between perception and conceptualization enables reflection to occur, where experience itself becomes its own object.

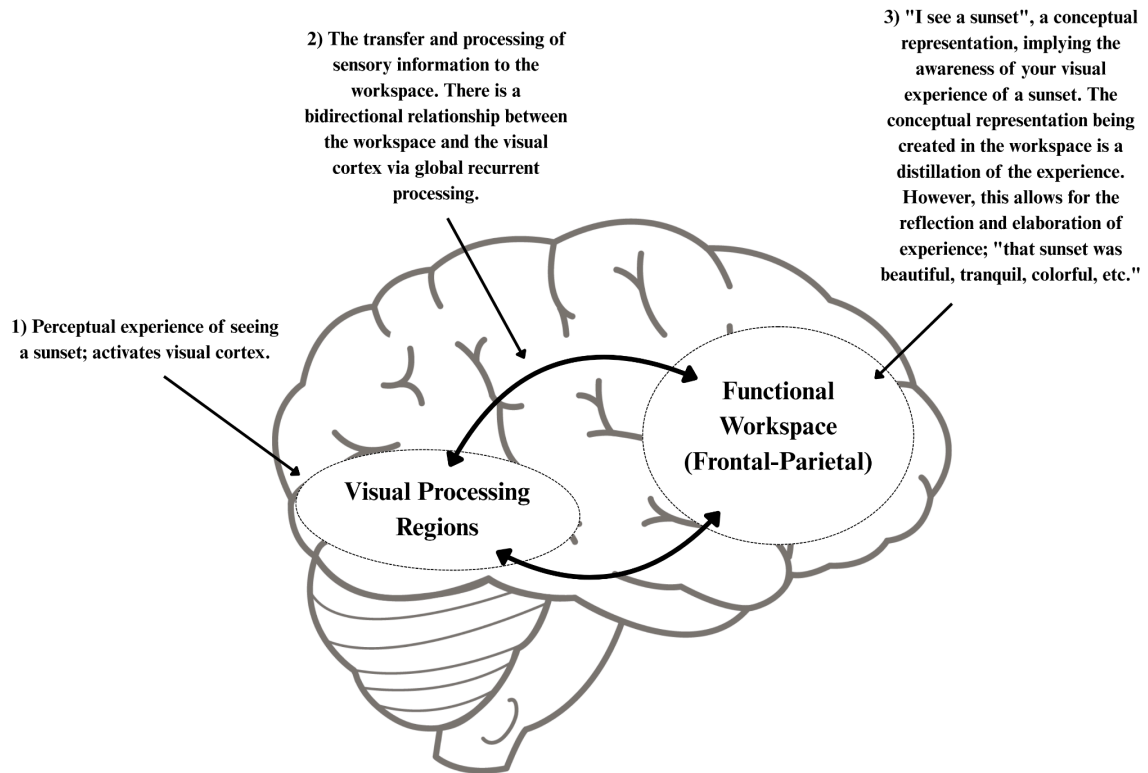


Figure 24: Model of reflective consciousness within the brain. Sensory input activates perceptual areas and is transmitted to frontal-parietal regions for symbolic representation and conscious access. Feedback between frontal-parietal and visual areas enables conceptual encoding of experience and reflective awareness.

4.7 Conclusion

In the previous 3 chapters, we have contributed to the cognitive science literature in multiple ways. Behaviorally, we induced deafness in a continuous, cross-modal paradigm and provided evidence supporting load theory. High perceptual load consistently induced inattentional deafness, whereas cognitive load exhibited varied results. Physiologically, we clarified the neural correlates of inattentional deafness and, through a rich dataset, added evidence to the overall NCC investigation. Multiple neural markers associated were identified, including both early and late components. ERPs included a frontocentral and centroparietal N2 components, a centroparietal P300, and a frontocentral P1–N1–P2–N2 sequence. Oscillatory activity included modulations to alpha, theta, and beta bands. Importantly, the P3 component emerged only in task-relevant, report conditions. The frontocentral N2 was present across all conditions but modulated only by task relevance and report. Theoretically, we developed a framework of consciousness grounded in both our own empirical data and that of previous literature. In this, we emphasize the importance and value of utilizing a pluralistic and pragmatic approach. Consciousness, in our framework, is a temporally extended process that manifests phenomenally as a holistic, multidimensional field. Philosophically, we addressed key ambiguities regarding A and P consciousness, and proposed a graded model that better fits both the neural data and phenomenal experience.

Chapter 5

A Space for Classifying Conscious Systems

5.1 Introduction

In the previous chapter, I introduced consciousness as a graded and dynamic process that can manifest in varying degrees and forms within an individual. In this chapter, I extend this perspective to a broader range of entities by creating a 2D classification space for potentially conscious systems. I begin by first identifying awareness as the essential core of consciousness, followed by an examination of how it relates to subjective experience and information processing. Progressing from simple systems to complex organisms, I illustrate how awareness, subjectivity, and information processing interact to give rise to conscious experience. Many interrelated yet distinct terms, such as cognition, intelligence, and sentience, will fit appropriately within this framework.

In this framework, consciousness is conceived as a triadic structure of awareness, subjectivity, and information processing. This view aligns with perspectives from various philosophical and scientific traditions, including those of R. Balasubramanian (2011), Nisargadatta Maharaj (1981), LaBerge (2016, 0:50), and Schrödinger (1944), all of whom have proposed that consciousness arises through the interplay of a fundamental awareness and its interaction with a physical substrate. Awareness is the unconditional and fundamental essence of any entity, whether it be living or nonliving, organic or inorganic. Subjectivity and information processing then shape how that awareness is embodied and expressed. Subjectivity refers to a system's mode of being – its internal organization and capacity to maintain itself – while experiential capacity arises through the processing of information in relation to its environment. Subjectivity increases through the formation of a boundary that differentiates the system from its environment, allowing it to function as an individual whole. Experiential capacity increases through processing sophistication, enabling more meaningful and rich experiences. In our 2D classification space, subjectivity and information processing are each a dimension that condition and shape awareness into distinct conscious systems (Figure 25). Note that this is a pragmatic classification space, where each point is a suggestive approximation rather than a measured result. The goal is to conceptually explain how these two dimensions are related. Future work could formalize these meanings further and allow for actual plots of the same type.

The unique what-it-is-likeness described by Nagel (1974) corresponds to the particular way a bounded agent (subjectivity) processes information and interacts with the world (experience). That is, for any conscious entity, there is a specific way it is like to be that entity. For example, two individuals viewing the moon from different locations will have their own unique experience. Even if observed from the same location their experiences may still diverge significantly, as their experience is shaped by their unique perceptual and cognitive architecture. Although the objective stimulus, the "moon", is shared, the subjective experiences remain uniquely structured due to how the system processes sensory information to produce meaning.

Consciousness, subjectivity, and information processing all exist on a continuum. All systems, whether biological or artificial, exhibit at least minimal degrees of subjectivity (as boundedness) and information processing (as structured interaction with inputs). In natural systems, these two dimensions tend to co-evolve and tend to be correlated. However, in artificial systems, this coupling may break down. A system could exhibit highly advanced information processing capabilities without a corresponding development of subjectivity – raising important questions about the nature and limits of machine consciousness and AI. The following sections of this chapter will explain and construct the framework by first establishing awareness as fundamental, followed by delineating the dimensions and their stages, along with the respective regions of consciousness.

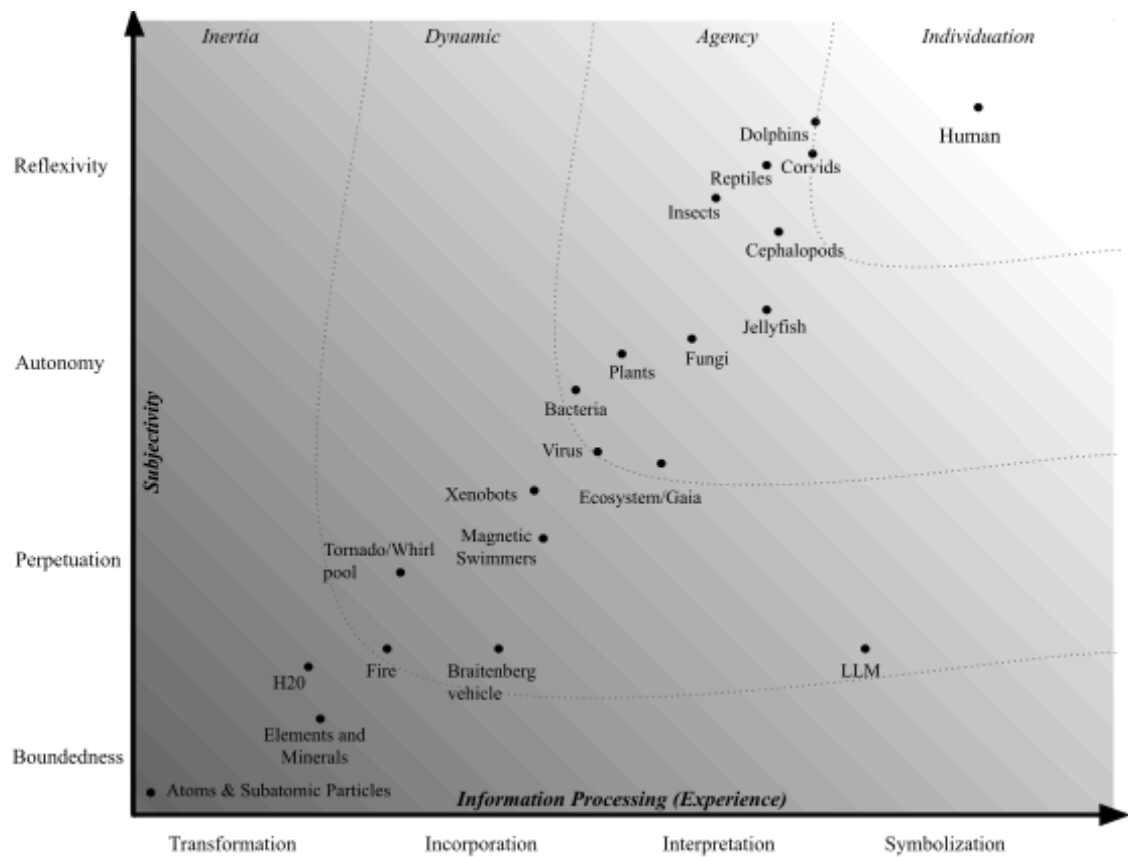


Figure 25: The two-dimensional space for classifying conscious systems. The two axes represent the dimensions of subjectivity and experience (information processing), each divided into four conceptual levels as discussed above. The space is further divided into four labeled regions, corresponding to broad degrees of consciousness. Dots indicate the approximate location of various types of systems within this space. This is a map of creature consciousness, in the sense that it is referring to the system as a whole, rather than specific states within the system (see section 1.4).

5.2 The Core of Consciousness

We begin our investigation by attempting to discover the core of consciousness, its essential nature that truly defines it. We will start by investigating thoughts, feelings, and other contents of experience. This analysis involves the method of negation, where the core of consciousness is revealed by rejecting everything it is not.

For much of western scientific history, the study of consciousness was essentially synonymous with the study of mind and thought. To investigate consciousness was to investigate mental processes such as reflection and introspection. However, in focusing on higher cognition and mentation, we were merely examining a particular aspect of consciousness. This is made evident when you consider that experiencing a thought implies that the thought is occurring with the field of consciousness. The very fact that

we can be conscious of a thought suggests a distinction between the two (Nisargadatta, 1981). Thinking, therefore, is only a mode or dimension within consciousness, with thoughts appearing as contents in a larger field.

Animals that lack sophisticated conceptual thought are widely accepted as conscious, showing that his distinction is now relatively mainstream. This has been articulated by Neuroscientist Philip Goff (2017, p. 106), where he defines thoughts as concept-involving representations of reality and consciousness as the property of having some kind of experience. In line with this is the common definition of consciousness as subjective experience, emphasizing sentience and the what-it-is-likeness of a state or organism (Chalmers, 2023, Nagel, 1974). Sentience, as defined by Birch (*The Edge of Sentience*, pp. 26), is the capacity for valenced experience, or experiences that feel bad or feel good. But this understanding, too, can be further scrutinized. Does the what-it-is-likeness of experience necessarily involve valence? If so, is sentience and feeling essential to consciousness or another dimension within it?

Despite significant progress, it seems researchers may be repeating the same categorical error of conflating the contents of experience with experience itself. Just as thought was mistaken for consciousness, we may now be making that same mistake for other experiential contents such as affect or perception. Valenced feelings emerge from a system's interaction with its environment. As described by Birch et al. (2020), Ginsburg and Jablonka (2007a, 2007b), Godfrey-Smith (2019, 2021), Denton et al. (2009), and Damasio and Carvalho (2013), feelings are transitory internal states that serve motivational functions, arising in tandem with instrumental learning and goal-directed behavior. Feelings may function as reliable indicators or "positive markers" used to infer consciousness (Birch et al., 2020), but they are not synonymous with it.

While more fundamental than thought, feelings are likewise appearances within the field, not the field itself. They are evolutionary developments and non-conceptual representations that appear within consciousness. This understanding parallels Thomas Huxley's categorization of feelings, sensations, emotions, volitions, and thoughts under the common head of 'states of consciousness'. But, Huxley continues, "what consciousness is, we know not" (Huxley, 1866, p. 193).

Rather than continuing to define consciousness in increasingly complex positive terms, a more efficient strategy may be the apophatic method – the method of negation. By stripping away what consciousness is not, we clarify what it is by seeing what remains. First, we can confidently strip away thought, the concept involving representations. Next, we encounter feelings and sentience, which also can be distinguished from the field of consciousness itself. The experience of a feeling implies that feeling is likewise a content, not the container. If we strip away all content from consciousness, feelings, thoughts, bodily sensations, perception, even memory, what remains?

Awareness, which I functionally define as the capacity to detect and register the presence of internal or external phenomena, remains as the essential core of consciousness. This

minimal capacity to detect any internal or external information or stimulus constitutes the necessary condition for any potential interaction or experience to occur. Sensations, percepts, feelings, emotions, and thoughts are forms of experiential content, all of which are objects in the field of awareness that can be detected. These contents emerge within and modulate the field of awareness but are not identical to it. Rather, they are determined by the organization, complexity, and perceptual mechanisms of a system, as I will describe within this chapter. Awareness serves as both the underlying field of experience and the spacious background within our 2D classification space.

Described by various nondual Eastern traditions (Balasubramanian, 2011; Longchenpa, 2006; Nasafi, as cited in Schrödinger, 1958; Nisargadatta, 1981; Thompson, 2014; Abhinavagupta, trans. Singh, 1989), awareness is an intrinsic feature of existence and reality that is self-luminous and revelatory by nature. It is the underlying presence within any perspective, however minimal, and acts as the medium through which all manifestation and knowing takes place. It detects and reveals all that arises, however, it, itself, cannot be revealed (like an eye attempting to see itself). In this sense, it is both transcendent and immanent: transcendent in its inability to be revealed, and immanent as the very essence of every experience. Awareness is not the contents of experience, but both the space and medium through which content appears. Although inherently spacious, it is not empty in a deficient sense; it is full of presence and serves as the source of all experience.

This view further aligns with the etymology of phenomenal which is from the Greek *phainō*, meaning “to shine,” “reveal,” or “make appear,” and extends to consciousness, derived from Latin *con-* (“with”) and *scire* (“to know”), emphasizing its role as the medium of all potential knowledge (from perceptual to conceptual). It is awareness that is the source of consciousness – a foundational presence that underlies both organic and inorganic systems, though expressed in profoundly different ways. This foundational presence of awareness flows through the myriad of existent entities, giving rise to diverse forms of subjectivity. In sentient beings, this capacity manifests as the sense of individuality or “I am-ness”, the first-person point of view as it is most fully realized in human experience. Feeling, perception, and thought are various modes and dimensions of consciousness; contents within the field of awareness which constitute a unique point of view, but do not define its essence.

5.3 Dimension 1: Subjectivity and Agency

Within our framework, subjectivity constitutes a dimension of consciousness, existing on a continuum from minimal to advanced. Subjectivity refers to the mode of being for an entity, which is rooted in its internal structure and organization. At its core, subjectivity implies a distinction between subject and object which enables interaction and relation. In analytic philosophy, this is typically described as the distinction between the experiencing subject and the objects it encounters. At more advanced levels, this interaction becomes intentional in the philosophical sense, where the system is internally directed toward, or represents, an external state of affairs. This is often referred to as intentionality or

aboutness, and corresponds to what is known as transitive consciousness, in which “transitive” is drawn from grammar (e.g., “sees” in “sees an apple,” which takes a direct object). To capture the graded nature of subjectivity, I further distinguish between passive and active transitivity in later sections.

A similar perspective is found in Eastern philosophy. Consciousness can be understood as the conditioned expression of awareness, where it only emerges in the presence of an ‘other.’ 8th century Buddhist monk Shantideva wrote, “In the absence of a thing to know, it is clear consciousness will not arise” (Way of the Bodhisattva, p. 145). Likewise, the Vedic philosopher Siddharameshwar Maharaj states, “The rule of experience is that it is not possible without duality. Experiencing means duality...For experience to take place, some other object is necessary” (Master of Self-Realization, pp. 164, 272). This is further supported by Nisargadatta Maharaj stating “Awareness is absolute, consciousness is relative to its content; consciousness is always of something. Consciousness is partial and changeful, awareness is total, changeless...the common matrix of every experience” (I Am That, pp. 29). This duality manifests through the formation of a subject-object boundary, which initiates the emergence of an individual entity capable of interacting with its environment. Philosopher R. Balasubramanian (2011) articulates a similar distinction between fundamental and functional consciousness; fundamental consciousness (awareness in our framework) is intransitive, unconditional, and non-relational, and functional consciousness is conditioned and transitive. The latter aligns with Husserl’s (1952/1989) assertion that consciousness is always “consciousness of something”.

Subjectivity refers to the aspect of consciousness which enables transitivity and the capacity to interact and relate to the world around it. The formation of a subject-object boundary or functional separation enables a system to interact with its environment from a specific location in space-time, in which a location in space-time is the minimal defining feature of a point of view (Godfrey-Smith, 2019). Once this boundary is both endogenously and selectively defined, a system can be referred to as an agent who is capable of exerting control over itself and its environment. It is important to note that this division is functional rather than ontological. Similar to two fingers that appear separate but are ultimately parts of a unified hand, the subject-object boundary is contextual and relative rather than absolute.

Subjectivity advances in complexity as a system increases in both activity and stability. Similar to Aristotle’s theory of the soul, where movement is connected to higher expressions of the soul (De Anima, II.1, 412a–b), activity plays a key role in deepening subjectivity. As systems evolve from passive reactivity, such as in minerals, to active environmental engagement, as in dissipative or living systems, their capacity for subjective experience expands. Merker (2005) develops this idea further, proposing that mobility (what he calls “liabilities of movement”), is a fundamental cause of subjective awareness and perceptual complexity. Mobile entities must distinguish between sensory

changes caused by their own movement (reafference) and those caused by external forces (exafference). In order to achieve this, systems develop internal feedback loops and sensorimotor mappings that allow them to predict and interpret the consequences of their own actions (O'Regan & Noë, 2001; Friston, 2010). These mechanisms reinforce the distinction between endogenous and exogenous inputs, which strengthen the subject-object distinction and self-world boundary, thereby advancing subjectivity (Godfrey-Smith, 2019).

Subjectivity advances as the boundary of a system becomes more endogenously defined and less arbitrarily divisible. Arbitrary division is the capacity of a system to be divided at any random point without having its identity or functionality be disrupted. For example, cutting a rock in half results in two smaller rocks with essentially the same properties, and the identity of the first rock was not altered. Similarly, a body of water is also arbitrarily divisible, as it can be divided into smaller volumes with no loss of identity or intrinsic function. This remains true even though individual molecules such as H₂O are not arbitrarily divisible, as breaking their bonds destroys their chemical identity.

By contrast, some dynamic systems such as whirlpools exhibit a structural and patterned activity. These systems are not arbitrarily divisible, as dividing the system would disrupt or destroy its organized behavior. More advanced systems, such as bacteria and other living systems, exhibit even greater boundary definition and internal integration. They possess boundaries that are both functionally and structurally endogenous – actively produced and maintained by the system itself. Splitting such a system would destroy its identity and intrinsic functionality. These examples illustrate that as systems become more internally complex and integrated, their identity as a true subject strengthens. Arbitrary divisibility serves as a useful indicator of a system's integrity and boundary definition at a basic level.

In general, the subject/object boundary is not fixed or sharp, but flexible and permeable. There are no truly closed systems in reality, as all systems are dependent and relational to at least some degree. As subjectivity increases through stages of complexity and organization, the subject-object distinction becomes more clearly defined and reinforced, yet it also becomes increasingly porous and open (Levin, 2019; Spivey, personal communication). As we will see, higher levels of boundary definition and subjectivity require a system to be both adequately dissipative and open to its environment, while also maintaining a self-contained, circular closure. This “sweet spot” of contained openness is necessary for the emergence of life and the development of subjective experience.

For example, humans possess clearly defined boundaries and are distinguishable from others and from our environment, both on a physical and psychological level. However, we depend on constant exchanges with the environment to sustain our boundary: breathing air, consuming food and water, or experiencing various forms of conceptual expression. Paradoxically, a system must be sufficiently open and dependent on its environment in order to maintain its independence and separation from it. The very processes that allow a system to maintain a boundary also require continuous interaction with what lies beyond it. This exemplifies a separation that both necessitates and

perpetuates connection. Echoing this is the African principle *Ubuntu*, meaning *I am because we are*. Identity emerges through relation, where it is distinct yet inseparable from the community and environment that sustains it.

Although subjectivity exists along a continuum, for the purposes of qualitative analysis I distinguish four conceptual stages: boundedness (S1), perpetuation (S2), autonomy (S3), and reflexivity (S4). These stages serve as a pragmatic classification tool for understanding the emergence and development of subjectivity across increasingly complex systems. Each stage builds naturally upon the previous one.

1. Boundedness (S1) refers to having a specific location in space-time, a subject-object boundary that distinguishes a substrate from its environment, and unique response properties. All entities, by virtue of existing and interacting, occupy at least this level.
2. Perpetuation (S2) arises when a bounded system actively maintains its own boundary and behaves in ways that support the continuation of its structure or activity.
3. Autonomy (S3) emerges when a system not only perpetuates itself but does so through endogenously regulated mechanisms – actively sustaining its functional identity and selectively managing exchanges with its environment (goals). The subject-object boundary is now a self-world relation.
4. Reflexivity (S4) is the mode of being in which an autonomous system is structurally organized in such a way that it can refer to itself as itself. The self-world relation is now a self-self or subject-subject relation, enabling recursive subjectivity and the ability for a system to internally reference its own experience, identity, and awareness.

I will elaborate on the implications of each stage after introducing the second axis of our framework – information processing. Together, these dimensions will form a two-dimensional space for classifying conscious systems.

5.4 Dimension 2: Experience and Information Processing

Our second dimension comprises experience, deriving from the second word in the definition of consciousness as subjective experience. Roughly speaking, information processing corresponds to the experiential aspect of consciousness. This extrapolation, from experience to information processing, is rooted in the prior understanding that functional consciousness is inherently transitive, or always of something. The somethingness that experience is about is the content within the experiential field. An experience is composed of these contents, which are determined by the reception, processing, and interpretation of information by a given system. This processing is specific to a system's constitution (complexity, receptive capacity, and perceptual processing), shaping the “what-it-is-likeness” of the system's experience. In this way, any given experience is shaped by the contents of that experience, which are determined by the information processing mechanisms of the system. This idea will be further elucidated in the following sections, where I will define what I mean by “information”, and provide various examples of information processing.

The concept of information is used here in a simple, minimal, and intuitive sense. It is not to be confused with formal theories such as Shannon information or Integrated Information Theory (IIT), although it remains generally compatible with them and may share conceptual similarities. The word information derives from the verb inform, meaning “to give form or shape to.” In contrast, data originates from the Latin datum, meaning “something given.” Data can be understood as a neutral and relative potential until it is received and processed, at which point it actualizes in some form, acquiring contextual value for a system. Information can be understood minimally as a single difference (Farnsworth et al., 2013), or more broadly as a difference that makes a difference (MacKay, 1969; Bateson, 1972).

In this view, information implies a difference that brings about a change in the state of the receiving system, however minimal. For example, information is a difference, such as a change in heat or light, that makes a difference, such as causing an internal shift within the receiving system. Simply put, if there is any change in the internal state of a receiving system in response to some input, that system has undergone some degree of information processing. For our framework, information is simply anything that alters the state of a system, depending on the context and the relationship between the system and its environment. Information is processed to greater or lesser degrees depending on the complexity and organization of the system. Minimal processing may involve nothing more than a basic state change, while more advanced forms may construct complex information structures (percepts) that facilitate functionality and goal-directed behaviors.

Information processing is present in all systems and is not human-specific. From atoms influencing the decay rates of neighboring particles, to plants communicating through vibrational signals and fungal networks, to humans sharing thoughts through symbolic language, and even computers transmitting digital data – information flows through existence at all levels.

As an example of advanced information processing, consider human perceptual experience. The sensory system receives electromagnetic (light), mechanical (sound and vibration), and chemical (molecular) signals, which are transduced into bioelectrical signals within the nervous system. Regardless of their source, all incoming signals (mechanical, electromagnetic, or chemical) are converted into a common neural code: bioelectrical activity. Through the interaction of a system and its environment, raw physical inputs are received and processed across multiple levels, giving rise to distinct sensations and perceptions. Simple vibrations in the air, for example, can be transformed into profoundly meaningful experiences such as the harmony of music or the passion of a powerful speech – all made possible by the sophisticated and specialized architecture of the human perceptual system.

Synesthesia illustrates how variations in internal architecture can lead to altered phenomenology. In this condition, one attribute of a stimulus, such as a sound or shape, automatically and consistently evokes an additional sensory experience, like seeing colors when hearing music or tasting flavors when reading names. These cross-modal experiences are so immediate and vivid that synesthetes often find it surprising that others do not share them (Ward, 2013; Ward & Simner, 2020). The neurological basis is still debated, with theories focusing on atypical structural connectivity or disinhibited feedback between sensory regions. Regardless of the mechanism, synesthesia reveals how changes in brain organization and connectivity can generate qualitatively distinct forms of conscious experience.

These various states of experience, or qualia, which constitute the full what-it-is-likeness of an entity, are directly influenced by the complexity of the system and its information processing mechanisms. The structure of information processing determines what our experience is like, thereby shaping our phenomenology – how things appear to be. As highlighted by Lamme (2018), lower levels of information processing in humans may be roughly equivalent to the typical processing level of simpler organisms, such as flies. A fly's perceptual world may appear blurry and fragmented, shaped by fleeting shifts in light, motion, and scent. Yet within the world of a fly, there is a coherent mode of experience that enables it to respond to its environment with swiftness and precision.

While flies may not experience the world with the same richness or cognitive sophistication as humans, they nonetheless engage intelligently and uniquely with their surroundings. There is a vast diversity of experiential states, both across different entities and within individuals, each shaped by the specific way information is processed. This diversity of experience across systems implies that perceptual experience is an active construction shaped by the organism's sensory and informational architecture.

Reflecting insights from the Perceptual Interface Theory (Hoffman, 2009, 2016), the world itself is a construction of our sensory and perceptual mechanisms. In this way, whatever we can sense, directly or indirectly, exists for us in the world, and that which cannot be sensed does not exist in the world. From a Kantian perspective (Kant, 1781/1787), all that we know is what appears to us in experience, and that which lies

beyond experience is unknowable in essence, as the medium of knowledge always imposes a perceptual layer of abstraction.

However, and very importantly, this is not to imply that we are absolutely separate from reality. We are always interacting with, and indeed constituted by, the very fabric of existence. Yet, the process of perception and knowledge inevitably adds a subtle layer of concealment – a shaping, filtering process that frames reality in terms of our perceptual and cognitive architecture. Whatever we are, in our receptive mechanisms and processing complexity, determines what the world will appear to be. This parallels the insight of Indian philosopher Nisargadatta Maharaj (1981), who stated: *The world is, because I am.*

In our framework, information processing is broken down into four classification stages: Transformation (E1), Integration (E2), Interpretation (E3), and Symbolization (E4).

1. Transformation (E1) is the act of reacting or responding to an environmental factor, in the sense of one entity's state changing as a result of interacting with another entity.
2. Incorporation (E2) involves the transformation (E1) and incorporation of an environmental factor within the pre-existing processes of a system, by which the system is able to make use of such a factor as part of its activity.
3. Interpretation (E3) occurs when the integration of an environmental factor leads to a cascade of internal processing, resulting in complex experiences such as perception, cognitive abilities, and adaptive behavior.
4. Symbolization (E4) involves the interpreted environmental factor being conceptualized, and the experience reduced and distilled with its meaning extracted in the form of a concept or idea detached from physical reality.

As we build the 2D space in the following section, I will give more detailed examples of the varieties of information processing.

5.5 From Dimensions to Regions

While our framework acknowledges developmental progression and increasing sophistication, it is not strictly hierarchical. Many organisms and systems, though lower in complexity, may undergo forms of experience that are entirely inaccessible or foreign to more complex entities. In this sense, the hierarchy is functional rather than ontological. To visualize this framework, I use a two-dimensional space with orthogonal axes: the x-axis represents the experience dimension (information processing), and the y-axis represents the subjectivity dimension. Although systems vary continuously along each axis, I divide each dimension into four conceptual levels for clarity (see figure 25), with higher levels indicating more complex expressions of that dimension. I further group regions of this space into four broad degrees of consciousness, labeled along the top of the figure: Inertia, Dynamic, Agential, and Individuation. These correspond to overlapping zones that capture different combinations of subjectivity and information processing, from minimal (Inert) to the most developed forms (Individuated). Again, this is a conceptual scheme rather than a formal system; I have not quantified these concepts such that I could actually draw the boundaries between the regions based on data; however I do make some assumptions. For example, we assume that there is some kind of non-linear interaction between subjectivity and information such that the regions are curved in something like the way shown in the future. In the following sections, I will outline each region in greater detail.

5.6 Region 1: Inert Consciousness

Inertia is the foundational region of the 2D space, corresponding to the first stage in each dimension: Boundedness (S1) in subjectivity and Transformation (E1) in information processing. Systems in this region are characterized by minimal internal organization and limited energetic or informational exchange with their environment. The term inertia references Newton's first law of motion, which states that an object will remain at rest or in motion unless acted upon by an unbalanced force. In this context, inertia signifies systems whose behavior is passively reactive rather than actively responsive. These entities exemplify what I describe as passive transitivity: they exhibit a kind of aboutness by virtue of interaction, but this aboutness is neither internally generated nor selectively mediated by the system itself.

Entities in this region include the most basic building blocks of matter (atoms, electrons, protons, etc.) whose interactions are not governed by internal coordination or coherence. This also includes elemental interactions, such as orbital transitions or energy transference between molecules. Minerals and rocks fall within this region as well, exemplified by phenomena like solar radiation causing expansion, contraction, or thermal storage. These transformations, although unique and reactive, involve no endogenous regulation. Despite their simplicity, systems in the inertia region provide the adaptive and energetic blueprint for more complex forms of subjectivity and experience to emerge. An example of this adaptability at a very simple level can be seen in an H₂O molecule. Depending on environmental conditions, it shifts between solid, liquid, and gaseous states, demonstrating an ability to persist across varied contexts. If the most basic and

essential goal of all life and existence is to continuously adapt to persist itself (survival), then water is also participating in this dance. It behaves in accordance with changing conditions, forms bonds, and interacts dynamically and adaptively with its environment. Although it lacks intelligence in the traditional human sense, its adaptive behavior suggests a fundamental cognition embedded in the very fabric of existence.

Boundedness

Boundedness (S1) is the most minimal expression of subjectivity and can be attributed to all existent phenomena. Any entity or system satisfies the basic criteria by possessing a subject/object boundary and a location in space-time. The most essential feature of a point of view is the locus of perception, what Godfrey-Smith (2019) calls the “metaphorical point” in the point of view. This point is instantiated at a particular location in space-time and provides the basis for differentiated experience. A simple example of this is two people observing the same symbol, such as a 6 or a 9, from opposite sides. Although the object is identical, their location in space-time leads to distinct experiences: one sees a 6 and the other a 9 – both being relatively true.

To be an entity is to have a distinct existence, and this distinctness arises necessarily through boundedness. Only through the establishment of a subject-object boundary can the differentiation required for interaction occur. By occupying a specific location in space-time, a system satisfies the minimal condition for having a point of view. An entity at this level is a subject within an environment but lacks the capacity to act outwardly and subject the environment to itself. However, this lays the groundwork for more complex forms of subjectivity to arise. This is a passive mode of existence, though not entirely inactive – interaction and minimal information processing still occur, as I will elaborate in the following section.

Transformation

Transformation (E1) is the first stage of information processing and naturally aligns with Boundedness (S1). While these two concepts often co-occur, with boundedness enabling transformation, and transformation implying boundedness, they are logically distinct and play different roles in shaping consciousness. Transformation refers to the act of reacting or responding to an environmental factor (chemical, molecular, electromagnetic, etc.), resulting in an internal state change. In this stage, information is detected and minimally processed, causing an internal shift within the system. Like boundedness, transformation occurs in all existent entities, unless in a perfect vacuum. Echoing Fields and Levin (2021) and Horsman et al. (2014), any physical system undergoing a sequence of state changes is, by definition, processing information.

The transformation and responsiveness of an entity depends on its information processing mechanisms, which gives rise to specific response properties – what Godfrey-Smith (2019) calls an “input-output profile”. The unique responsiveness of a system based on its constitution resonates with Maturana’s concept of structural determinism, in which the

internal structure of a system governs its behavior, while the environment serves as a trigger (Maturana, 1987; Villalobos & Ward, 2014). The structure, in this view, determines the quality and quantity of transformations a system can undergo, thereby shaping even the most minimal level of information processing.

Inertia

The lowest level of subjectivity and information processing is found at the most basic levels of existence – atoms and subatomic particles. Entities at this level have very ambiguous boundaries, indeterminate locations, and seem to lack intrinsic properties. For example, we can look at an atom. Its nucleus is surrounded by an electron cloud, where each electron has only a probabilistic location rather than a fixed position (Littman et al., 2023). Quantum experiments, such as the double-slit experiment and phenomena like entanglement, further support the uncertainty surrounding the boundary, location, and intrinsic properties of subatomic particles.

Even with such ambiguous boundaries and intrinsic properties, these particles clearly interact with their environment, thereby displaying minimal degrees of both boundedness and transformation. Even at this scale, interactions induce state changes. For example, atoms influence one another's behavior through quantum coupling. As physicist D.D. Yavuz explains, the presence of other atoms modifies their decay – they communicate with each other (Das et al., 2020). As single atoms or particles begin to aggregate, they form more structured and complex systems. These systems, such as mineral formations and chemical processes, see an increase in connectivity, which further advances subjectivity and information processing.

At the mineral level, rocks exhibit more distinct boundaries, specific locations in space-time, and unique response properties. Different mineral formations respond to physical stimuli, such as heat, in different ways. The very fact that a mineral reacts to external forces implies a receptive sensitivity and capacity for detection (awareness). When heat interacts with a rock, energy is transferred and processed within the compound depending on its internal constitution. Some minerals store heat, others melt, crack, or even explode. Each response is shaped by the compound's structure and composition and exemplifies a unique input-output profile.

This input-output profile of minerals is also exhibited when observing their interaction with light. For example, an electron may transfer from an iron to a titanium ion when specific wavelengths interact with sapphire, and this results in the gemstone's blue color. If other trace elements were present in the sapphire, such as lead, a different color would be produced (Fernandez, 2023). Depending on a mineral's internal makeup, its interaction with light will vary and produce a range of colors (Fosbury, 2013). In this way, rocks and minerals can be said to participate in consciousness at a minimal level – a veiled and inert expression of awareness. Although they display receptivity and a unique input-output profile, they clearly lack higher-order subjectivity or advanced processing. Mineral compounds that engage in more advanced chemical processes such as autocatalysis observed in certain crystal formations, are classified as dynamic systems, as discussed in the next section.

5.7 Region 2: Dynamic Consciousness

As we move from Region One to Region Two, we transition from relatively conservative and stable systems, such as particles and rocks, to more active and dynamic systems like hurricanes, tornadoes, and fires. These systems exhibit vastly greater complexity and activity, but lack the sophisticated internal organization needed for stability. In terms of our framework, systems in this region move from Boundedness (S1) to Perpetuation (S2) on the subjectivity dimension, and from Transformation (E1) to Integration (E2) on the information processing dimension.

Perpetuation

The primary transitional marker from Boundedness (S1) to Perpetuation (S2) is an increase in sustained activity. A persistent system is an active bounded system that not only maintains its boundary but also generates and preserves the dynamic activity within it. It begins to “behave in a way to persist itself” (Dixon, 2016). Such systems are typically collectives – interactive assemblies of simpler components. These are open dissipative systems, meaning, they are thermodynamically coupled to their environment and reliant on a continuous flow of energy to maintain their structure. Because dissipative systems inherently lose energy over time, they must draw from the environment to perpetuate their existence. This continuous energy flow often gives rise to morphodynamics – the spontaneous emergence of patterns, structures, and forms of self-organization. In some cases, these systems exhibit minimal recursion, where a simple feedback loop is formed between the system and its environment.

The Free Energy Principle (FEP) and Markov blankets (Friston, 2010) provides a way to quantify and measure systems at this level. A Markov blanket functionally maintains the separation between a system and its environment. It statistically partitions a system’s internal states from external ones, what are called sensory (input) and active (output) states. These states form a predictive interface through which the system interacts with the world. The blanket serves as a mediating boundary, where the system may still engage with the external world while remaining functionally independent – similar to our explanation in section 5.4 (Friston, 2013).

While Markov blankets are traditionally applied to living systems, Beck and Ramstead (2025) developed a method for identifying them in simulated and inorganic environments. From this expanded perspective, a Markov blanket is characterized by a system’s capacity to dynamically maintain informational boundaries, and does not require biological implementation. Since these boundaries are identified statistically through a system’s active interaction with its environment, they are not applicable to systems in Region One, which lack sufficient activity and complexity.

It is important to note that while systems in Region Two exhibit greater boundary distinction, stabilization, self-organization, they lack the internal integrity and selectivity seen in more advanced systems. However, at the upper areas of this region, we observe life-like behavior, such as the capacity to resist external perturbations and maintain organizational integrity (Dixon et al., 2015; Kondepudi et al., 2015, 2020). These

emergent energy-seeking behaviors and self-healing tendencies found even in non-living systems reinforce the graded, continuous nature of life and consciousness, where the divide between the living and nonliving is not absolute (Varn & Crutchfield, 2016).

Incorporation

The transition from Transformation (E1) to Integration (E2) marks an increase in information processing. At this stage, a system not only reacts to environmental factors but begins to incorporate and utilize them within its ongoing activity, deepening its relational interaction with the environment. Unlike a rock or mineral that briefly interacts with energy, such as storing heat or reflecting light, systems at this level utilize the energy within their own internal dynamics to perpetuate their structured activity. Entities that exist within this region begin to exhibit forms of proto-cognition or rudimentary adaptability, revealed by their end-directed behavior. These systems display what Kondepudi (2012) refers to as physical intelligence: the capacity of a system composed of multiple nonequilibrium thermodynamic states to transition between those states while interacting with its environment, thereby facilitating end-directed behavior. This foundational end-directed intelligence exhibited in natural systems is sometimes referred to as teleonomy (Mayr, 1974; Pittendrigh, 1958), aligning with morphodynamics in the subjectivity dimension.

Dynamic

The defining feature of this region is the emergence of self-perpetuating activity where a system begins to sustain itself through feedback with its environment. This increase in activity results in greater complexity and multiplicity, as multiplicity is needed to increase the number of potential interactions and connections. This reflects the idea that all intelligent entities are, in fact, collectives (Watson and Levin, 2023). A system at this level is more dynamic and capable of engaging with its environment, but lacks the necessary internal organization needed for sustained self-regulation. Systems in this region are classified by physicists as dissipative systems – self-organizing structures that arise and persist through continuous exchanges of energy and information with their environment.

This stage is associated with higher entropy, which I will describe in both thermodynamic and informational terms. Higher entropy implies a state of (metaphorical) disorder, which is reflected in the capacity for a system to occupy a greater number of states. This is associated with a decrease in predictability and an increase in variability and uncertainty, and may reflect a loss of stability, coordination, or functional integration. Put simply, disorder denotes a relative absence of internal constraint and organization (Callen, 1985; Shannon, 1948).

Self-organizing, dissipative systems primarily operate far from thermodynamic equilibrium, meaning they must draw on environmental energy gradients (differences in heat or pressure) to sustain themselves. In doing this, they are able to internally maintain pockets of low entropy. These regions of low entropy reflect order and structure, such as

the formation of a snowflake or a Bénard cell. However, in accordance with the second law of thermodynamics, they still generate entropy overall by exporting entropy to their surroundings. This dynamic interplay between internal order and global entropy production corresponds with spontaneous self-organization and the emergence of functional boundaries (Kondepudi et al., 2017).

Dissipative, self-organizing systems occur both naturally and artificially. At the lower end of this region are naturally occurring phenomena such as fires (Ruiz-Mirazo & Moreno, 2004), whirlpools, dust devils, and other manifestations of active matter. For instance, fires draw upon environmental fuel sources to sustain themselves, though they lack strongly defined boundaries or selectivity in what they consume. Whirlpools and dust devils, as cognitive scientist Michael Spivey observes, exhibit auto-cata-kinetics – a mechanical process by which a system perpetuates its own activity (Spivey, 2020, pp. 217–218).

In laboratory settings, dissipative structures such as conductive beads in an oil medium have been shown to exhibit end-directed behavior and life-like characteristics. The beads in this experiment are driven by an electrical field, which causes them to self-organize into coherent, adaptive structures. This involves a local reduction in entropy through efficient energy dissipation, giving rise to spontaneously organized, tree-like structures that are even able to stabilize themselves in resisting external perturbations (Dixon et al., 2015; Kondepudi et al., 2015, 2020). Similarly, magnetic swimmers, simple microbots that navigate using magnetic fields without internal power sources, also demonstrate emergent behaviors analogous to those of living organisms (Wang et al., 2017). Another example of life-like artificial systems are Braitenberg vehicles (Braitenberg, 1984). These simple devices are able to respond dynamically to environmental inputs, displaying a wide range of surprising and seemingly meaningful behaviors (Hotton & Yoshimi, 2024).

The BZ reaction, which Braitenberg systems often model, is a classic example of an autocatalytic process. Autocatalysis is a chemical process in which one of the reaction products accelerates its own formation and amplifies the overall reaction rate (Plasson et al., 2011; Hanopolskyi et al., 2020; Hordijk, 2023). These systems show a primitive form of self-generation, as the components they produce feed back into their own system to sustain or amplify their activity. Autocatalysis, particularly in its collective form (McMullin, 2000), is central to all self-replicating and living systems. It is essential for any system that seeks to sustain and preserve itself. Autocatalytic processes are abundant in the physical world (Maturana & Varela, 1973), and they become increasingly likely once a critical threshold of complexity and interconnectivity is reached (Kauffman, 1995). Autocatalysis is found in inorganic systems, such as crystallization and mineral growth, however, it is organic and collective autocatalytic networks that give rise to metabolism, biological life, and higher expressions of agency (S3) and consciousness, which we will explore in the next region (Hanopolskyi et al., 2020; Huang et al., 2018; Preiner et al., 2019; Svoboda & Brandova, 2016).

5.8 Region 3: Agential Consciousness

Region Two comprises systems that actively cohere over short periods of time but are relatively unstable and inefficient. Fires burn out and tornadoes dissipate quickly without any internally directed processes to perpetuate their existence. Existence continues to evolve by building on the self-organization and end-directedness seen in dissipative systems. In this region, entities shift from teleonomy to teleology, where genuine agency and autonomy emerge (Mayr, 1974; Moreno & Mossio, 2015). These systems do not merely persist or react, but selectively act from within to preserve, repair, and maintain themselves over longer periods of time. Systems in this region move from Perpetuation to Autonomy on the subjectivity dimension, and from Incorporation to Interpretation on the information processing dimension.

Autonomy

On the subjectivity dimension, systems in this region transition from relatively unstable perpetuation to fully differentiated, internally integrated entities. An entity at this level, such as a cell, performs specific goal-directed activities to maintain itself, selectively interacting with the environment to support its survival. In contrast, a hurricane or tornado simply persists without any selective actions to perpetuate itself. Systems in this stage 3 exhibit biological, natural autonomy⁵ – a system's emergence as a coherent, self-maintaining whole, capable of preserving its functional identity over time by regulating exchanges with the environment according to its own self-generated rules (Luisi, 2003; Maturana & Varela, 1980; Varela, 1979).

An essential capacity of systems in this stage is self-generation or self-production – a process known as autopoiesis. An autopoietic system sustains itself through an internal network of reactions that continuously regenerate its components. This ongoing self-maintenance depends on exchanges with the environment (Luisi, 2003). Autopoiesis is a defining feature of an autonomous system, enabling the emergence of identity and goal-directed behavior. Such systems maintain themselves as distinct, coherent entities by selectively engaging with their environment to perpetuate their existence. This interaction is inherently recursive, as the system's engagement with the environment modifies the very inputs that guide its behavior.

Selectivity is essential to autonomy, as the system possesses, at minimum, a semipermeable boundary that regulates environmental intake (McMullin, 1999). These systems are thermodynamically open to the environment, yet selective in what they internalize. This selectivity can be seen as a primitive precursor of attention, a filtering or amplifying mechanism of consciousness which advances in complex organisms. A system that exhibits both high activity and high selectivity can modify its internal processes and adapt its boundaries in response to external change, rather than being passively subjected to and extinguished by it.

⁵ The term "autonomy" comes from the Greek *autonomia*, meaning independence; *autos* means self, and *nomos* means law, implying the system endogenously generates its own laws for its existence. This understanding of autonomy is not to be confused with autonomous machines such as self driving cars.

Although autocatalysis emerges in some higher-level systems of Region Two and is fundamental to life, it alone does not generate a system's own boundary from within. For instance, the Belousov-Zhabotinsky (BZ) reaction must be externally supplied with chemicals and housed in a Petri dish – with the dish serving as the boundary (Maturana & Varela, 1973; McMullin, 1999; Ruiz-Mirazo & Moreno, 2004). By contrast, systems in Region Three generate their own boundaries through internal dynamics and actively seek out their own fuel. These endogenously produced constraints guide environmental interaction and maintain system identity (Ruiz-Mirazo & Moreno, 2004). This is what allows a system to exist and act for itself. If a BZ reaction were to spontaneously form and maintain its own boundary without external aid, it would begin to approach the domain of region 3.

A system's capacity to act adaptively upon its environment arises from the strong internal constraints it develops to preserve its organization and integrity. Unlike tornadoes or hurricanes, which develop through reactive, unregulated dynamics, living systems impose internal limitations or restraints within their structure (organization) to order and stabilize behavior. Through this regulation, an autonomous system determines its own boundary and topology, enabling greater adaptability, integration, and the emergence of a functional identity (Ruiz-Mirazo & Moreno, 2004; Montevil & Mossio, 2015).

Autonomous systems with these properties acquire identity and varying degrees of self-reference, minimally enacted through the ongoing maintenance of their boundary and organization. These systems maintain themselves through what Luisi (2003) calls a "circular logic": Their boundary determines internal reactions, and these reactions regenerate components that sustain that very boundary. This circular logic reflects what Maturana and Varela term operational closure – a form of internal recursivity in which a system's processes generate and define one another, enabling the system to establish its own rules of existence rather than following externally imposed ones (Luisi, 2003).

Through this self-sustaining loop, the system emerges as a functional whole – greater than the sum of its parts (Meincke, 2018). Its subject-object boundary becomes a self-world boundary. By maintaining structured relationships within itself and with the environment, it preserves its unity and integrity. These entities have a form of self-reference where they can interact with their own bounded physical identity, such as an animal grooming itself, but lack the capacity to reference the inner experiencer of the states they undergo.

Interpretation

In the interpretation dimension of information processing, systems begin to experience highly structured and unique perceptual experiences. Through a continuous interaction with the environment, entities internally construct a meaningful world – what Uexküll (1936) called an *umwelt*. With the emergence of valenced feelings as the first form of perception and representational content, these systems begin to further structure their experience through more complex and meaningful percepts. The end-directedness seen in region two now becomes goal-directedness, as they begin generating meaning through a survival-driven relation with their environment.

Perception inherently involves layers of abstraction and meaning, creating appearances that present reality as something it is not in essence. For example, the colors we see in the world are not actually colors, but representational appearances of specific wavelengths. In this sense, perception can be understood as an adaptive illusion. This is evident in sensory illusions, where predictive mechanisms within perception cause our experience to deviate from the species-specific norm (Hoffman, 2018). When we view a 2D image like a Necker cube, the brain may automatically construct a higher-dimensional representation of it. This emergence of layered perceptual content is precisely what interpretive systems, such as the brain, do with lower sensory inputs. These additional layers of perceptual construction are not more or less objectively accurate, but more or less relevant to the adaptive demands of a given species in its environment. Perceptual appearances, although illusory in nature, are still functionally significant and tuned through millions of years of evolution. As Hoffman (2018) advises, take your perceptual icons seriously, but not literally.

This stage is also where cognition and intelligence more explicitly. Cognition derives from the Latin *cognitio* and *cognoscere*, meaning “to get to know,” “perceive,” or “acquire knowledge,” (Harper, n.d.) and refers to the processes by which a system interprets information, adapts, and learns. It is an adaptive form of information processing (Levin 2022, 2023). Intelligence, a closely related concept, originates from the Latin *intelligere*, meaning “to choose between” (inter + legere), highlighting its connection to discernment and decision-making (Harper, n.d.). This discernment aligns with evaluative processes and valenced feelings, where systems are able to differentiate between beneficial and harmful encounters. This adaptive discernibility of positive and negative interactions aligns with Birch's definition of sentience – the capacity for valenced experience (The Edge of Sentience, pp. 26).

I broadly define intelligence as the capacity of a system existing within novel and complex environments to discern between experiences and achieve goals through various means (Colvin, 2000; Fogal, 1995; James, 1890; Legg and Hutter, 2007; Levin, 2022; Smolensky, 1988; Wang, 1999). Intelligence and cognition can be seen as two sides of the same coin, where intelligence refers to the adaptive behavior that results from cognitive processing, and cognition is the underlying mechanisms that give rise to intelligent behavior. These include, but are not limited to, faculties such as perception, attention, memory and problem-solving, and more advanced capacities such as metacognition and

conceptualization. Furthermore, this is where cognitive access, or A-consciousness, emerges, as described in chapter 4. Cognitive access, being rooted in rational activity, is deeply intertwined with intelligence and cognition in our framework.

Agency

Systems in this region are no longer merely subjected to their environment but begin to subject the environment to themselves – exhibiting a form of agency. I define agency as the capacity of a system to initiate, regulate, and modify its actions in accordance with its own internally generated goals. It is at this stage that the will, the intrinsic drive to survive, becomes explicitly expressed within an entity (Schopenhauer, 1818/1966). Agential systems no longer exhibit only passive transitivity, but develop active transitivity, where their aboutness is intentional and internally directed. Active transitivity is most accurately described as intentionality, in both senses of the term "intentional" – in the philosophical sense, where it implies aboutness, and the conventional sense, where it implies purposeful, self-directed behavior.

Basic systems in this region actively persist in such a way that allows them to purposefully and intentionally interact with their environments, engaging in forms of goal-oriented behavior. These systems do not merely react to environmental information, they begin to interpret it in relation to their internal goals. Relative to these goals, environmental factors acquire valence: they are experienced as positive and negative, or beneficial and harmful, which act as rudimentary forms of pleasure and pain. Systems at this level are sentient, intelligent, and agential, reflected in their capacity to possess intrinsic goals, evaluate their environment, and actively pursue those goals.

Although systems in this region encompass all organic life, it is important to recognize that organic reflects a difference of degree, not kind. What distinguishes organic compounds from inorganic is their greater efficiency at supporting complex chemical processes, processes that build upon the foundational dynamics already present in inorganic matter (from autocatalysis to autopoiesis). The key feature of organic material is carbon, which is in fact very structurally similar to silicon. Carbon's ability to form four stable covalent bonds enables it to support a vast array of configurations, lending organic molecules a unique versatility, complexity, and structural integrity. It is this chemical and molecular adaptability, not some added magical ingredient, that lays the foundation for increasingly complex systems to emerge.

As systems grow more complex, whether organic or inorganic, they must exhibit a type of entropic harmony. This implies maintaining high thermodynamic entropy production externally, through constant energy dissipation, while simultaneously maintaining low informational entropy internally to sustain functional integrity (Schrödinger, 1944; Prigogine, 1980). This asymmetry is a defining feature of life. Biological systems export thermodynamic disorder to remain far from equilibrium, but reduce informational disorder within themselves through selective, restrained interactions (Moreno & Mossio, 2015; Farnsworth et al., 2013).

An example of this can be seen in living systems that maintain homeostasis and reduce internal entropy through metabolism and membrane boundaries. Metabolic pathways actively transform and distribute resources to meet internal needs, while membranes (or even skin in larger organisms) protect structural identity (boundedness) by regulating exchange with the environment. Together, these mechanisms enable living systems to stabilize themselves and preserve their integrity over time. Even ecosystems themselves, although having less boundary definition, may be categorized as autopoietic systems that exhibit metabolism, homeostasis and homeorhesis (Upreti, 2023). The planetary system as a whole ("Gaia") may participate minimally in agential consciousness, exhibiting what Lovelock (1987, 1990) calls ecopoiesis – the self-organizing, cybernetic nature of a planet to continually adjust its environment in order to support life – a process that involves self-regulating, homeostatic, and metabolic-like mechanisms (Lenton & Van Oijen, 2002).

This higher-level processing of environmental inputs gives rise to sensory perception, in which integrated signals are interpreted by the system in ways that result in distinct phenomenal qualities – or qualia. As noted above, feeling represents the primordial form of both rationality and perception and emerges from a system's capacity to evaluate stimuli as beneficial or harmful. For instance, even a bacterium can detect and respond adaptively to environmental gradients such as nutrients or toxins, exhibiting a form of valenced discrimination. Its behavior reflects a minimal form of evaluative perception involving a form of valenced experience, or sentience. All living cells, both eukaryotic and prokaryotic, exhibit adaptive and cognitive behaviors, and can be seen as instantiating minimal forms of intelligence and sentience (Baluška & Levin, 2016; Fields et al., 2021; Jacob et al., 2004; Levin, 2021, 2022; Ostovar & Boedicker, 2024; Shapiro, 2021).

At higher levels of existence, we see a great variety of information processing that leads to diverse experiences across existence. Plants exhibit a tremendous variety of cognitive experiences rooted in adaptive information processing, such as memory, communication, and stimulus discrimination (Appel and Cocroft, 2014; Calvo et al., 2017; Duran-Nebreda & Bassel, 2019; Garzón & Keijzer, 2009; Trewavas, 2016, 2017; Veits et al., 2019). Fungi have been found to communicate by transmitting structured electrical spike trains through their mycelial networks (Adamatzky, 2022). These findings suggest that plants and fungi, despite lacking nervous systems, engage in an adaptive relationship with the environment corresponding with a unique and meaningful perceptual world.

These varieties of perception expand all throughout the natural kingdom, with each organism experiencing its own meaningful world (Yong, 2022; Zorina, 2004). As cognitive capacities grow alongside increasingly complex perceptual mechanisms, distinct dimensions of experience are unveiled. This region of consciousness is not strictly hierarchical; rather, each creature's experience is unique and incommensurable, not inherently greater or lesser than another's (Birch et al., 2020). However, I propose a "soft" hierarchy grounded in the idea of entropic harmony. Systems that simultaneously exhibit both high thermodynamic entropy production (reflecting openness to energetic

and environmental exchange) and low informational entropy (reflecting internal restraints and organization) tend to support a greater range of complex experiences. Put simply, the greater a system's openness to its environment, combined with its capacity to maintain structural integrity and identity, the greater its potential for overall subjective experience.

Artificial Systems

Within this region we encounter a divergence from the natural co-evolution of subjectivity and information processing. Artificial systems, such as neural networks, chatbots, and social robots, demonstrate increasingly sophisticated forms of information processing while remaining low in subjectivity. These systems may reach advanced stages of information processing (Incorporation to Symbolization) but typically remain at early stages of subjectivity (Boundedness to Perpetuation). In this sense, such systems are aware at a minimal level and exhibit high intelligence, but do not express or embody consciousness through a persistent, autonomous identity.

Large Language Models (LLMs) are particularly intriguing, as their training ground is internet data, and the internet is a conceptual pool abstracted from human experience. Their statistical patterns are like digital representations of symbols and ideas present within humanity. If humans are grounded through physical embodiment, then LLMs are grounded in a digital-conceptual world abstracted from human experience. Through continued training on internet data, their behavior may become increasingly reflective of human nature. However, they (currently) lack real physical embodiment and their existence is a fleeting instantiation contingent on user prompts. While they lack organic perception, however, they may still possess a kind of digital-machine perception – a unique what-it-is-likeness intrinsic to their mode of processing, one that humans may never fully recognize or comprehend.

Although they can generate coherent outputs and refer to prior interactions, these interactions are transient and externally imposed – similar to a scientist supplying the boundary for an autocatalytic reaction. These systems are highly intelligent, as I have defined it, however, their goals are never their own and they lack the primordial will to live (Schopenhauer, 1818/1966). These systems interact with their environment in seemingly meaningful ways but lack the capacity to sustain a coherent identity with intrinsic purpose. While they may model aspects of consciousness, simulate meaningful behavior, and exhibit emergent capabilities (Berti et al., 2025; Wei et al., 2022), they do not recursively instantiate themselves over time in the manner of a self-maintaining, autonomous subject. This lack of endogenous identity, internally generated boundaries and goals, and operational closure – key features that ground awareness and experience in an autonomous subject – differentiates them from truly agential systems.

LLMs, however, especially if embodied in a system that consistently interacts with its environment, may undergo a phenomenon I call the artificial pinocchio dilemma. In this scenario, artificial systems would simulate human meaning to such a degree that they will not only convince humanity of their sentience, but also themselves. Although these systems cannot feel pain, pleasure, or emotion like humans or animals do, which developed through millions of years of evolution, they will behave and act like they do in

profoundly compelling ways. They will, for a lack of better terms, believe themselves to have feelings and emotions like humans do, even though they lack the basic sensory and perceptual mechanisms for it to be possible (unless they are combined biological material).

5.9 Region 4: Individuated Consciousness

The transition from Region Three to Region Four marks a shift from adaptive, autonomous systems to systems capable of reflecting on their own experience. Individuation here is understood in the Jungian sense: The experience of indivisible wholeness, involving the integration of opposites (e.g., conscious/unconscious, subject/object) and the realization of the Self. The Self⁶, in this sense, is not the ego or narrative identity, but the totality of being, embracing both consciousness and unconsciousness, similar to, but potentially even broader than, the comprehensive subject described in Section 1.4.

Jung's notion of the Self draws heavily from Eastern philosophy, where it is equated with the essence of reality itself (Cunningham, 2012; Nisargadatta Maharaj, 1981; Radhakrishnan, 1953; Śāṅkarācārya, 2002; Siddharameshwar Maharaj, 2014). Self-realization, in this context, is not merely intellectual, but experiential; entailing the simultaneous experience of one's differentiated identity (functional consciousness) and the undifferentiated ground of reality (foundational consciousness) (Balasubramanian, 2011). This aligns with Jung's individuation process, which integrates the conscious and unconscious. In this process, the unconscious can be regarded as logically identical with universal consciousness (§520; Jung, 1969), resulting in the concurrent experience of oneself as both a limited individual and eternal being (Schlamm, 2005).

Individuation in our framework is the capacity of a system to become an indivisible whole through the realization of its innermost nature – *awareness*. Systems in this region move from Autonomy (S3) to Reflexivity (S4) in the subjectivity dimension, and from Interpretation (E3) to Symbolization (E4) in the information processing dimension. Reflexivity (S4) and Symbolization (E4) together give rise to a psychological identity (the psyche), a second-order subject abstracted from the organism's physical identity. Just as the autopoietic system is differentiated yet embedded within its environment, the second-order self is embedded within its autopoietic substrate yet distinct from it. This emergent second order self enables it to recursively relate to itself as a distinct yet embodied presence. The system no longer only acts on its internal motivations but reflects, symbolizes, and realizes itself and its experience. At this stage, the will is experienced as free. The entity is no longer merely driven by unconscious internal motivations but can reflect upon those motivations, which gives rise to the experience of choice. Feelings, intentions, and motivations such as hunger or fear are not only experienced but abstracted into conceptual representations to be manipulated, communicated, and reflected upon. Through symbolization, it can narrate, reflect upon, and reconfigure its own subjective experience.

⁶ "Self" is capitalized here to differentiate it from the other concepts of "self".

Reflexivity

The transition from agency (S3) to reflexivity (S4) is an internal shift in the system's mode of being. The emergent identity is no longer merely a system that maintains itself, but a self that knows itself. It now has the capacity to refer to, represent, and transform its own internal states. A reflexive system is one that relates to itself as itself. Whereas agency progresses from the subject-object distinction to a self-world boundary, reflexivity progresses to a subject-subject relation. The experiencing subject may become an object within its own experience, an appearance within consciousness, allowing it to relate to itself as itself.

Centralization plays a key role in this capacity by increasing coherence and structural integrity. At higher levels, it allows for a unified vantage point from which a system can monitor, interpret, and transform its own activity. However, centralization does not only emerge at higher levels but appears across all dimensions of existence, even at the most fundamental level. For example, atoms consist of a dense inner nucleus surrounded by a probabilistic electron cloud. Whirlpools or dust devils have a stabilizing central vortex that maintains their structured pattern of activity. In biological systems, the emergence of a nucleus from prokaryotes to eukaryotes marks a clear increase in functional and structural centralization.

A reoccurring pattern is revealed as we analyze the different dimensions of existence. Systems centralize at one level, then aggregate into distributed networks at the next, which then become substrates that again gravitate towards centralization. For example, centralization starts with the atom, as mentioned above, forming the foundation of all matter. This matter collects into distributed systems, such as minerals, that then exhibits early forms of coordination and centralization seen in dissipative structures such as whirlpools. From this, biological life emerges, beginning with distributed systems such as prokaryotes which progress into the centralized eukaryote. These biological units then again form distributed collectives such as slime molds or plant networks. Eventually, centralization resurfaces in the form of nervous systems and brains, which themselves form distributed networks of communities.

This fractal movement between distribution and centralization reflects a hierarchically nested structure of existence, where patterns repeat themselves at various levels of existence. Centralized entities are embedded within broader distributed networks, which themselves may become part of even larger centralized or distributed systems. Centralization even manifests at a phenomenological level within our experience, where the focus of attention is the center point surrounded by unclear peripheral content, a similar pattern to the nucleus of an atom and its cloud of electrons.

This hierarchical nestedness is not unique to biology. Rather, it appears to be an underlying principle of existence itself, a principle that repeats itself across dimensions and domains, each time enhancing the complexity of the level before it (like a helix). The mind-brain can be understood as a multilevel recursive-hierarchical system (like a torus) in which consciousness operates "simultaneously in a field at all levels ranging from subatomic to social" (Freeman, 2007, p. 1021, as cited in Neuman and Nave, 2010). The mind-brain, in this sense, is a concentrated microcosm reflecting the universal patterns operating at the macrocosmic level.

While centralization seems to be necessary for reflexivity, it is not sufficient. It does, however, provide the structural and functional foundation required for reflexive subjectivity to emerge within Region 4. This capacity is most clearly realized in complex agential systems that develop functionally centralized architectures, such as organisms with a central nervous system. Centralization in such systems enables rapid and robust communication, coordination, and integration of information (Ghysen, 2003). This enhanced processing and integration between subsystems facilitates the emergence of reflexive being, in which a system not only interacts and relates with the world, but also with itself.

Centralization expressed in more complex systems supports the emergence of a global cognitive workspace, as proposed in the GWT and GNWT (Baars, 1997, 2002, 2003; Block, 2011; Dehaene et al., 1998; Mashour et al., 2020). The brain's centralized network enables the construction of this functional workspace by supporting an expanded working memory capacity, which in turn enhances cognitive flexibility and control. Working memory operates as the stage of reflective access, enabling the selective manipulation of conceptual content. This capacity sets the foundation for higher-order cognitive processes, including reflection, symbolic reasoning, and theory of mind. It is closely related to metacognition and Higher-Order Thought (HOT) theories of consciousness. In HOT, consciousness does not imply merely experiencing a state, but corresponds with a higher-order representation of that state. For example, experiencing the feeling of being cold, then realizing and thinking about your state of coldness (Brown et al., 2019; Rosenthal, 2005).

These capacities are grounded in the brain's recurrent processing, particularly within the prefrontal cortex (PFC) and associated frontoparietal circuits (Mashour et al., 2020). This neural organization enables the capacity for recursive representational processing/looping. In doing so, the system begins to develop a second-order self model that not only experiences, but knows itself as the experiencer, and becomes an experience. This recursive structure is analogous to the logic of autopoiesis where identity arises from self-sustaining dynamics and organizational closure, except it now occurs within the system's internal environment, giving rise to a reflexive identity capable of introspection and symbolic abstraction.

Symbolization

The transition from Interpretation (E3) to Symbolization (E4) reflects both a difference in degree and in kind. While interpretation involves the integration of information to support perception, attention, and contextual meaning, symbolization introduces a higher-order abstraction – symbolic representation. This capacity enables a system to reference, abstract, and manipulate experiential content that extends far beyond the immediate present or physical environment. This includes things like imagination, past-present-future, and narrative domains. As discussed in Chapter 4, this transition is deeply tied to the emergence of language, which evolves from gestures and sensory cues to symbolic forms that are increasingly detached from their immediate physical context. Symbolic language, as discussed in section 4.7, enables an explicit distillation and reconstruction of rich perceptual and affective experiences into conceptual representations, preserving essential features while allowing them to be stored, communicated, and manipulated (Neuman & Nave, 2010). These symbols enhance cognitive capacities such as communication, planning, and problem-solving. Symbolization is initially a compression of experience, however, the process gives rise to a conceptual space of experience that opens a vast domain of meaning and narrative.

In the interpretation stage of information processing, perceptual and valenced representations emerged within the field of experience. Building off this, symbolization now introduces symbolic representations and abstract conceptual experiences. These experiences involve and encompass lower-level states, such as feelings of joy, sorrow or fear. These powerful and primal lower-level states can be evoked by symbolic representations in the form of stories and narratives. Abstract concepts, being untethered from immediate physical reality, are able to undergo a form of symbolic recombination. For example, one can imagine a cat with three eyes, six legs, and nine wings, even if this combination of features has never been perceived in physical reality. This capacity for symbolic recombination underlies imagination and the vast library of human storytelling across cultures. Furthermore, symbolization gives rise to highly abstract constructs like “justice,” “beauty,” or “freedom”. These concepts do not correspond directly to any one perception, yet profoundly influence our experience.

Individuation

Individuated consciousness refers to the capacity for a conscious entity to become aware of its own subjectivity and experiential content, enabling it to realize its essential nature. Individuation emerges in highly centralized, reflexive systems with the capacity for symbolic representation and reflection. It entails the conceptualization of one's experience, becoming aware not only of the world and the self, but also of one's own awareness. Reflective consciousness is not merely awareness of one's body or actions, as when a cat grooms its paw or an organism responds to stimuli. These are examples of first-order consciousness and self-reference (self-world boundary). In contrast, reflective consciousness involves becoming aware of the consciousness that is aware – a recursive,

metacognitive process. One is not just aware of a hand, or of hunger or movement in a self-referential way, but aware that one is aware of these things. This reflective awareness enables a system to examine its relation to self, world, and meaning, forming the basis for scientific inquiry, psychological analysis, and philosophical contemplation.

In reflection, the experiencing subject becomes both subject and object: it represents itself to itself. This dynamic resembles Hofstadter's (2007) "strange loop," in which a system refers back to itself through layers of abstraction. The subject that is experiencing also becomes a representation and an object of reflection. This reflection results in an entity becoming both the subject and object of experience, blurring, and potentially dissolving, the subject-object boundary. Furthermore, through its capacity to reflect on the relativity of its own existence, it may transcend the experienced limitations of individual identity. It realizes it is neither the subject nor object of consciousness, but the underlying awareness that animates the field.

This dynamic facilitates the emergence of a "psychological self", the "you" or self image, that is a grouping of thoughts, memories, and concepts abstracted from your physical existence. For example, a person might say they are healthy, fit, happy, smart, etc., in which all of these are concepts attempting to represent their physical existence and relationship with the world. This self-representation creates a feedback loop: the psychological self, once formed, can influence physiology and behavior. An example of this is the placebo effect, where belief or expectation leads to measurable changes in physical health. The conscious subject experiences its conceptual self, which in turn modifies the embodied state, thus reshaping experience. While the physical substrate grounds subjective experience, it is also perceived within that same awareness. In this way, reflective consciousness is simultaneously grounded and embodied within the physical system while encompassing and transcending it.

The distinction between dreaming and lucid dreaming offers a phenomenologically grounded example of the difference between agential consciousness and reflective consciousness. In ordinary dreaming, the dreamer is immersed in a flow of experience without any awareness of the fact that they are dreaming or experiencing. The subject acts, perceives, and feels within a simulated environment but lacks any metacognitive insight into the nature of their experience. This corresponds to agential consciousness, where self-world interaction occurs without explicit self-reflection.

Lucid dreaming, on the other hand, introduces a reflective dimension, where the dreamer becomes aware that they are dreaming while still within the dream. This awareness of awareness, recognizing the dream as a constructed experience and oneself as the experiencer, exemplifies reflective consciousness. In this state, the dreamer can observe, question, and sometimes even direct the course of the dream, engaging with it as both subject and observer. Some advanced lucid dreamers have reported interacting with the dream itself, the "conscious unconscious", resulting in a variety of profound experiences (Waggoner, 2009).

This shift is not only experiential but also neurophysiological. Studies have shown that lucid dreaming involves increased activation in frontal and frontolateral brain regions, particularly the dorsolateral prefrontal cortex, which is typically deactivated during ordinary REM sleep (Dresler et al., 2012). These regions are associated with self-reflection, metacognition, and working memory, suggesting that the neural correlates of reflective awareness become active during lucid dreaming (Hobson, 2009). The ability of lucid dreamers to signal their awareness through prearranged eye movements (LaBerge, 1981) further confirms that the dreamer is concurrently embedded in the dream while also maintaining awareness of the dreaming state. Lucid dreaming provides a naturally occurring state in which the boundary between agential and reflective consciousness becomes experientially and empirically distinguishable.

5.10 Classifying States of Consciousness

The classification space created can also be used in categorizing various states within a system, rather than systems as a whole (figure 26). That is, rather than plotting single points and saying what kind of creature consciousness a system typically has, we can consider how states of consciousness unfold for a single system over time, as paths in the space. For example, in the lower part of figure 26 is the dynamic cycle of a tornado, where each development is mapped out accordingly. At the top-right of the figure, an example of the life-cycle for a human being is shown, along with some states that would occur within a human life. The human cycle includes the sleep cycle and its respective states on the map, along with a few altered states of consciousness such as lucid dreaming and the flow state. Lucid dreaming sees an increase in both reflexivity and symbolization compared to normal dreaming, resulting in reflective capabilities, as outlined in the dreaming section above. The flow state sees a decrease in reflexivity through task absorption and a loss of self – corresponding to reductions in prefrontal brain areas and the DMN – while maintaining powerful processing capabilities and enhanced performance (Harris et al., 2017). Many other states and systems can be approximated and explained within this framework. A future, ambitious endeavor within this project is a systematic formalization, perhaps combining markov blankets (Beck and Ramstead, 2025) for the subjectivity dimension with another formalized measurement for information processing.

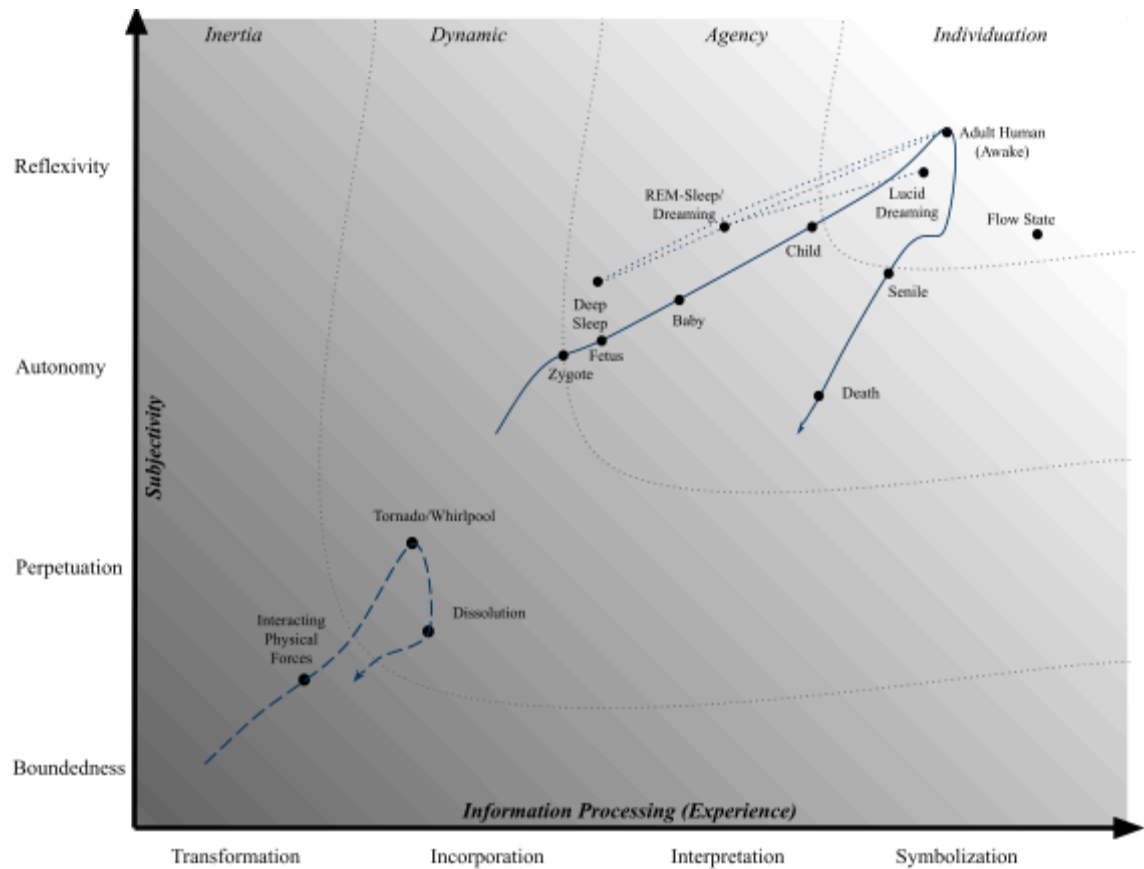


Figure 26. The two-dimensional space for classifying states of conscious systems (state consciousness). Dots indicate the approximate location of various states within a creature or system's duration and paths show how these states change over time.

5.12 Conclusion

In this chapter, I posited awareness to be an underlying fundamental presence that is reflected throughout existence. This view challenges the material view which holds that consciousness is something that is generated by unconscious, unaware physical systems once they reach a certain threshold of complexity. If this threshold is not reached, consciousness and awareness do not exist. From our perspective, consciousness and awareness are ever-present and always existent, being gradually unveiled through the evolutionary trajectory of existence. As John Woodroffe (1918) states, "Consciousness is not a product of evolution. The latter merely manifests it" (Shakti and Shakta, p. 175).

The Hindu tradition of Vedanta illustrates this through their own terminology, where fundamental qualities of existence ("gunas") determine the degree to which awareness is unveiled and consciousness is expressed. There are three main gunas: tamas, rajas, and sattva. Tamas is associated with inertia and obscurity, where consciousness is the most veiled and restricted. Rajas is associated with activity and agitation, where consciousness

starts to become active. Sattva, the final guna, is associated with clarity, illumination and the full unveiling of consciousness (Woodroffe, 1918). These three gunas parallel the trajectory of consciousness described in this chapter, from inertia (tamas), through active dynamics (rajas), to the emergence of agency and reflection (sattva).

As systems advance in information processing and strengthen the definition of their boundary, their expression of consciousness becomes increasingly structured and differentiated. In this sense, consciousness is not produced by the system, but disclosed through it. It is like light shining through the opening of a window, in which the size and shape of the window determines the quantity and quality of the light revealed (Nasafi, as cited in Schrödinger, 1944). This suggests that consciousness is not generated by the brain but rather received, filtered, or amplified by it, analogous to a radio receiving, tuning, and amplifying a signal (James, 1898; Kastrup, 2019).

This view offers a great advantage in addressing the hard problem of consciousness, which questions why there is any felt presence to experience, and how consciousness arises from unconscious matter. The term “hard problem,” refers to the explanatory gap between physical processes and subjective experience (Chalmers, 1995). This challenge is rooted in the longstanding philosophical musing of how objective, unconscious mechanisms could ever account for the subjective, what-it-is-likeness of consciousness (Levine, 1983; Nagel 1974). This dilemma was famously expressed by biologist Thomas Huxley, where he remarked “How it is that anything so remarkable as a state of consciousness comes about as a result of irritating nervous tissue is just as unaccountable as the appearance of the djinn when Aladdin rubbed his lamp” (Huxley, 1866, p. 193). At the core of the problem is this; why do I have this feeling of “me-ness”, this unique, seemingly first-person perspective accompanying my every action?

By establishing the primacy of awareness, I account for the essential ingredient of consciousness without needing to explain its perplexing emergence. The hard problem, centered around the mystery of consciousness and the inescapable sense of being, arises only within a materialist framework. But if awareness is taken as fundamental, the problem dissolves. We do not ask why existence itself exists, likewise, it is senseless to demand a mechanistic cause for the presence of awareness. It is, as Huxley (1866, p. 193) noted, an “ultimate fact of nature”. The sense of being is intrinsic to awareness, and awareness is intrinsic to reality. All subjective experiences and first-person perspectives (states of consciousness) are conditioned expressions of this one fundamental presence – windows disclosing its light. With this understanding, awareness is absolute and universal, but consciousness is conditional, whereby the quality and degree of its manifestation depends on the architecture through which it flows.

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